



Essays on the Causes and Demographic Consequences of Employment Uncertainty

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Vincent Jerald R. Ramos

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Advisors

Co-Advisor

Prof. Mark Hallerberg, PhD
Hertie School

Co-Advisor

Prof. Dr. Michaela Kreyenfeld
Hertie School

Third Advisor

Prof. Anna Matysiak, PhD
University of Warsaw

Fourth Advisor

Prof. Daniele Vignoli, PhD
University of Florence

Executive Summary

Labor market positions are simultaneously determined and deterministic. During and after episodes of economic shocks, it is not uncommon to see spikes in both macro- and individual-level measures of employment uncertainty including unemployment, fixed-term employment, and informality, among others. At the same time, these experiences of employment uncertainty have profound implications for demographic behavior and other life course outcomes. During the peak of the COVID-19 pandemic, for instance, birth rates fell in most countries, and whether these short-run fluctuations translate to long-run and cohort-specific effects remains to be seen. Recognizing these interrelationships, this dissertation is a collection of essays exploring narrower questions around the concept of employment uncertainty.

What do we mean by “*employment uncertainty*”? **Chapter 1** summarizes insights from seminal and contemporary empirical literature, revisits existing typologies and operationalizations of this concept, and demonstrates the overarching framework of this dissertation. This chapter also presents updated estimates of macro-level associations between measures of employment uncertainty (e.g., unemployment, vulnerable employment, and self-employment rates) and total fertility rates using dynamic panel models.

How did stringent COVID-19 restrictions affect informal employment? **Chapter 2** leverages the regional variation in the imposition of extreme COVID-19 lockdowns in the Philippines to estimate its effects on informal employment. Using a difference-in-differences design on pooled Labor Force Surveys, the paper shows that extreme lockdowns increase the probability of informal employment for women more than men. This effect is most pronounced for mothers with minor children, highlighting how some women during the pandemic played a dual role as workers and carers of first resort. In the analysis, a conceptual and empirical distinction is made between compositional and survivalist informalization.

How do initial (uncertain) employment conditions affect subsequent fertility behavior? **Chapter 3** leverages inter- and intra-cohort differences in employment conditions to estimate its effects on fertility behavior within the first ten years after labor market entry. Using the German Socio-Economic Panel, the paper identifies initial employment uncertainty in two ways— working on a fixed-term contract and starting during a recession. To partly address endogenous selection into initial conditions, the paper employs a two-step identification strategy combining an optimal full matching step and an event history modeling step using the matched data. The paper demonstrates that women who enter with a fixed-term contract and males who enter during a recession experience lower first birth hazards up to a decade after entry.

Are there class-based differentials in the transition to second-order births? **Chapter 4** analyzes the association between occupational class membership and second-birth transitions in Germany, testing whether workers from higher social classes possess an “economic advantage” for fertility decisions. Using event history models on data from the German Socio-Economic Panel, the paper reveals this to be the case. Men and women in service classes, as well as those who experience upward career mobility post-first birth, experience strongly elevated second-birth rates.

How do future caregiving responsibilities and employment uncertainties affect fertility decisions? Departing from the past- and present-orientedness of the preceding chapters, **Chapter 5** analyzes the effects of *future* caregiving and employment scenarios and narratives on current ascribed fertility intentions. Using a factorial survey experiment on a sub-sample from the German Socio-Economic Panel, respondents are randomly exposed to hypothetical scenarios of couples with varying levels of future caregiving responsibilities and employment uncertainties. Results show that couples without these responsibilities and uncertainties have higher fertility intentions relative to when these are high. Further testing for heterogeneities by the respondents’ gender and own employment markers, results lend support to a gendered difference in the effect of future caregiving responsibilities, whereas neither objective nor subjective measures of respondents’ employment markers moderate the effect of future employment uncertainty.

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The Hull—Base of the Ship

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Chapter 1

Introduction

“The concept of class is organized around one common principle—it is the kind of chances in the market that determines the common conditions of individuals’ fate.”

-Max Weber

Waters et al. (2010) translation

What affects people’s labor market positions and how do these positions shape subsequent demographic behavior? A broad and relevant question that spanned centuries of thinkers, its re-examination is warranted in an era of post-crisis recovery, pronounced demographic changes, and scientific advances in analyzing causal relationships. Yet such a question warrants a more nuanced contextualization—there is perhaps no uniformity neither to our understanding of what constitutes employment uncertainty nor its centrality in shaping life course outcomes. The recent COVID-19 pandemic revealed that people’s labor market positions (e.g., whether they are unemployed, in short-time work, informally employed, etc.) are, on the one hand, partly determined by the severity of the pandemic and its accompanying policies and, on the other hand, significant determinants of demographic outcomes such as fertility and parenting intensity (Sobotka et al. 2023; Naujoks et al. 2022; Maurizio et al. 2023; Chodorow-Reich and Coglianese 2021).

However, the directionality and magnitude of these relationships likely vary based on context and subpopulation groups. In prior work analyzing how recessions affect

fertility, women in cohorts that entered the labor market during the 1998 Asian Financial Crisis in South Korea experienced lower completed fertility (Choi et al. 2020), while women labor market entrants during the 1980 recession in the US eventually had higher completed fertility rates relative to comparable non-recession cohorts (Schwandt and von Wachter 2020). This difference can be attributed to various non-mutually exclusive reasons. One could point to a cultural or contextual explanation that women in the US and South Korea during those times might differently perceive the importance of economic factors in childbearing decisions. Another reason could be that the magnitude of the two recessions isn't exactly comparable—the negative effect of the 1997-98 Asian Financial Crisis on the South Korean economy far outweighed the negative effect of the 1980 recession on the US economy, at least when looking at GDP growth rates. Finally, it could also simply be an empirical reflection of the competing income and substitution effects in the economic analysis of fertility (Becker 1960). Recognizing the thick strand of research in this regard, this dissertation builds on the idea that more often than not, employment conditions matter even for non-labor market outcomes such as childbearing decisions.

Core to this dissertation is the argument that employment uncertainty is both a **determined** and a **deterministic** phenomenon. It is determined in the sense that most measures of employment uncertainty exhibit countercyclical patterns—unemployment, informality, fixed-term employment, short-time work, and subjective job worries, among others, tend to increase during economic downturns. This is not a new idea—standard aggregate demand-aggregate supply models in economics posit that as a response to an exogenous shock in demand (for example, due to the COVID-19 lockdowns), aggregate output is depressed and this ripples through lower demand in input markets including for both capital and labor. When firms demand *less* labor, how and why does employment uncertainty rise? One simple and obvious answer is the rise in unemployment due to layoffs. Unemployed people, by definition, are willing and available to work but cannot look for one and, consequently, are in an uncertain labor market position. Among developing countries, however, another risk-coping strategy during recessionary periods is to engage in informal work instead, another type of uncertain employment. Given the difficulties of finding opportunities in the formal sector, some people are constrained to work informally in temporary, irregular, low-paid, and sometimes precarious conditions. This “survivalist motive” also partly explains how uncertain employment conditions are themselves outcomes **determined**

by exogenous shocks and crises.

Employment uncertainty is likewise **deterministic**— future family formation and labor market outcomes are in part a function of past, present, and future employment uncertainty. For instance, the Global Financial Crisis, which unambiguously increased rates of employment precarity at least in the short-run, affected not only individual labor market trajectories but also family formation outcomes, especially in Europe and the US. Looking at population-level measures, Comolli et al. (2021) show that there has been a steeper decline in marriage rates and a halt on increasing fertility rates in Nordic countries during and after the 2008-09 crisis. In a comparative context, Comolli and Bernardi (2015) find a permanent positive effect of the recession on childlessness among women in their late 30s in the US but Caltabiano et al. (2017) does not find this effect to hold for Italy. Apart from this, the relationship between business cycles and fertility has also been extensively studied in the literature from both a macro and a micro perspective (see Caltabiano et al. 2017; Hofmann et al. 2017; Comolli and Bernardi 2015; Kreyenfeld and Andersson 2014, for example). Recently, more attention has been devoted to the explicit framing of economic uncertainty as an idea or perception of a “future” narrative (Vignoli et al. 2020a; Lappegård et al. 2022).

Understanding employment uncertainty as both determined and deterministic, this dissertation is a collection of four stand-alone chapters broadly on its causes and demographic consequences. The rest of this chapter motivates the importance of this topic and the selection of specific contexts as well as presents broader trends and patterns of employment and fertility. In Chapter 2, informal employment is demonstrated to increase as a result of extreme lockdown enforcement in the Philippines but only for women, particularly mothers with minor children. This evidence highlights women’s dual role as workers and carers of first resort. In Chapter 3, initial employment conditions are demonstrated to affect subsequent fertility behavior— female fixed-term entrants and male recession entrants have lower first birth hazard rates within the first 10 years of entry, relative to their permanent and non-recession counterparts. These findings imply the persistent nature of initial conditions that affect later life outcomes. In Chapter 4, occupational class positions are shown to be a relevant gradient of higher-order births such that men and women in relatively economically advantaged service classes experience elevated second birth rates relative to workers in lower occupational classes. Finally, in Chapter 5, future employment uncertainty and forthcoming caregiving responsibilities are demonstrated to negatively affect fertility

intentions, implying that multiple and simultaneous experiences of uncertainty hamper and threaten the enabling conditions of initiating parenthood.

1.1 Operationalizing Employment Uncertainty

Subtly highlighted in the preceding section is that employment uncertainty is best understood as an umbrella term that comes in various shapes and forms. By textual definition, “*uncertainty*” simply involves imperfect, incomplete, or unknown information. An informally employed worker, for instance, has no certainty that s/he will have a job the next day nor that s/he will receive a salary by a certain date absent any job contract. Similarly, a fixed-term worker has no certainty that s/he will receive a contract extension or a permanent position upon the completion of the current contract. The commonality between these two workers is that their current labor market positions inherently involve uncertainties that, say, their formally employed and permanently employed counterparts do not. In some spaces in the labor literature, uncertainty is interchangeably used with “*precarity*”, both denoting a certain degree of *inferiority* and *involuntariness* relative to non-uncertain positions. While the social stratification literature has thoughtfully threshed out and defined varieties and levels of precariousness within the labor market (see Standing 2012, for example) dating back to Weber et al. (1922) and other earlier thinkers, there still seems to be no collective or universal definition of what constitutes employment uncertainty.

Among the more recent attempts to schematize uncertainty is that of Blossfeld et al. (2006) who characterize three types of (overall) uncertainty—economic, temporal, and employment. The first refers to the uncertainty of one’s employment and educational circumstances, the second refers to the growing preference against long-term binding commitments, and the last refers to one’s job stability owing to being in the public sector (versus private) and with a permanent contract (versus fixed-term). While this schema appropriately constructs the conditions of what makes a certain socio-economic position uncertain in a hyper-globalized era, it can be argued that it is mainly limited by its dimensions. The types of uncertainty in this schema are far too few to capture its other manifestations. For instance, in most developing countries, informal working arrangements are more of the norm rather than the exception and these carry way more uncertainty than formally employed workers with automatic enrollment and access to social security. Further, uncertainty may also emanate from

occupational class positions as skilled classes benefit from more opportunities and better working conditions as economies transform and increasingly become service-oriented.

Thus, this section attempts to expand our collective understanding and description of uncertainty, capturing not only positional markers of labor market status (e.g., employed vs. unemployed vs. inactive) but also objective markers of employment arrangements and subjective indicators of uncertainty. Table 1.1 summarizes the various forms and typologies specifically of employment uncertainty and each is subsequently discussed in more detail.

Table 1.1: Forms and Typologies of Operationalizing Employment Uncertainty

Type of Uncertainty	Forms and Categories
Positional	- Employed
	- Unemployed
	- Inactive
Durational	- Fixed-term
	- Permanent
Statutory	- Informal
	- Formal
Recessionary	- Recession
	- Non-Recession
Class-Based	- Occupational Schemas (e.g. EGP)
Subjective	- Worried
	- Not worried
Caregiving	- Caregiving
	- Non-caregiving

Positional Uncertainty. In official labor market statistics, the working-age population is typically classified into one of three states— employed, unemployed, and inactive. These individual states are mutually exclusive labor market *positions*, of which some inherently possess more uncertainty relative to others. From internationally harmonized definitions, unemployment refers to people willing and available to work, and looking for work but cannot find one. In comparison with people who want to and are available to work and are currently working (i.e., employed) and people who neither prefer nor are able to work and are not looking for work (i.e., inactive), unemployed people are relatively in more uncertain positions. Positional uncertainty in this dissertation thus encompasses the inherent uncertainty of the job search process experienced by being in a state of unemployment. In prior literature examining the

unemployment-fertility nexus, some have used macro-level measures of unemployment rates at the national or subnational level (Sobotka et al. 2011; Currie et al. 2014; Matysiak et al. 2021a; Schneider 2015), whereas some used individual-level states of unemployment (Kreyenfeld 2010; Andersen and Özcan 2021; Adsera 2011).

Durational Uncertainty. While many “typical” employment contracts have an open-ended and unlimited duration (with some countries offering strong protections from layoffs), a sizable segment of the employed population works in time-bound or term-limited arrangements without any guarantee of renewal, extension, or regularization upon its expiration. Paradoxically, having a fixed-term contract carries less uncertainty on one hand in that there is clarity and precision as to one’s employment duration but, on the other hand, more uncertainty in terms of future job prospects and, in some contexts where contractual workers have less or no coverage, access to certain entitlements (e.g., parental leaves, health insurance, etc.). Kalleberg (2000) argues that fixed-term workers in the US contend with elevated turnover rates and heightened job instability, thereby perpetuating a sense of insecurity, financial strain, and anxiety among fixed-term workers. In Germany, fixed-term workers have been shown to have lower earnings, lower subsequent wage growth, and higher risks of future unemployment relative to permanent contract workers (Giesecke 2009). The same wage penalties for fixed-term workers within a decade post-entry have likewise been demonstrated in Australia (Fauser and Mooi-Reci 2024). In this dissertation, durational uncertainty thus refers to the inherent perceived and actual instability and precarity of being in a fixed-term contract. A durational operationalization of employment uncertainty using fixed-term contracts is common and has been done in prior literature investigating its effects on subsequent fertility behavior (Auer and Danzer 2016; Schmitt 2012, 2021; Gebel and Giesecke 2009; Vignoli et al. 2020b).

Statutory Uncertainty. A majority of workers in low and lower-middle-income economies are in informal employment arrangements that are either not covered or governed by employer-employee contracts or not declared to tax and public insurance authorities or both. Take for example the case of casual street vendors or freelance output-based workers—these are classified as employed in official statistics but do not receive the same types of protection and entitlements as formally employed workers do. There are various definitions of informal employment across countries and to date, there is no universally recognized operationalization of who is considered informally employed. Benanav (2019) offers a historical account of this lack of harmonization and

points to the ILO’s evolving constructions and definitions of “informal employment” and “employment in the informal sector”. The absence of contractual arrangements and proper statutory declarations and registrations (e.g., tax filings) of informally employed workers makes them inherently susceptible to less protection and access to entitlements (e.g., insurance coverage), if at all existent. In terms of the personal repercussions of informal employment, Karabchuk (2020) shows that these workers express lower fertility desires while Ducanes et al. (2023) find that they have lower wages and longer work hours, and are more likely to have no rest days and work below minimum wage rates.

Recessionary Uncertainty. Employment uncertainty likewise has a cyclical element in that the timing of labor market entry per se matters. All other employment conditions held constant, recession cohorts have been demonstrated in the literature to have lower wage trajectories (von Wachter 2020b,a; Oreopoulos et al. 2012), and different family formation patterns (Hofmann and Hohmeyer 2016; Raymo and Shibata 2017; Choi et al. 2020). The uncertainty experienced by recession entrants is multi-faceted. First, there are fewer job opportunities during periods of economic downturns, thereby potentially lengthening job search and lowering initial wages. Second, depending on the magnitude and severity of these downturns, the negative effects on subsequent wage and employment outcomes may persist up to decades after entry, thereby generating cohort-specific “scarring” phenomena (Rothstein 2021; Huckfeldt 2022). Beyond labor market entrants, the fertility decisions of individuals are negatively affected by other indicators of macroeconomic uncertainty including Google searches of the term “spread” (Comolli and Vignoli 2021) and media coverage of economic news (Guetto et al. 2023b).

Class-Based Uncertainty. Emanating from the Weberian distinction of types of stratification between a prestige-based “*Stände*” (Status) and material or resource-based “*Klasse*” (Class) (Weber et al. 1922), decades of succeeding literature has operationalized class schema using occupations and demonstrated inter-class differentials in both attitudinal and behavioral outcomes including employment, political beliefs, cultural consumption, and family formation (Erikson et al. 1979a; Chan and Goldthorpe 2007; Goldthorpe and McKnight 2006; Baizan 2021; Kreyenfeld et al. 2023b). Central to contemporary operationalization of social class is its anchoring on occupational categories, assuming that people within the same class are relatively similar in terms of their material resources and exposure to economic and political

cycles. In Erikson et al. (1979a, 1983), class positions are classified into nine-, seven-, five-, and three-fold schemas, where distinctions are made between the “service” class that encompasses workers in managerial and professional occupations and the “unskilled” class representing workers in manual and routine occupations including farm work. Thus so, this dissertation also adopts a class-based operationalization of employment uncertainty wherein workers in lower-class positions are exposed to greater levels of economic insecurity and susceptibility to economic trends and fluctuations (e.g., offshoring, recessions, automation, etc.) that service-class workers may not similarly face.

Subjective Uncertainty. Perhaps the most straightforward measure of uncertainty is one’s subjective worries about their labor market positions. Rather than making class-based or category-based imputations of uncertainty that presuppose intra-class homogeneity, subjective measures of job worries are direct and able to capture individual-level differences. Many publicly available representative surveys contain questions including “*How concerned are you about your job security*” (German Socio-Economic Panel) or “*How satisfied are you about your job security*” (UK Household Longitudinal Survey). Responses to these questions are typically measured on a Likert scale, thus allowing straightforward comparisons. People with relatively higher subjective job-related worries have been demonstrated in the literature to negatively correlate with family formation in the form of lower fertility intentions or delayed childbearing (Kreyenfeld 2015; Matera et al. 2022; Van Wijk et al. 2021; Hofmann and Hohmeyer 2013).

Caregiving uncertainty. While all the preceding types and operationalization of employment uncertainty pertain to differences within *paid* employment categories, there are of course uncertainties emanating from unpaid work as well. Caregiving responsibilities, either toward children or elderly parents, are strongly gendered and cause observable penalties in employment outcomes and subjective well-being. Kleven et al. (2023) demonstrate that across the world, motherhood penalties in terms of employment rates are pronounced and persistent, although these are smaller for low-income countries. Understandably so, intergenerational coresidence being more common in these contexts and the abundance of local domestic workers and caregivers may cushion the employment impacts of motherhood. In terms of overall caregiving work, Schäper et al. (2023) shows that women perform around twice more than men do. The presence of caregiving responsibilities, therefore, generates uncertainty in that

caregivers face greater time demands outside of paid employment than non-caregivers. This consequently affects their employability, available work hours, health outcomes, and even fertility decisions (e.g., having a child/another child). It is important to note that while caregiving responsibilities arguably increase burden of workers and their personal precarity, it is also inherently **uncertain**. As in the case of caring for a sick parent, for instance, there is genuine uncertainty about when this need would arise, how much caregiving is required, and how long this caregiving responsibilities persist.

Note that this schema is an extension of the schematic illustration of uncertainty in prior work. While Blossfeld et al. (2006) characterize uncertainty as either economic, temporal, or employment, this section narrowly focuses on employment uncertainty, expands its typologies and markers, and incorporates different terms that may be more appropriate descriptions. For instance, what Blossfeld et al. (2006) coin as “economic” is called “positional”, whereas “employment” is termed “durational”.

1.2 Employment Uncertainty as a Global Phenomenon

Employment uncertainty is a pervasive global phenomenon explained by non-mutually exclusive factors including globalization, technological change, business cycles, and concentration, among others. Indeed, the COVID-19 pandemic highlighted the fragility of global labor markets, causing widespread displacement estimated to be approximately 255 million full-time jobs (Berg et al. 2021). Regardless of *how* employment uncertainty is operationalized, it can be argued that it is generally a countercyclical phenomenon that increases during periods of economic shocks. This section shows historical trends and patterns of employment uncertainty and a closer look at the Philippine labor market, one of the world’s most significantly impacted countries during the COVID-19 pandemic.

1.2.1 Trends in Employment Uncertainty

Despite schematically outlining the typologies of uncertainty in the preceding section, the greater challenge lies in measurement. Whereas some measures have internationally agreed-upon definitions and harmonized estimates (e.g., unemployment rate, identifying recession years, etc.), all others either do not have common definitions

(e.g., informal employment) or are not regularly and consistently measured (e.g., fixed-term employment, class schemas, subjective worries, and caregiving). Thus, to illustrate global trends we rely on ILO-harmonized historical estimates of two measures of employment uncertainty— (1) unemployment rate (as a share of the total labor force); and (2) vulnerable employment rate (as a share of the employed population). While the former straightforwardly refers to the share of the labor force looking for, available, and willing to work but can’t find one, vulnerable employment is an ILO-formed concept referring to the share of the employed population who are contributing family workers and own-account workers. To some extent, this definition from the ILO captures what we mean with the term “statutory uncertainty” in the preceding section (informally employed vs. formally employed workers). This ILO measure captures workers who are in undeclared, informal, or piece-rate work (e.g., street vendors, domestic workers, etc.) and has the advantage of being harmonized across many countries.

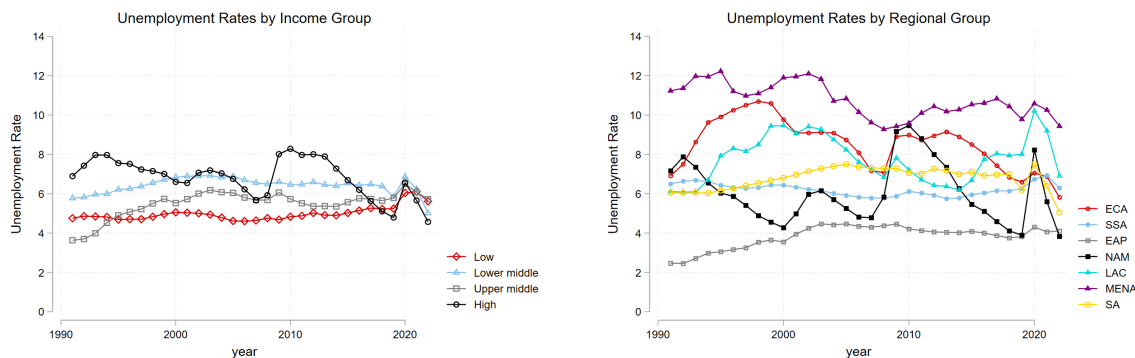


Figure 1.1: Global Trends in Unemployment Rate (1991-2022)

Note: Income groups are based on the latest classification by the World Bank using per capita Gross National Income while regional groups are based on the World Bank’s administrative divisions. Harmonized ILO estimates of unemployment rates are used to ensure comparability.

In Figure 1.1, it is interesting to note how the cyclical nature of unemployment rates is more pronounced among high-income economies— most notably, the spike in unemployment for this group during the Global Financial Crisis in 2008-09 which was not reflected among low and lower-middle income economies. The same observation can be made when comparing regional groups. On the contrary, the COVID-19 pandemic did cause unemployment spikes across all regions and income groups, albeit in different magnitudes. Interestingly, low-income countries have, on average, lower unemployment

rates than high-income countries. Since the latter typically have more extensive social welfare systems, including income replacement programs and other forms of social assistance, they can also lead to higher reported unemployment rates compared to low-income countries. In these contexts, displaced workers in low-income countries may either not be counted as part of the labor force if they are not actively seeking work, consequently shrinking the size of the labor force and the reported unemployment rate or they may transition towards some kind of informal work since they cannot afford unemployment, thereby minimizing observable transitions to unemployment.

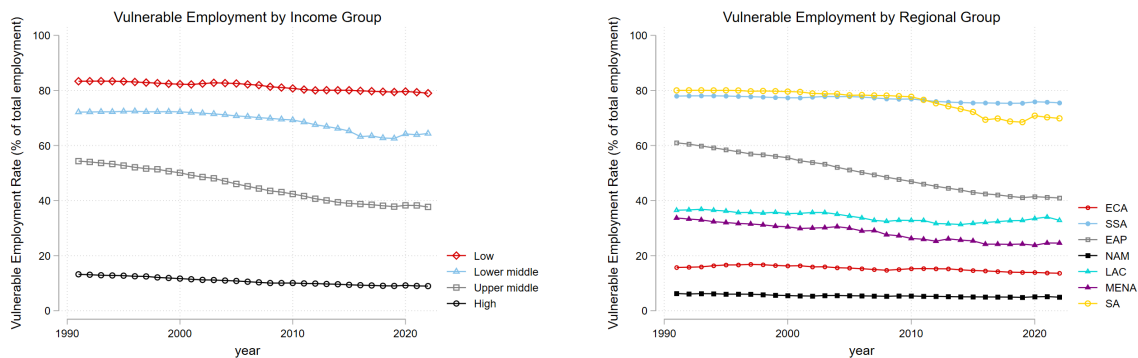


Figure 1.2: Global Trends in Vulnerable Employment Rate (1991-2022)

Note: Income groups are based on the latest classification by the World Bank using per capita Gross National Income while regional groups are based on the World Bank's administrative divisions. Harmonized ILO estimates of vulnerable employment rates are used to ensure comparability. This rate covers contributing family workers and own-account workers as a percentage of total employment.

With regard to the size of the vulnerably employed population, Figure 1.2 shows stark differences across income and regional groups, as expected. Whereas high-income and developed economies, in general, have persistently low shares of vulnerable employment, a vast majority of workers in low and lower-middle-income countries are in vulnerable arrangements. In South Asia and Sub-Saharan Africa, they comprise about 80% of the employed workers and have remained at that level for over 3 decades. This category encompasses various forms of informal employment and other precarious arrangements that leave workers vulnerable to exploitation, poverty, and social exclusion. By definition, vulnerable workers have limited access, if at all, to social safety nets, including unemployment insurance, disability benefits, or pension schemes, making them all the more susceptible to shocks.

Looking at the figure more closely, it is interesting to note that by and large, the vulnerable employment rate (VER) is not that sensitive to business cycle fluctuations. Even among high-income economies where unemployment spikes are pronounced during past periods of crises, VER has been relatively stable. This implies that even if the employed population shrinks, its composition (between vulnerable and non-vulnerable workers) remains stable. However, there seem to be some exceptions to this pattern. Among lower-middle-income economies and in South Asia, East Asia and the Pacific, Sub-Saharan Africa, and Latin America, there seems to be a very small increase in the VER in 2020.

1.2.2 Country-in-Focus: Philippines

These observations from Figure 1.2 on the VER dynamics in 2020, at the height of the COVID-19 pandemic, necessitates the importance of decoupling the effect of a certain shock (e.g., COVID-19) on unemployment and on the size of vulnerable (or informal or precarious) employment. Indeed, one can observe a “compositional” increase in VER even if the employed population shrinks if the decline in non-vulnerable is larger than the decline in vulnerable employment. The COVID-19 pandemic and its ensuing lockdowns present a unique opportunity to better understand the dynamics of vulnerable and/or informal work in the presence of an exogenous shock.

What happens to informal employment in developing countries during economic downturns? Looking at historical accounts could be an insightful starting point. There is some consensus that informal employment increases as a response to economic shocks (Albertini et al. 2019; Colombo et al. 2019). During and after financial crises episodes in Latin American countries in the 1990s, the Argentine peso crisis in 2001, and its spillovers in Uruguay in 2002, informal employment rates increased and remained high for years succeeding the crises (Colombo et al. 2019). This implies that during a recession when demand for labor is depressed, involuntary employment serves as a *cushion* for job displacement which workers can do rather than be unemployed or inactive. Since then, there have been other, more severe, crises that emerged such as the COVID-19 pandemic.

The second chapter of this dissertation takes a closer look at how informal employment in the Philippines responded to the imposition of extreme lockdowns in select regions. The Philippines is widely considered to have imposed one of the world’s longest and strictest lockdowns that were enforced in part through military

action (Reuters 2020; Olanday and Rigby 2020; See 2021), making it an interesting and relevant case study. The central thesis in this dissertation is that (statutory) employment uncertainty, operationalized using informal employment, is **determined** by both economic uncertainty (e.g., the COVID-19 pandemic or the Asian Financial Crisis, both of which depressed aggregate demand) and the ensuing state responses (e.g., the imposition of extreme lockdowns or the adoption of austerity policies).

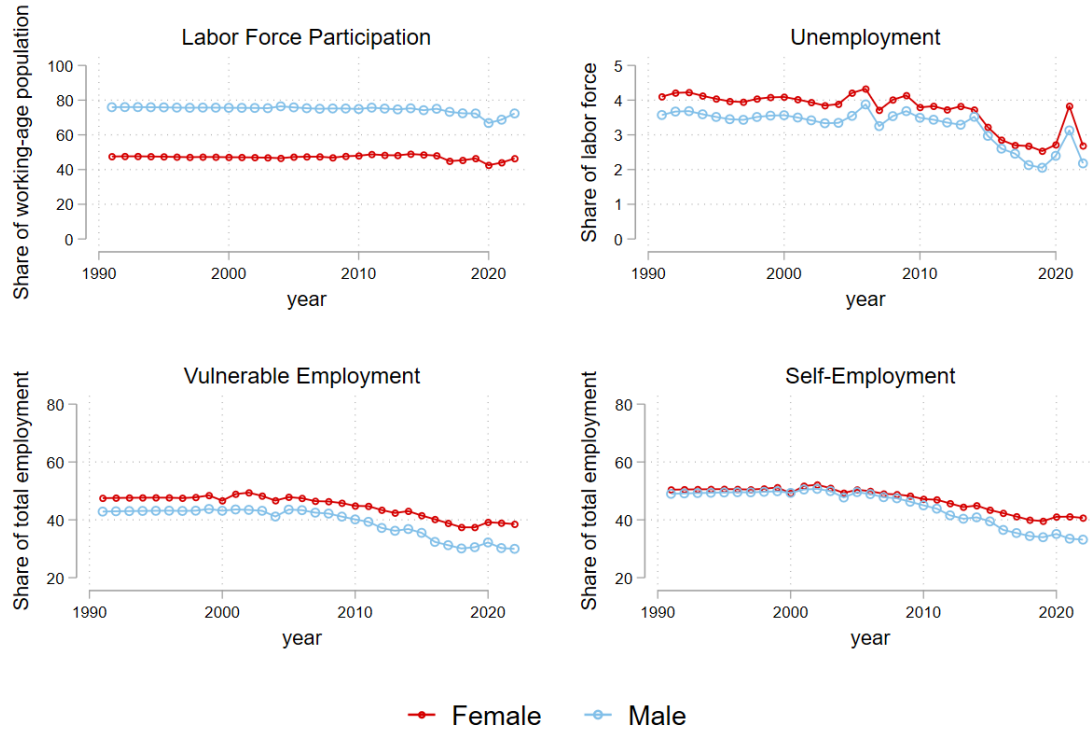


Figure 1.3: Labor Market Indicators in the Philippines, by Gender (1991-2022)

Note: These indicators are all based on ILO estimates.

As is the case in most other contexts, the Philippine labor market is visibly gendered and, in select indicators, has been increasingly so in recent years. Figure 1.3 shows 4 indicators— labor force participation, which is the sum of employed and unemployed people as a share of the working-age population, unemployment, vulnerable employment, and self-employment rates. Most notably, women have persistently lower labor force participation rates and higher unemployment rates, vulnerable employment rates, and over the last decade, also self-employment rates. Thus, more women are situated in uncertain labor market positions. The gendered

effect of the COVID-19 pandemic is barely visible in these descriptive figures— while the spike in unemployment in 2020 was larger for women than men, the increases in vulnerable and self-employment rates do not seem substantially larger for either sex.

There are limited possibilities to infer causal claims from these descriptive figures. For one, these estimates are aggregated at the national level, while government responses to the pandemic, such as the imposition of lockdowns, have been region-specific. Thus, this dissertation tests the question of whether employment uncertainty, measured through informal employment, is indeed **determined** by workers' exposure to extreme and militarized lockdowns and, if so, whether such effect is gendered.

1.3 Demographic Repercussions of Employment Uncertainty

1.3.1 Seminal Theories on the Role of Economic Conditions

Malthusian proposition. The idea that demographic behavior, and population dynamics more broadly, is a function of employment conditions is recurrent and historically well-examined. Malthus (1798) contends that there is a cyclicity (or “oscillations” as he puts it) between population growth and economic conditions and, considering that (geometric) population growth outpaces (arithmetic) food production growth, this cycle would eventually lead to poverty and famine. Touching on the interrelationships between scarcity, labor productivity, and family formation, central to the Malthusian proposition is that when a population increases and there is more supply of labor than the “proportion of work in the market”, wages fall. As a consequence of these low-waged employment conditions that necessitated longer work hours, marriages and births fall such that the population will stall, exerting upward pressure on wages and improvements in economic conditions. Family formation then becomes conducive and the population will increase, ultimately repeating the dismal cycle (Malthus 1798).

Malthus (1798) proposes solutions or “checks” to limit population growth and prevent famine and hardships— positive and preventive checks. The former are events that increase mortality (and decrease population growth) including famine, disease, war, and other natural disasters. These consequently increase death rates in a population, helping balance the size of the population and the available resources. These checks were regarded as inevitable to prevent population growth from exceeding

the carrying capacity of the environment. However, Malthus also recognized the accompanying suffering caused by these events. On the other hand, preventive checks are individual or social measures to consciously control population growth and limit births including late marriage, celibacy, and contraception. Unlike positive checks that affect mortality, preventive checks reduce population growth through a decline in fertility. Malthus (1798) believes that preventive checks are preferable because they offer a more humane means of controlling population growth. Ultimately, moral restraint and self-discipline are central to limiting family size. The dismal predictions of widespread famine and poverty due to unchecked population expansion have been recurrent, most notably with Ehrlich (1968) illustrating scenarios of future food insecurity and mass starvation as a result of the rapid population growth in the 20th century. Broadly, the Malthusian observation that family formation is negatively affected by a decline in wages implies that births are directly affected by labor market conditions.

Boserupian response. Contrary to the rather pessimistic implications of the Malthusian proposition, Boserup (1981) emphasizes human ingenuity, technological innovation, and resilience towards the challenges posed by population growth. Specifically, she argues that historically, societies innovated and developed new farming techniques such as crop rotation, irrigation, terracing, and animal cultivation to meet the increased demand for food. Therefore, increasing population densities serve as sufficient *incentives* for technological change rather than a precursor to eventual hardship. Another difference of the Boserupian argument is the role that social institutions play in facilitating innovation that causes some societies to adopt better, whereas Malthus' observations contain little to no nuance based on the type and structure of a society. In relation to employment conditions, Boserup (1981) shows that growing societies historically responded to increased demand for food particularly by employing more labor-intensive farming methods and technologies. Therefore, another implication of her work is that population growth could lead to increased employment opportunities in the agricultural sector. Contemporary scholars have carried on this argument—“human ingenuity” is an abundant and fast-growing resource that could innovate society out of scarcity (Simon 1981). Similarly, Lam (2011) contends that despite the seeming success of the increase in world food production, some regions, particularly Africa, have been less successful, owing to both social institutions and resource constraints.

Economic models. The economic analysis of fertility in a rational choice framework emanates from the seminal work of Becker (1960, 1981) where childbearing decisions are modeled similarly to consumption goods and the trade-off occurs between the quantity and quality of children. Thus, changes in income and relative prices ultimately determine fertility behavior. As a response to an improvement in income (or employment conditions), the “income effect” captures its positive relationship with fertility such that having higher incomes and more resources is an enabling condition for family formation. On the other hand, a “substitution effect” as a response to an income increase captures how individuals may reduce fertility decisions due to the increased opportunity cost of childbearing. Empirically, however, Becker (1960) observes that there is a lack of a strong income effect in the income-fertility nexus and explains this by introducing a “quality” component in the model. Simply put, an increase in income might be spent either on another child or additional investments in existing children. Overall, his work highlights the role of economic incentives and constraints in influencing fertility behavior and, since then, subsequent literature has extended this theory (see Wrede 2011, for example, where hyperbolic discounting is introduced in the model.).

Relatedly, Easterlin (1976), best known for the “Easterlin Paradox” of income levels and happiness, posits in the same seminal paper that whereas there is no relationship between those two factors, there is a (positive) relationship between income levels and fertility. He observes that fertility rates tend to be higher during periods of economic prosperity and lower during downturns, as shown by the post-World War II baby boom in the United States, which coincided with a period of economic growth and increased job opportunities. On the contrary, settlements with lower land availability for farming are shown to be associated with lower fertility.

In economics, there seems to be a shift in the theoretical underpinnings of studying fertility choices. An extensive review by Doepke et al. (2023), who studied under Gary Becker (Doepke 2015), highlights the “new economics of fertility” where evolving family structures and gender roles play a more central role. Doepke et al. (2023) emphasizes that whereas fertility used to decrease with income and female labor force participation in the past, now, among high-income economies, the association of these two variables with fertility is positive. He argues that part of what explains this is the shift from the quantity-quality tradeoff to a career-family tradeoff— women in high-income countries at present aspire for both career AND family stability. With this

development, the importance of labor markets becomes even more central since high-unemployment societies might convert temporary caregiving breaks of mothers to long-term unemployment and/or inactivity. This “new economics” brand thus recognizes the important role of both labor market and family policies (e.g., parental leave, subsidized childcare, and family planning, among others) in explaining fertility dynamics. In prior work, Doepke et al. (2019) shows that parenting intensification, where parents increase time and material investments towards their children, plays a role in the lowering of fertility rates. The more recent economics literature on employment conditions and fertility has therefore evolved into one that better takes into consideration social policies and changing norms.

Demographic Transition Theories. While economic fluctuations explain fertility decisions, these theories (at least the older ones) are incomplete in that they do not offer nor incorporate the role of ideational change in demographic developments such as cohabitation and secularization (Lesthaeghe 2010; Zaidi and Morgan 2017). The first and second demographic theories (FDT and SDT, respectively) incorporate both economic and labor market conditions on one hand and changes in ideas and norms on the other, to provide a fuller account of why and how societies transitioned from high fertility and mortality to low fertility and low mortality ones (FDT) and from low fertility societies to “lowest-low” fertility ones (SDT). In explaining the demographic changes in Western Europe from the 18th to the 20th century, Notestein (1945, 1952) shows that societies changed from high to low fertility and mortality in part because of significant improvements in healthcare and sanitation as well as industrialization and urbanization. Another important element during the FDT is the shrinkage of family sizes, shifting towards smaller nuclear families. These reasons, however, are perhaps insufficient to explain the shift of low fertility societies to even lower levels of fertility.

The SDT (Lesthaeghe and Van de Kaa 1986; Lesthaeghe 2010) thus emphasizes three components— structure, culture, and technology as the main drivers of modernization. In particular, the increase in individual autonomy, secularization, and female emancipation ultimately reduces fertility rates, in part through the childbearing postponement mechanism (Sobotka 2004). While most interpretations and empirical applications of the SDT highlight its ideation and culture component over employment uncertainty, Kotowska et al. (2008) argue that, in the case of Poland, labor market conditions and educational expansion (e.g., structure component of SDT) seem to be most pronounced. While originally formulated to explain changing demographic

patterns in developed Western societies, other work has likewise applied the SDT components to explain trends in other contexts. Lesthaeghe (2020) uses the SDT to explain increasing non-marital cohabitation in the Philippines, juxtaposed with strong stigmatization of this behavior by Catholics who comprise around 80% of the population.

Another phenomenon that occurs alongside the SDT is globalization and, relatedly, rising uncertainty in labor market positions (Blossfeld et al. 2006). As such, the processes alongside globalization, such as wage competition and a “race-to-the-bottom” approach, might partly explain the rise in employment uncertainty and, consequently, fertility postponement and/or reduction (Mills and Blossfeld 2013). Apart from an increased interconnectedness of markets, globalization likewise facilitates the flow of ideas, which in part explains the shift in cultural norms and values referred to by the SDT. Increased globalization and competition within labor markets necessitated countries to implement policies geared toward educational expansion. These have implications for fertility differentials across individuals (women, in particular) of different levels of educational attainment. In the UK, for instance, highly educated women have higher rates of childlessness, later ages at first childbirth, and a lower likelihood of meeting desired family size (Berrington and Pattaro 2014; Berrington et al. 2015).

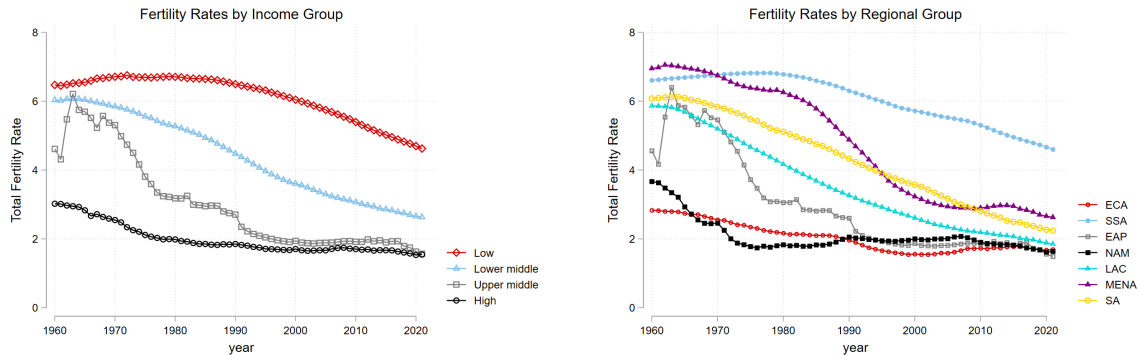


Figure 1.4: Global Trends in Total Fertility Rates (1960-2022)

Note: Income groups are based on the latest classification by the World Bank using per capita Gross National Income while regional groups are based on the World Bank’s administrative divisions. Total fertility rates (TFRs) estimate the average number of children per woman if she survives through her biological childbearing period and experiences the same age-specific fertility rates of that year.

Examining trends over time, total fertility rates have remarkably decreased over the last 60 years. Figure 1.4 shows that high-income countries, on average, saw a decline in fertility rates until 1990, which have plateaued since then. On the contrary, low and lower-middle-income countries have been experiencing a pronounced decrease in fertility rates. By regional groups, Sub-Saharan Africa has notably lagged in its fertility decline, at least compared to the Middle East and North Africa, South Asia, and Latin America, despite having about the same fertility rate in 1960.

1.3.2 Contemporary (Macro) Findings on Employment Uncertainty

Each operationalization of employment uncertainty in Table 1.1 has spurred separate strands of the literature, although some (e.g., positional and durational) are arguably more explored than others (e.g., class-based, caregiving) (Buh 2023). A meta-analysis of the effect of unemployment and fixed-term employment on fertility behavior in Europe reveals that this nexus is, by and large, negative but gendered (Alderotti et al. 2021). Notably, female fixed-term employment and male unemployment are found to negatively affect fertility, while the negative effect is most striking in Southern European countries where social protection systems and family policies are weaker (Alderotti et al. 2021). In each chapter of the dissertation, the literature on each type of employment uncertainty and its individual-level effects on fertility is examined in closer detail.

In lieu of focusing on the vast micro-level literature of employment uncertainty-fertility which the stand-alone chapters do so in detail, this section highlights macro-level patterns and trends. Macro (e.g., cross-country) approaches to studying the effects of unemployment, and macroeconomic conditions more broadly, on fertility rate have been extensively used. On one hand, fertility postponement was more prevalent in countries with high and persistent unemployment (Adsera 2005, 2011). Apart from postponement, unfavorable economic conditions in developed countries led to lower total fertility (Karaman Örsal and Goldstein 2018). Indeed, most studies in this area of research find fertility to be pro-cyclical—it typically decreases during periods of high unemployment and increases otherwise (Cazzola et al. 2016; Goldstein et al. 2013; Gozgor et al. 2021; Karaman Örsal and Goldstein 2018; Sobotka et al. 2011; Comolli 2017; Doepke et al. 2023).

Most of these studies model the country-level fertility rates as a function of lagged

unemployment rate (Goldstein et al. 2013; Karaman Örsal and Goldstein 2018; Gozgor et al. 2021; Comolli 2017; Bellido and Marcén 2019) (or at the sub-national level as done by (Matysiak et al. 2021a), while some papers employ multi-level methods combining individual-level fertility and country-level unemployment (Currie et al. 2014; Adsera 2005, 2011) or descriptive analysis of changing trends and associations (Sobotka et al. 2011; Doepke et al. 2023; Cazzola et al. 2016). These papers find a negative relationship between unemployment and fertility rates, although each paper carefully emphasizes heterogeneities such that the effect sizes are stronger for certain population subgroups like women in their 30s and middle-educated women (Comolli 2017) and in certain regions like Southern Europe (Goldstein et al. 2013).

On the other hand, fewer papers find that an increase in aggregate unemployment has a non-significant or even positive effect on fertility. Among this strand is that of Fernandez-Crehuet et al. (2020) that uses fractional integration and cointegration estimation methods and finds the absence of a long-run relationship between unemployment and fertility, as well as the absence of a short-run relationship in most of the developed countries in the sample. Further, unemployment has also been shown to increase fertility in England, suggesting that the substitution effect dominates the income effect, such that the low opportunity cost of childbearing during periods of high unemployment facilitates childbearing (Aksoy 2016). Similarly, Karaman Örsal and Goldstein (2018) and Doepke et al. (2023) find that among developed countries in the 70s and 80s, good economic times (e.g., high growth, low unemployment) are associated with lower fertility, implying that the direction of this association has indeed changed over time.

1.3.3 Extension I. Re-examining the Unemployment-Fertility Nexus

These contemporary findings of a macro-level relationship between unemployment and fertility suffer from two main limitations in scope and breadth. First, as it relates to scope, all the papers cited in the preceding section cover developed country contexts, mostly using OECD or European macro data. This constraint naturally arises because of the lack of long-run harmonized data on employment and fertility indicators in low and lower-middle-income countries. In effect, the trade-off that these papers faced was to use fewer countries (narrower geographic focus) with longer periods (more years) rather than more countries with shorter periods. While this approach strengthens the

internal validity of these papers by using reliable and consistent data, it also limits our understanding of the unemployment-fertility nexus to a very narrow group of countries. In other words, the “macro” unemployment-fertility papers we have to date are only macro to the extent of high-income countries. Recent work attempts to bridge this gap test the macro-level relationship between inequality and fertility in the Global South (Castro Torres et al. 2022), although this analysis does not cover labor market indicators.

In light of this limitation, this chapter re-estimates the unemployment-fertility nexus using a larger set of countries and data from the World Bank’s World Development Indicators. As a trade-off, this dataset only contains ILO-harmonized estimates of the unemployment rate for a vast majority of countries since 1990, although its fertility data dates back to 1961. As such, the unemployment-fertility regression models are estimated using panel data of at most 187 countries and territories from 1991-2022. In comparison, while this panel is still wider (in terms of countries) and longer (in terms of years) than the panels used in most other macro-level analyses (Goldstein et al. 2013; Matysiak et al. 2021a; Comolli 2017; Bellido and Marcén 2019), it is still not as long as other analyses that use data dating back to the 70s or 80s (Karaman Örsal and Goldstein 2018; Doepke et al. 2023).

Similar to most papers, the total fertility rate is modeled as a function of the lagged unemployment rate using fixed effects, thereby testing the hypothesis that the unemployment rate is negatively associated with fertility. Building on Goldstein et al. (2013); Karaman Örsal and Goldstein (2018), and Comolli (2017) by including additional macroeconomic and demographic controls, I estimate the model parsimoniously specified below:

$$F_{c,t} = \alpha + \gamma U_{c,t-1} + \delta D_{c,t-1} + \omega M_{c,t-1} + \theta Z_{c,t} + \epsilon_t, \quad (1.1)$$

where γ is the coefficient of interest, representing the effect of the lagged unemployment rate in country c in the past year on the fertility rate at year t . Further, D and M are vectors of demographic (population, age dependency, labor force participation, and infant mortality rates) and macroeconomic (real per capita GDP growth, inflation, and foreign direct investment inflows) time-varying lagged controls to capture the potentially confounding effects of the social and economic aspects of a country. Finally, Z is a vector of region and period fixed effects, whereas ϵ_t is the composite error term.

Table 1.2: Fixed Effects Regressions– Unemployment and Fertility

VARIABLES	(1) Base	(2) All w/DC	(3) w/MC	(4) Base	(5) Females w/DC	(6) w/MC	(7) Base	(8) Males w/DC	(9) w/MC
L.Unemployment Rate	-0.030*** (0.006)	-0.018*** (0.004)	-0.015*** (0.005)	-0.034*** (0.006)	-0.016*** (0.003)	-0.014*** (0.004)	-0.021*** (0.006)	-0.015*** (0.004)	-0.011** (0.005)
L.Infant Mortality Rate		0.011*** (0.002)	0.011*** (0.002)		0.013*** (0.002)	0.013*** (0.002)		0.010*** (0.002)	0.010*** (0.002)
L.LFPR		0.007 (0.005)	0.005 (0.005)		-0.002 (0.003)	-0.002 (0.004)		0.013** (0.006)	0.011* (0.006)
L.Total Age Dep Ratio		0.028*** (0.002)	0.029*** (0.003)		0.028*** (0.002)	0.028*** (0.003)		0.028*** (0.002)	0.028*** (0.003)
L.Log Population		-0.914*** (0.115)	-0.795*** (0.139)		-0.835*** (0.109)	-0.724*** (0.135)		-0.958*** (0.118)	-0.842*** (0.141)
L.Real GDPPC Growth			-0.002** (0.001)			-0.002** (0.001)			-0.002** (0.001)
L.Inflation CPI			-0.000** (0.000)			-0.000** (0.000)			-0.000** (0.000)
L.FDI Inflows			0.000 (0.000)			0.000 (0.000)			0.000 (0.000)
Year Dummies	Y	Y	Y	Y	Y	Y	Y	Y	Y
Constant	4.092*** (0.073)	15.263*** (1.955)	13.439*** (2.349)	4.160*** (0.071)	14.521*** (1.813)	12.760*** (2.245)	4.008*** (0.068)	15.380*** (2.006)	13.712*** (2.385)
Observations	5,610	5,370	4,648	5,610	5,370	4,648	5,610	5,370	4,648
R-squared	0.521	0.799	0.792	0.540	0.801	0.794	0.511	0.799	0.791
Number of centry_id	187	179	173	187	179	173	187	179	173

Robust SEs in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Note: These models use lagged regressors to account for conception time. “DC” refers to demographic controls that account for the confounding effects of population size and composition, whereas “MC” refers to macroeconomic controls that account for macro-level economic indicators. For female and male models, the variables on unemployment rate, infant mortality rate, and LFPR are all gender-specific.

1.3. DEMOGRAPHIC REPERCUSSIONS OF EMPLOYMENT UNCERTAINTY23

Table 1.2 shows the results from estimating this model. Columns (1) to (3) estimate the effect of *total* unemployment rate on fertility while the 6 subsequent columns replicate these models using female and male unemployment rates, respectively. Results show a robustly negative effect of unemployment on fertility across all specifications, although the magnitude of the female unemployment rate is slightly larger compared to the male unemployment rate. In my preferred specification (which has the complete demographic and macroeconomic controls), a one percentage point (pp) increase in the unemployment rate is associated with an average decrease in the TFR in the following year by 0.015 units (children per woman). For the gender-restricted measures of unemployment rates, a one pp increase in the female and male unemployment rates is associated with an average decrease in the subsequent year's TFR by 0.014 and 0.011 units, respectively.

As an additional test for heterogeneity, the negative relationship between unemployment and fertility might be different depending on whether a country is high or low income. In Table 1.3, the analysis is replicated for the higher and lower-income country samples, respectively. The former group is composed of upper-middle and high-income countries whereas the latter comprises low and lower-middle-income countries based on the World Bank's most recent classification. Indeed, results suggest that the effect size for the whole sample is the same for the higher-income sample (Column 2) but among lower-income countries, the magnitude is larger— a one pp increase in the lagged total unemployment rate is associated with a 0.029 units average decrease in TFR.

One of the more difficult barriers to the macro-level analysis of fertility dynamics is the challenge of incorporating “time” (Goldstein et al. 2013) and determining the ideal lag of unemployment or any macroeconomic indicator being used (Sobotka et al. 2011; Karaman Örsal and Goldstein 2018). First, prior empirical exercises reveal that fertility is a persistent (or path-dependent) (Gozgor et al. 2021; Luci-Greulich and Thévenon 2014) and non-stationary (Cazzola et al. 2016; Fernandez-Crehuet et al. 2020) measure. As such, the formal modeling of TFRs should ideally account for the “dynamic endogeneity” problem that arises—that is, fertility rates at time t are directly influenced by fertility rates in $t-1$. One approach to do so is using Arellano and Bond (1991)-type estimators, particularly Blundell and Bond (1998) and Windmeijer (2005) two-step System Generalized Method of Moments (GMM) which accounts for the dynamic endogeneity problem that arises when the lagged fertility rates are included

Table 1.3: Fixed Effects Regressions– Unemployment and Fertility, by Income Group

VARIABLES	(1) All	(2) Upper-Mid/High	(3) Low/Lower-Mid
L.Unemployment Rate	-0.015*** (0.005)	-0.015*** (0.002)	-0.029*** (0.009)
L.Infant Mortality Rate	0.011*** (0.002)	0.039* (0.021)	0.014*** (0.003)
L.LFPR	0.005 (0.005)	0.013 (0.008)	0.004 (0.008)
L.Total Age Dep Ratio	0.029*** (0.003)	0.019*** (0.004)	0.036*** (0.004)
L.Log Population	-0.795*** (0.139)	-0.723*** (0.146)	-1.065*** (0.348)
L.Real GDPPC Growth	-0.002** (0.001)	-0.006*** (0.002)	-0.003* (0.001)
L.Inflation CPI	-0.000** (0.000)	-0.003** (0.001)	-0.000** (0.000)
L.FDI Inflows	0.000 (0.000)	-0.000 (0.000)	0.002 (0.002)
Year Dummies	Y	Y	Y
Constant	13.439*** (2.349)	11.282*** (2.213)	17.902*** (5.734)
Observations	4,648	1,439	1,999
R-squared	0.792	0.706	0.838
Number of centry_id	173	51	77

Robust SEs in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Note: These models use lagged regressors to account for conception time and control for both demographic and macroeconomic variables. These employment measures are based on ILO-harmonized estimates. Income groups are based on the latest classification by the World Bank using per capita Gross National Income.

among the independent variables.

In cross-country studies that analyze the relationship between economic conditions and fertility, finding suitable instrumental variables is challenging (Karaman Örsal and Goldstein 2018), although other studies have employed a similar GMM approach (Gozgor et al. 2021; Luci-Greulich and Thévenon 2013, 2014). Since the time horizon of our panel is shorter relative to the number of countries, GMM estimation is also suited to address the Nickell bias (of obtaining inconsistent estimates) in “*small-T, large-N*” panels. Extending equation 1.1 by controlling for two lags of the unemployment rate to account for longer gaps for planning and conception, I estimate:

$$F_{c,t} = \alpha + \beta F_{c,t-1} + \gamma_1 U_{c,t-1} + \gamma_2 U_{c,t-2} + \delta X_{s,t-1} + \theta Z_{s,t} + \epsilon_t, \quad (1.2)$$

where γ_1 is the coefficient of interest, representing the effect of the lagged unemployment rate in country c in the past year on the fertility rate at year t . The rest of the vectors are similar to the earlier equation. More broadly, the contributions of this analysis are two-fold. On the methods front, it employs a dynamic panel data estimation method that addresses the dynamic endogeneity problem that arises because of the persistent and non-stationary nature of fertility rates. Indeed, previous work has recognized that this “more complex estimator” (Karaman Örsal and Goldstein 2018) generates consistent and efficient estimates under a properly specified model (Gozgor et al. 2021). On the substantive and thematic front, it contributes to the re-examination of the unemployment-fertility nexus that incorporates both the COVID-19 pandemic years and the full extent of the aftermath of the Global Financial Crisis.

Results in Table 1.4 reveal that the magnitudes of the main coefficient of interest (L.Unemployment Rate) are quite similar to those in the fixed effects models, demonstrating robustness. A one pp increase in the lagged total, female, and male unemployment rates are associated with lower TFRs by 0.016, 0.019, and 0.012 units, respectively. In addition, TFR is indeed persistent and non-stationary as shown by the lagged TFR being significant. Finally, the second lag of the unemployment rate is also robustly significant and positive, such that a one pp increase in total, female, and male unemployment is associated with a 0.017, 0.022, and 0.012 units increase in the TFR two years later. A plausible conjecture might be postponement— as a result of an increase in unemployment at time t , births that would have otherwise occurred at time $t + 1$ just occurred at time $t + 2$ instead, echoing the tempo-specific effects of macroeconomic conditions described in prior studies (Sobotka et al. 2011).

Table 1.4: GMM Estimates— Unemployment and Fertility

VARIABLES	All			Females			Males		
	Base	w/DC	w/MC	Base	w/DC	w/MC	Base	w/DC	w/MC
L.TFR	0.982*** (0.015)	1.037*** (0.016)	1.034*** (0.011)	0.985*** (0.015)	1.029*** (0.019)	1.023*** (0.012)	0.984*** (0.015)	1.035*** (0.015)	1.042*** (0.011)
L.Unemployment Rate	-0.013*** (0.004)	-0.016*** (0.004)	-0.016*** (0.004)	-0.016*** (0.006)	-0.019*** (0.006)	-0.019*** (0.006)	-0.012*** (0.004)	-0.016*** (0.004)	-0.012*** (0.003)
L2.Unemployment Rate	0.009* (0.005)	0.017*** (0.004)	0.017*** (0.004)	0.013* (0.007)	0.022*** (0.006)	0.022*** (0.006)	0.007 (0.005)	0.016*** (0.004)	0.012*** (0.004)
L.Infant Mortality Rate		0.001** (0.000)	0.000 (0.000)		0.001** (0.001)	0.000 (0.000)		0.001** (0.000)	0.000 (0.000)
L.LFPR		-0.004* (0.002)	-0.003** (0.001)		-0.003 (0.002)	0.000 (0.001)		-0.006*** (0.002)	-0.005*** (0.002)
L.Total Age Dep Ratio		-0.005*** (0.001)	-0.004*** (0.001)		-0.005*** (0.001)	-0.004*** (0.001)		-0.005*** (0.001)	-0.004*** (0.001)
L.Log Population		0.014 (0.011)	0.007 (0.004)		0.007 (0.011)	0.011** (0.005)		0.016* (0.010)	0.005 (0.005)
L.Real GDPPC Growth			0.001 (0.001)			0.000 (0.002)			0.001 (0.001)
L.Inflation CPI			0.000 (0.000)			0.000 (0.000)			0.000 (0.000)
L.FDI Inflows			-0.000 (0.000)			-0.000 (0.000)			-0.000 (0.000)
Year Dummies	Y	Y	Y	Y	Y	Y	Y	Y	Y
Income Group Dummies	Y	Y	Y	Y	Y	Y	Y	Y	Y
Regional Group Dummies	Y	Y	Y	Y	Y	Y	Y	Y	Y
Constant	0.007 (0.026)	0.196 (0.239)	0.204* (0.115)	0.001 (0.028)	0.104 (0.221)	-0.088 (0.090)	0.032 (0.031)	0.330 (0.205)	0.390*** (0.130)
Observations	5,423	5,191	4,532	5,423	5,191	4,532	5,423	5,191	4,532
Number of cntry_id	187	179	173	187	179	173	187	179	173
Number of instruments	54	88	112	54	88	112	54	88	112
Windmeijer SE correction	Y	Y	Y	Y	Y	Y	Y	Y	Y
Orthogonal + small SS corrections	Y	Y	Y	Y	Y	Y	Y	Y	Y
AR(2) test for serial correlation	0.997	0.684	0.932	0.949	0.639	0.751	1.000	0.778	0.916
Hansen test for overid. Instruments	0.000	0.000	0.004	0.000	0.002	0.010	0.000	0.000	0.004

Robust SEs in parentheses; *** p<0.01, ** p<0.05, * p<0.1

Note: “DC” refers to demographic controls that account for the confounding effects of population size and composition, whereas “MC” refers to macroeconomic controls that account for macro-level economic indicators. For female and male models, the variables on unemployment rate, infant mortality rate, and LFPR are all gender-specific.

1.3.4 Extension II. Novel Analysis Using Other Measures of Employment Uncertainty

The second limitation of contemporary macro-level findings summarized above, as it relates to breadth, is their reliance on the unemployment rate, or some variation of it (female/male/youth), as a proxy for labor market indicators. Recall that in the schematic classification in Table 1.1, positional uncertainty (e.g., unemployment vs. employment) is only one among many dimensions of employment uncertainty. In most of the developing world where unemployment is persistently low (see Figure 1.1), vulnerable employment seems to be the larger and more pressing problem (see Figure 1.2). Similarly, self-employment and part-time employment are alternative measures of employment uncertainty. How these indicators affect fertility rates has not been tested in the literature to date, to the best of my knowledge.

Following the exact estimation strategies in equations 1.1 and 1.2, I analyze the effect of three alternative measures of employment uncertainty: (1) vulnerable employment rate; (2) self-employment rate; and (3) part-time employment rate. One limitation of the WDI data is that the ILO-harmonized part-time employment rates cover fewer years than the other two measures. As such, the sample size is severely lessened in these models. For conciseness, only the results of the fully specified models are presented (including demographic and macroeconomic controls).

Results in Table 1.5 suggest a negative effect of vulnerable and self-employment. A one pp increase in the lagged VER and SER, as a share of the total employed population, is associated with 0.008 units of lower average TFR. Note that these two variables could be taken as indicators of how large a problem employment uncertainty is among the employed population, a metric that the unemployment rate simply does not account for. Further, part-time ER has a statistically insignificant effect, in part owing to the very small sample size in that model. On the other hand, GMM estimates in Table 1.6 do not find any of the three employment measures as significant determinants of fertility rates, thereby suggesting caution in interpreting the findings of the fixed effects models and warranting more and perhaps narrower analyses.

Taken together, these analyses lend evidence to the idea that (higher) unemployment rates are a robustly strong determinant of (lower) total fertility rates, whereas the effect of vulnerable and self-employment rates are also negative, but not robust to alternative estimation strategies.

There are other shortcomings of the “macro” approach that warrant future work.

Table 1.5: Fixed Effects Regressions– Employment Measures and Fertility

VARIABLES	(1) Vulnerable ER	(2) Self ER	(3) Part-Time ER
L.Employment Measures	-0.008** (0.004)	-0.008* (0.004)	-0.000 (0.001)
L.Infant Mortality Rate	0.012*** (0.002)	0.011*** (0.002)	0.005* (0.003)
L.LFPR	0.008 (0.005)	0.007 (0.005)	0.006 (0.004)
L.Total Age Dep Ratio	0.030*** (0.003)	0.030*** (0.003)	0.020*** (0.003)
L.Log Population	-0.818*** (0.139)	-0.820*** (0.139)	-1.008*** (0.135)
L.Real GDPPC Growth	-0.001 (0.001)	-0.001 (0.001)	-0.003** (0.002)
L.Inflation CPI	-0.000** (0.000)	-0.000** (0.000)	-0.001 (0.001)
L.FDI Inflows	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)
Year Dummies	Y	Y	Y
Constant	13.797*** (2.348)	13.846*** (2.357)	16.876*** (2.280)
Observations	4,648	4,648	1,734
R-squared	0.790	0.790	0.635
Number of cntry_id	173	173	147

Robust SEs in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Note: These models use lagged regressors to account for conception time and control for both demographic and macroeconomic variables. These employment measures are based on ILO-harmonized estimates.

Table 1.6: GMM Estimates– Employment Measures and Fertility

VARIABLES	(1) Vulnerable ER	(2) Self ER	(3) Part-Time ER
L.TFR	1.018*** (0.015)	1.018*** (0.015)	0.972*** (0.052)
L.Employment Measures	0.010 (0.010)	0.012 (0.012)	-0.000 (0.002)
L2. Employment Measures	-0.012 (0.010)	-0.014 (0.012)	-0.000 (0.001)
L.Infant Mortality Rate (Female)	0.001* (0.000)	0.001* (0.000)	-0.000 (0.002)
L.LFPR (Female)	0.001 (0.002)	0.001 (0.002)	0.002 (0.004)
L.Total Age Dep Ratio	-0.003*** (0.001)	-0.003*** (0.001)	-0.001 (0.004)
L.Log Population	0.018*** (0.007)	0.017** (0.007)	0.007 (0.015)
L.Real GDPPC Growth	0.004*** (0.001)	0.004*** (0.001)	0.005** (0.002)
L.Inflation CPI	0.000 (0.000)	0.000 (0.000)	-0.002 (0.004)
L.FDI Inflows	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)
Year Dummies	Y	Y	Y
Income Group Dummies	Y	Y	Y
Regional Group Dummies	Y	Y	Y
Constant	-0.181 (0.163)	-0.168 (0.171)	-0.219 (0.383)
Observations	4,532	4,532	1,499
Number of cntry_id	173	173	107
Number of instruments	112	112	113
Windmeijer SE correction	Y	Y	Y
Orthogonal + small corrections	Y	Y	Y
AR(2) test for serial correlation	0.902	0.873	0.341
Hansen test for overid. Instruments	0.007	0.007	0.606

SEs in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Note: These dynamic models account for the non-stationarity of fertility rates and are appropriate for “small T, large N” panels. The employment measures used are based on ILO-harmonized estimates.

First, total fertility rates are period-specific measures, which means that we cannot infer whether declining fertility rates are mainly explained by timing or quantum effects, or a combination of both. For this, cohort-specific fertility data might be more suitable as has been used by (Pifarré i Arolas 2017). However, this isn't available for most countries included in the extension analyses conducted above. Further, TFRs or crude birth rates as used in most macro analyses are also unable to distinguish between birth order, which means we cannot distinguish whether declining fertility rates manifest through a reduction in first births (or increasing childlessness) or through a reduction in higher-order births (or decreasing family size). While some countries have order-specific fertility data (Goldstein et al. 2013), this is unavailable for most countries around the world.

1.3.5 Country-in-Focus: Germany

Germany, like many other European countries, is undergoing profound demographic changes characterized by sub-replacement fertility rates, an increasing share of elderly in the population, and an increasing international migrant stock. Adapting to these demographic changes is core to securing Germany's economic future, as these developments have accompanying implications for labor supply, pension fund solvency, and healthcare systems. The literature has extensively analyzed the factors affecting fertility, migration, and life expectancy in Germany. While changing social norms that contribute to the postponement of family formation is a plausible explanation of fertility trends, employment uncertainty, political events (e.g., reunification), and policy reforms also play a role.

Among developed economies where the employment uncertainty-fertility nexus has been studied extensively, Germany is interesting for three non-exhaustive reasons. First, since the end of World War II, Germany never really had very high levels of fertility rates in comparison with other developed Western economies. Whereas the post-war period saw the TFR increase in the US up to around 3.6 in 1960, Germany's "peak" post-war fertility was around 2.5 in 1965 (see Figure 1.4). Compared to other high-income countries in the world, Germany's TFR has been consistently below the average of its income group (see Figure 1.4). Germany's TFR reached the replacement level of 2.1 in the early 70s and has remained in sub-replacement levels ever since.¹

¹Note that these observations are based on WDI data that combine East and West Germany, two countries with notably different fertility rates and patterns. See Goldstein and Kreyenfeld (2011).

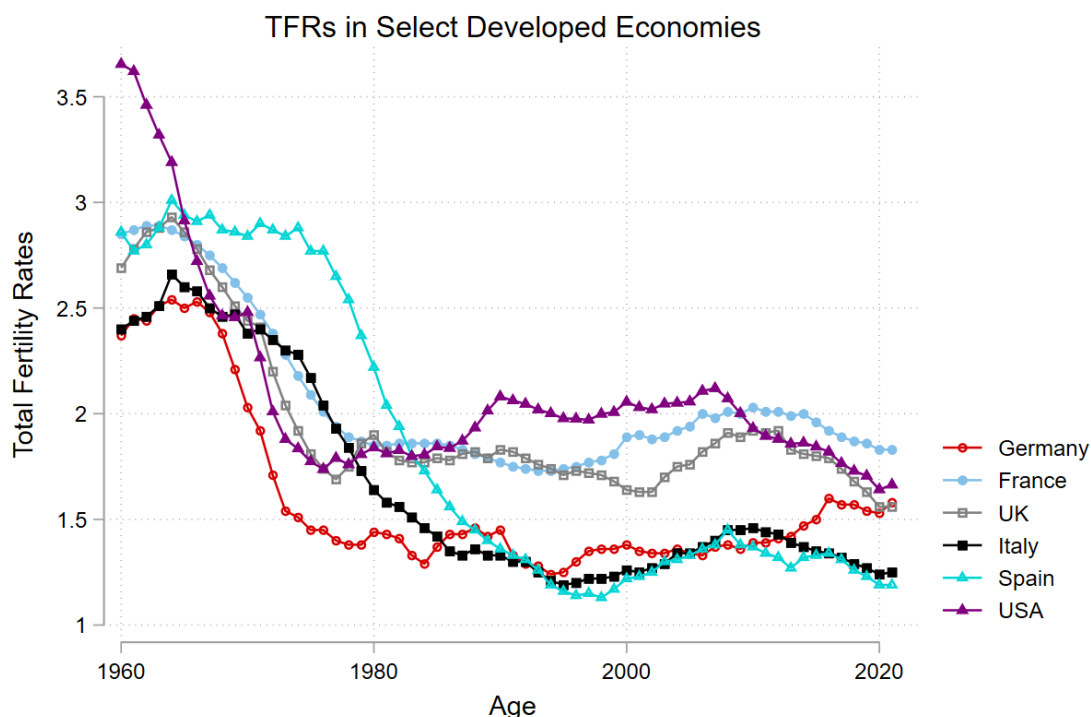


Figure 1.5: Total Fertility Rates in Select Developed Countries (1960-2022)

Second, in recent decades, it is interesting how Germany's fertility rates have hovered much closer to those of Southern European countries like Spain and Italy, despite having substantially different welfare regimes. Figure 1.4 shows that Germany only broke apart from Spain and Italy's fertility levels in the early 2010s. Finally, it is of course notable that Germany was divided into East and West for around 4 decades, with both countries having distinct fertility rates as shown below.

Looking at period fertility rates is instructive in that it offers a comprehensive snapshot of childbearing in a country but it is also limited as it does not reveal the age distribution of mothers, which varies over time. Figure 1.6 shows the age-specific fertility rates (ASFR) in East and West Germany in 1956 and 2016, a 60-year gap with the reunification occurring in the middle (1990). As the TFR is simply a sum of the ASFRs, the plot shows the age composition of TFR by region between these two periods. Notably, the peak of both curves has gradually shifted rightward over time, implying postponement from about the early 20s in 1956 to around 30 in 2016. Further, the area of the curves has also shrunk considerably (implying lower TFR). While East and West German fertility rates have arguably somehow converged, there

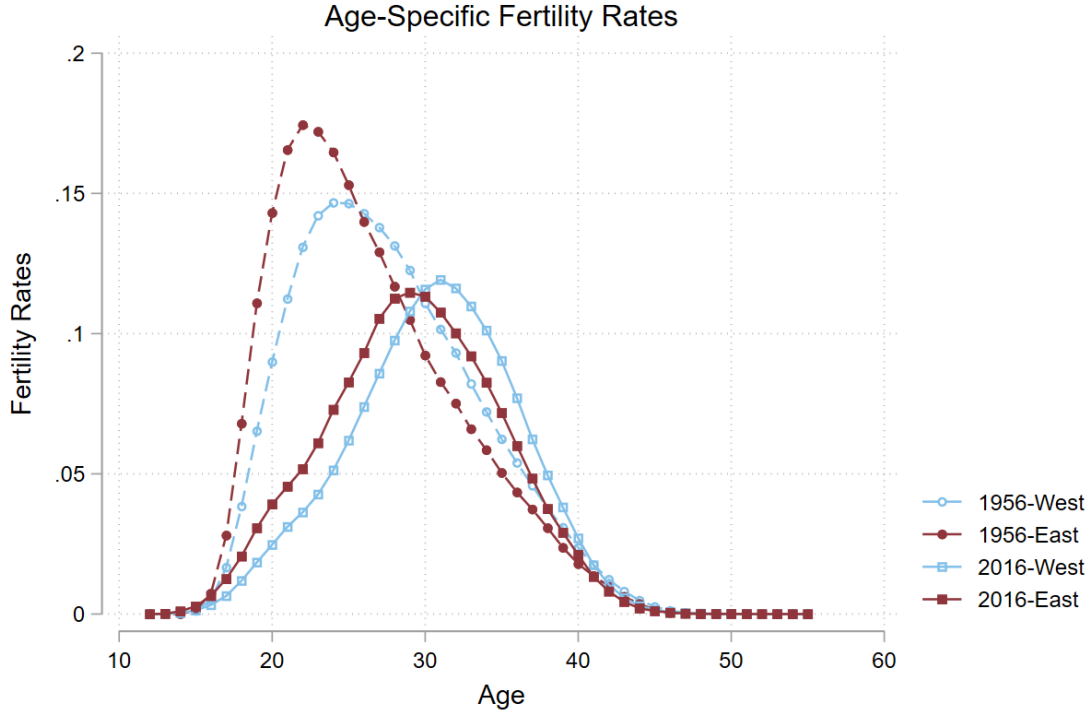


Figure 1.6: Age-Specific Fertility Rates in Germany: A 60-year Comparison

still seems to be a slight difference, especially in terms of the median age of first birth.

These interesting patterns and developments of fertility in Germany warrant a closer examination. In Chapter 3, fertility is analyzed using the event and timing of childbearing within 10 years after entering the labor market. In Chapter 4, fertility is operationalized through higher-order births (i.e., having second and third births). Finally, Chapter 5 measures fertility intentions, arguably a precondition of fertility behavior.

1.4 Methodological Approach and Summary of Chapters

1.4.1 Causal Identification

A common thread of these four stand-alone chapters is the application of both established and emerging causal identification strategies in quantitative social science

research. The “causality revolution”, where closer attention is given to establishing causal relationships, emerged as researchers developed novel statistical techniques and used larger and more comprehensive datasets that mimic experimental designs as closely as possible. A core tenet of this set of methodological tools is the use of counterfactuals and potential outcomes (Rubin 2005; Winship and Morgan 1999), which involves comparing observed outcomes to what would have happened under different conditions. Where randomized control trials (RCTs) are not feasible or ethical, quasi-experimental methods leverage on approximations of experimental conditions typically using observational data. Survey experiments, where certain survey components are randomized across respondents, have likewise emerged as a novel methodological tool to estimate causal effects.

These identification strategies are common in economics and, increasingly, in political science and sociology (Gangl 2010). The use of quasi-experimental strategies also seems to be fitting for questions that are of interest to demographers. For instance, at the individual, how does the experience of unemployment at time t affect a person’s fertility decisions at time $t + n$? Or at the macro level, how does a certain event (e.g., COVID-19 pandemic) affect overall fertility rates? Bhrolcháin and Dyson (2007) make an interesting point—in 1966, the year of the Fire Horse, crude birth rates in Japan fell by 25% compared to the previous year owing to the folk belief of bad luck for childbearing. Without resorting to any advanced quasi-experimental models, observants describe this phenomenon as causal, using arguments that are comparable to the intuition of formal models—it is pronounced, temporary, proportional, and non-recurrent (Bhrolcháin and Dyson 2007). The same argument can be made in other notable “kinks” in demographic patterns—in contemporary China, annual population growth rates were negative in 1960 and 1961, directly as a consequence of the Great Chinese Famine emanating from the Great Leap Forward campaign. In these cases, demographers were able to intuitively argue the underlying causal mechanisms of a demographic event. Beyond intuition, there is an increasing use of causal identification strategies in demographic research (Huinink and Brüderl 2021; Engelhardt et al. 2009; Wunsch and Gourbin 2020; Kreyenfeld 2021; Mize and Manago 2022), some of which are employed in this dissertation—namely, difference-in-differences (DiD), matching, and survey experiment designs.

These methods are employed on two main data sources. For Chapter 2, merged quarterly Labor Force Surveys (LFS) of the Philippines from 2016 to 2020 are used.

The LFS is a quarterly survey of around 40,000 households that is nationally and regionally representative. Absent register and panel data, the LFS is the most widely used data source for most countries for studying labor markets. For Chapters 3, 4, the German Socio-Economic Panel (GSOEP) (Goebel et al. 2019b) is used. It has been an annual household panel since 1984 and is a widely-used longitudinal data source, comparable to the Panel Survey of Income Dynamics (PSID) in the US and the UK British Household Panel Survey (BHPS), among others. In Chapter 5, we develop a novel survey experiment and fielded it in the GSOEP’s Innovation Sample (GSOEP-IS), a subsample of the main GSOEP intended for the implementation of innovative and/or experimental survey instruments.

1.4.2 Summary of Chapters

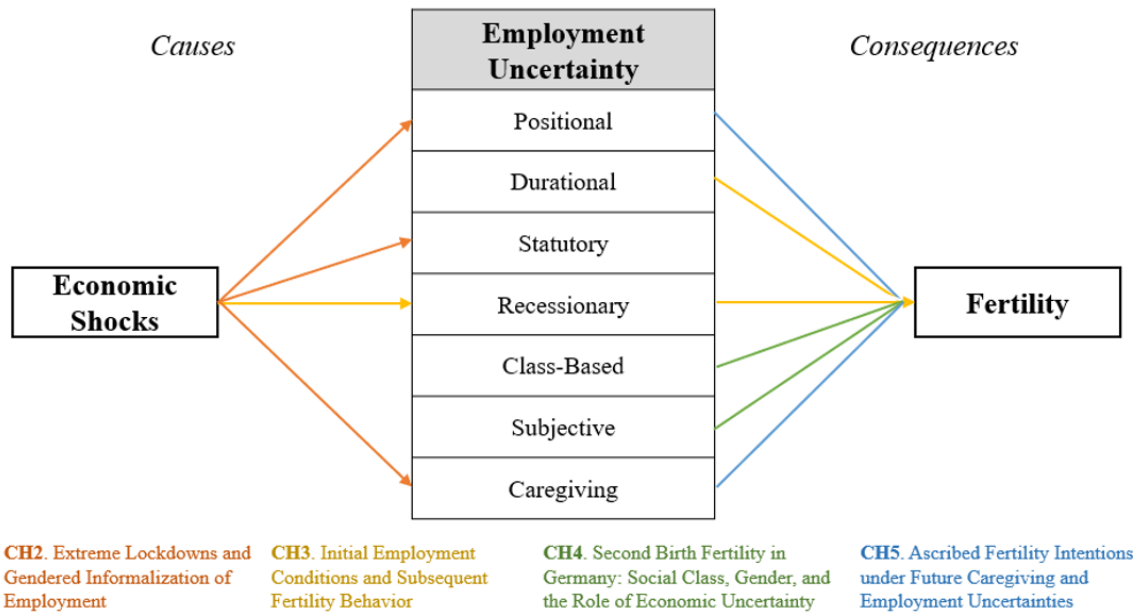


Figure 1.7: Dissertation Framework and Chapters

Figure 1.7 presents the overarching substantive framework of this dissertation, where employment uncertainty is central and is depicted as both a determined and a deterministic phenomenon. It is determined in the sense that it is in part a function of periods of economic uncertainty, broadly defined, including recessions, pandemics, and their accompanying policies. While it is deterministic of a broad set of life course

outcomes, this dissertation specifically focuses on fertility behavior and intentions which are a core demographic outcome.

In **Chapter 2**, employment uncertainty as an outcome *determined* by economic uncertainty and intervening state policies is unpacked in the case of the Philippines. Leveraging on the regional variation in the imposition of extreme and militarized lockdowns at the onset of the COVID-19 pandemic, this chapter employs a two-way fixed effects DiD design to analyze its effects on the probability of being informally employed. Previewing the results, the article finds that the lockdowns increased the probability of informal employment for women but not for men, and is most pronounced among mothers with minor children. This *informalization* phenomenon is demonstrated to have compositional and survivalist explanations and highlights the dual role that women play as workers and carers of first resort.

In **Chapter 3**, the relationship between *initial employment uncertainty and subsequent fertility*, a subject that has received extensive attention in the literature, is revisited using a twostep identification strategy in Germany. Operationalizing initial uncertain employment conditions using durational (fixed-term employment) and recessionary (recession year entry) typologies (see Section 1.1), the article examines their effect on first birth hazards within ten years after entry. Using the German Socio-Economic Panel (GSOEP), the article employs a two-step combination of propensity score matching and survival analysis, a method that has been used in the epidemiological literature, to address the issues of selection into first jobs. Previewing the results, fixed-term employment upon entry is demonstrated to negatively affect subsequent fertility for women but not for men, whereas recession entry negatively affects fertility for men.

In **Chapter 4**, the relationship between *current employment uncertainty and fertility behavior* is examined. Operationalizing employment uncertainty using a class-based lens, the article distinguishes the social classes of men and women based on their occupational categories into upper service, lower service, skilled manual/higher-grade routine nonmanual, and semi-/unskilled manual/lower-grade routine nonmanual classes. Using the GSOEP, the article uses survival analysis methods to unpack the relationship between social class and higher-order births. The focus on higher-order births is warranted in that it takes a certain amount of time for individuals to reach a certain level of “occupational maturity”. In their seminal work, Erikson et al. (1979a, 1983) uses people’s occupational positions in mid-30s to construct social class

Table 1.7: Employment Uncertainty Types, by Dissertation Chapter

Type of Uncertainty	Chapter			
	2	3	4	5
Positional	X			X
Durational		X		X
Statutory	X			
Recessionary		X		
Class-Based			X	X
Subjective			X	X
Caregiving	X			X

schemas. Thus, this work speaks to the strand of literature examining class differentials in completed fertility (Neyer et al. 2017; Dribe and Smith 2021) and higher-order births (Baizan 2021; Ekert-Jaffé et al. 2002) by focusing on the case of Germany and incorporating a gender and economic uncertainty perspective. Results demonstrate the elevated second birth rates of individuals in service classes and the role that upward career mobility after first birth plays in subsequent fertility decisions.

In **Chapter 5**, the effect of *future employment uncertainty and caregiving responsibilities on ascribed fertility intentions* is analyzed using a survey experiment in Germany. This article adopts a “future narrative” framework (Vignoli et al. 2020a), where the survey’s treatment conditions are explicitly framed to be future-oriented, departing from the two previous chapters that look at past and present experiences of employment uncertainty. This article also incorporates multiple types of uncertainty, namely, subjective and caregiving work (see Section 1.1), by constructing a hypothetical couple’s future job loss worries and forthcoming caregiving responsibilities. Previewing the results, having no forthcoming caregiving responsibilities and no employment uncertainty for both partners increases ascribed fertility intentions by 2.8 and 1.9 units, respectively, on an 11-point scale. This evidence speaks to the individual perceptions of the enabling factors for initiating parenthood.

Taken together, each chapter of the dissertation covers different aspects or types of employment uncertainty, as summarized in Table 1.7. Although each article is different in scope and methodology from the rest, there are overlaps in the treatment of employment uncertainty in some. For instance, class-based heterogeneities (between low/unskilled and service class occupations) are covered in Chapters 4 and 5, whereas caregiving responsibilities of females are covered in Chapters 2 and 5, although the former focusing on the effect of caring for a minor child whereas the latter focusing

on caring for a sick parent. These contrasts provide a nuanced and context-specific treatment of employment uncertainty. Finally, the concluding **Chapter 6** summarizes the overarching and salient takeaways from these articles, potential policy implications, limitations, and recommended directions for future research.

Chapter 2

Extreme Lockdowns and the Gendered Informalization of Employment

ABSTRACT

The adverse effects of COVID-19 on labour market outcomes are amplified by and partly attributable to the imposition of extreme mobility restrictions. While gendered disparities in job losses and working hours reduction are demonstrated in the literature, is an informalization of employment observed, and is this phenomenon likewise gendered? This article analyses the Philippines, a country that imposed one of the world's longest and strictest lockdowns, specifically how its imposition affected informal employment. A conceptual and empirical distinction between compositional and survivalist informalization is proffered—the former referring to informality induced by changes in the size and composition of overall employment, and the latter on informality induced by the need to work absent sufficient welfare support and precautionary household savings. Examining the regional variation in lockdowns as a quasi-experiment, results demonstrate that extreme lockdowns increase the probability of informal employment among employed women but not for men.

Keywords: COVID-19, gender, informal employment, survivalist motives, compositional factors

2.1 Introduction

The COVID-19 pandemic is unique in that broad and stringent mobility restrictions, which were not present in previous crises, were imposed. While general labour market outcomes (e.g. employment, labour force participation) have been analyzed in the growing strand of the literature on the effects of COVID, the impact of extreme lockdowns on employment, independent of the pandemic's blanket economic effects, have received little attention. Indeed, the pandemic caused what can be considered a “She-cession”—job losses disproportionately affected women across geographical and socio-economic contexts. In Latin America, 56 percent of job losses involved women and in most low and lower-middle income countries, working hours decreased disproportionately more for women (Cucagna and Romero 2021; Maurizio 2021). Among developed countries, job loss and decrease in working hours were disproportionately higher among mothers, at least in part due to their increased care responsibilities (Del Boca et al. 2020; Dias et al. 2020; Qian and Hu 2021). In many developing countries where informal employment constitutes a sizeable fraction of the employed population, informalization as a result of COVID-19 is thus far understudied (Jaga and Ollier-Malaterre 2022).

Lockdowns were common at the onset of the pandemic. Absent sufficient information about the transmission channels and mortality incidence of the disease, many countries erred on the side of caution and imposed extreme mobility restrictions that were instrumental in taming initial COVID outbreaks (Pajaron and Vasquez 2023). However, what exactly is restricted and how much these restrictions are enforced largely vary across countries. This article looks into the Philippines—widely reported by international media organizations as having imposed one of the world's longest and strictest lockdowns, with severe mobility restrictions enforced by military and police forces (Reuters 2020; Olanday and Rigby 2020; See 2021). In lockdown areas, all forms of public transport were stopped, offices and schools were entirely closed, and recreational activities, such as going for a walk, have also been disallowed. As an archipelagic country, not all regions in the Philippines were affected by COVID at the same time and with the same severity. Thus, widespread regional lockdowns, solely determined by the national government, applied only to some regions, whereas others underwent less restrictive containment measures. The labour market implications of this spatial variation in the design and enforcement of lockdown policies are thus far barely explored.

Among many labour market phenomena, “informalization” is particularly relevant for two reasons, apart from that it constitutes the majority of the globally employed population. First, informal workers generally experience greater precarity and uncertainty, leading to worse financial satisfaction and overall well-being (Ferrer-I-Carbonell and Gërzhani 2011; Webb et al. 2020).. Second, economic response (e.g. “pump-prime”) policies during periods of crises may have unwarranted consequences on labour market formalization. State deregulation succeeding the Global Financial Crisis is positively associated with the size of the informal economy and is partly responsible for its growth in some countries (Kus 2010). Despite COVID-19 lockdowns directly affecting an estimated 1.1 billion informally employed workers heavily concentrated in vulnerable and developing countries (ILO 2020), our understanding of informalization remains largely descriptive, with little evidence of its underlying determinants.

Looking back at previous economic downturns in developing countries is suggestive; there is some consensus that informal employment is countercyclical (Albertini et al. 2019; Colombo et al. 2019). Immediately after periods of banking crises in Mexico, Brazil, Costa Rica, and Bolivia in 1994, as well as after the Argentinian peso crisis of 2001 and its 2002 spillover crisis in Uruguay, informal employment increased and remained persistently high for succeeding years (Colombo et al. 2019). The same is true after natural disasters (Mendoza and Jara 2020; Pecha Garzón 2017).

However, little is said as to *why* we observe such informalization and this article conceptually and empirically fills this gap by distinguishing two channels. First, compositional factors, namely the size and composition of overall employment, shapes informal employment rates. When overall employment decreases and formal decreases more than informal does, this results in a “mechanical” increase in informal employment. Net of these “compositional effects”, however, the informal sector has long been pointed in the literature to play a cushioning role especially in developing country contexts. When economic uncertainty is high and formal job opportunities are scarce, workers would rather work informally rather than be unemployed or inactive. More broadly, the informal sector becomes part of one’s “survival strategy” in the absence of formal employment opportunities and government support in times of need Macdonald (1994).

Regarding gendered differences, women are disproportionately more likely to work informally during and after recessions. First, there are generally more limited opportunities for older and lower-educated women in the formal sector, which are

even more scarce during downturns (Floro et al. 2009; McCaig and Pavcnik 2015). Second, the flexibility of working conditions in informal employment allow women, who bear the brunt of care work, to better balance work and family responsibilities (Finnegan and Singh 2004; Rodin et al. 2012)). Indeed, during and after the 1997 Asian Financial Crisis, it was mostly older women in South Korea who were pushed into informal employment since prevailing gender and family norms concentrated formal work opportunities to men and younger women (Floro et al. 2009). Still, even in normal times, informal employment in developing countries is predominantly female-dominated (ILO 2018).

In the earlier months of 2020, it was not only the severity and contagiousness of COVID that was uncertain but also the length and tightness of ensuing lockdowns. Lockdowns triggered larger disruptions in economic activity—the length (Coccia 2021) and stringency (König and Winkler 2021) of lockdowns have a negative effect on aggregate economic output. To contribute to a broader understanding of the labour market effects of lockdowns in countries with sizable informal sectors, this article analyses whether extreme lockdowns increase the probability of being informally employed and whether this effect varies by gender. Using 16 pooled quarterly Labour Force Survey (LFS) data from the Philippines since 2016, the article employs a two-way fixed effects difference-in-differences (TWFE DD) approach and compares the probability of informal employment in lockdown and non-lockdown regions before and during the lockdown.

The article’s contributions are three-fold. First, it makes a conceptual and empirical distinction between compositional and survivalist factors as the underlying causes of employment informalization during the lockdown—a distinction that also applies to explaining informal employment dynamics in general. Second, it fills the gap in the literature on how state actions during crises enable gendered informalization (Hammer and Ness 2021) by isolating the “lockdown effect” from the other “recessionary” effects of the pandemic. Finally, this article broadly speaks to the literature on how informally employed women bear the brunt of labour market consequences during periods of crisis and how women play a dual role as both workers and carers of first resort.

2.2 Literature Review

Expansive and stringent lockdowns were periods of increased social and economic uncertainty that were inherently different from the uncertainty brought about by previous financial crises or recurrent natural disasters and weather shocks. There was unpredictability not only in terms of the contagiousness and severity of COVID but also in the length and stringency of extreme lockdowns. This section elaborates our expectations of the patterns of informalization and its gendered dimensions. In so doing, a conceptual distinction is made between compositional factors and survivalist motives that drive the informalization process (Kalleberg and Vallas 2017; Lee et al. 2020; Rogan et al. 2017).

Compositional factors encompass movements in the size and composition of overall employment—a reflection of the labour and capital structure where large swaths of the employed population are constrained in informal arrangements given low absorption in and intense competition for formal jobs. Survivalist motives on the other hand refer to the financial security that one gains from participating in the “shadow economy”, which encompasses informal work, rather than being unemployed or inactive. Unsurprisingly, in many macro-level analyses, informal employment follows a countercyclical pattern—it increases when economic output falls as it absorbs some people who lose jobs in the formal sector (Colombo et al. 2019; Feridhanusetyawan and Gaduh 2000; Fernández and Meza 2015; Floro and Dymski 2000; Nguyen and Su 2021).

The idea that the probability of working informally rises as a result of lockdowns cannot be simply taken as given, considering that the size, nature, and regulations governing informal work vary across developing countries. Contrary to earlier findings, Maurizio et al. (2023) demonstrate that during the COVID-induced contraction phase in six Latin American countries, informal employment exacerbated the decline in overall employment and did not play its usual countercyclical role. Further investigations in other contexts, as well as those that make a more nuanced distinction between the effects of lockdowns from that of COVID, would offer a more refined take on this idea including *why* informal employment responds as such.

2.2.1 How Extreme Lockdowns “Informalize” Employment

There are multiple channels through which extreme lockdowns may affect informal employment including both compositional factors (macro) and survivalist motives (micro-level behaviour). On aggregate, changes between employment and unemployment as well as formal and informal work affect the overall size and composition of the workforce. It is reasonable to expect that stronger lockdowns, likely due to higher incidence of COVID-19, induce larger contractions of overall employment (Qian and Fan 2020).

Suppose that due to an exogenous shock, the formal sector is more adversely hit than the informal sector. In this case, the decrease in formal employment outweighs the decrease in informal employment such that informal employment (as a share of the workforce) increases¹ even if the number of informally employed people does not necessarily increase. One example is the case of Latin American countries where lockdowns amplified the decrease in regular employment in terms of working hours and actual number of jobs, but did not have the same effect on informal employment (Maurizio 2021).

Since lockdowns impose mobility restrictions, those who are working home-based informal jobs such as piece-rate and freelance workers may find it easier to retain jobs rather than formal workers whose jobs require physical presence (e.g. retail services, hospitality and tourism, etc.) (ILO 2020; Maurizio 2021). Meanwhile, self-employed individuals who typically rely on working outdoors (e.g. street vendors) face the risk of arrest if caught outside since these activities are prohibited during a lockdown. This “chilling effect” of extreme lockdowns compositionally increases the probability of working informally if employment losses are more concentrated among the formally employed.

On the other hand, involuntarily displaced workers in the Philippines face a massive constraint—social assistance mechanisms are largely absent and savings are low. Under these conditions, one cannot afford to be unemployed and would rather work informally, a mechanism classically referred to in the sociology literature as survivalist motive (Macdonald 1994; Lee et al. 2020) and in the economics literature as the “luxury unemployment hypothesis” (Udall and Sinclair 1982). Indeed,

¹Consider a simple computation of informal employment rates (IER) as follows: $IER = I / (F + I)$ where I and F denote the number of informally and formally employed individuals respectively, such that $I=1$ and $F=2$ yields an informal employment rate of 0.33. Suppose that F decreases to 1.5 and I remains at 1, such that the decrease in F is larger than the decrease in I , IER increases to 0.4.

widely-accessible insurance programs are absent in the Philippines, although some institutionalized assistance programs are present.² While there was a one-time cash transfer for poorer households at the onset of the pandemic, this was plagued by delayed distribution (Cervantes, 2020) and a relatively low payout.³ Ordinary households are also beset by having insufficient savings due to low average savings rate (5%) of disposable income (Terada-Hagiwara 2009)).

The inability of many households to weather episodes of involuntary job loss in developing countries means that workers in low-income households cannot afford to be unemployed and would rather take a job even if conditions are poor or informal. The survivalist motive also captures the flexibility advantage of informal work—since these are usually occupations with no fixed hours, workers can plan around a schedule that allows for balancing work and family responsibilities (Sarker 2021; Singh and Kaur 2022). Figure 4.1 summarizes both compositional and survivalist informalization—two non-mutually exclusive mechanisms of employment informalization. In light of these factors, our prior expectations point to a lockdown-induced informalization warranting a test of this hypothesis:

H1: Extreme lockdowns increase the probability of informal employment, conditional on being employed.

²From the Department of Labor and Employment, TUPAD is a cash-for-work program for displaced and disadvantaged workers; CAMP is a one-time PHP 5,000 (USD 90) financial assistance for displaced workers; and AKAP is a one-time PHP 10,000 (USD 180) financial assistance for displaced migrant workers. The Department of Trade and Industry (DTI) Livelihood Seeding Program provides in-kind capital/inputs for micro, small, and medium enterprises. Finally, the Department of Agriculture (DA) Rice Farmers Support Program is a PHP 5,000 (USD 90) cash aid for rice farmers. Estimates shown in Appendix Table A.9 demonstrate low coverage with some programs covering less than one-tenth of one percent of households. These are consistent with World Bank estimates showing low coverage of social protection programs (Cho and Johnson 2022). Among households with a non-working head, coverage of Active Labour Market Policies is likewise low—all programs have a coverage of less than 0.2 percent.

³The Social Amelioration Program in 2020, a one-time cash transfer (disbursed in two tranches) that was supposed to aid households in lockdown regions faced multiple distribution bottlenecks. In some instances, it took local government units 1-2 months from the start of the lockdown for the beneficiary households to receive the aid. This also corresponds to a one-time payment per household of 8,000 pesos (USD 140), which is slightly below the national poverty line income of 8,022 pesos. Therefore, this program is likely insufficient to deter paid employment activity during the lockdowns, especially for poorer households who need immediate assistance.

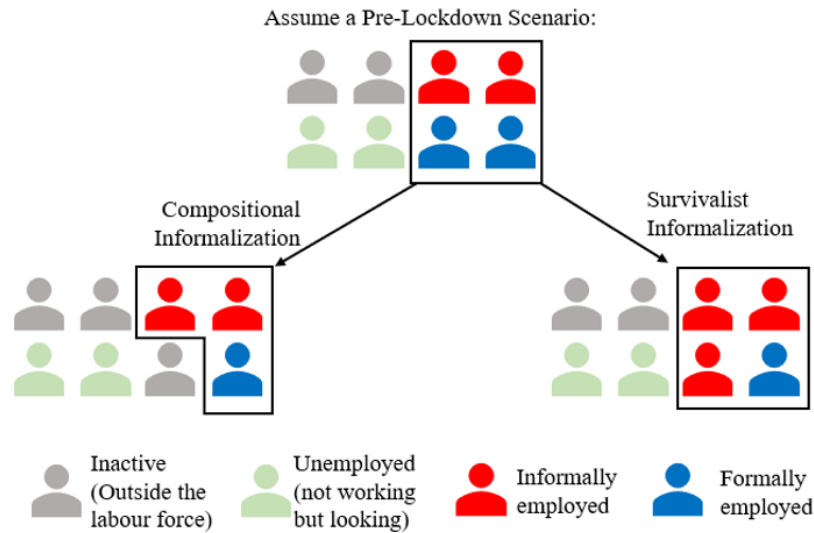


Figure 2.1: Distinction between compositional and survivalist informalization

Note: In this hypothetical scenario, informal employment (as a share of total employment in the black box) pre-lockdown is 0.5. A pure compositional effect increases informal employment to 0.66 and a pure survivalist informalization increases informal employment to 0.75. These are two non-mutually exclusive mechanisms of employment informalization.

2.2.2 How "Informalization" is a Gendered Phenomenon

The literature on the COVID-19 “She-cession” demonstrates the gendered effects of the pandemic across geographical and socio-economic settings. In both developed and developing countries, job losses and decreases in working hours were disproportionately higher for women (Cucagna and Romero 2021; Del Boca et al. 2020; Dias et al. 2020; Fan and Moen 2022; Maurizio 2021; Yavorsky et al. 2021; Ducanes and Ramos 2023).. Among the most prominent explanations is that despite the increased involvement of fathers in housework during the pandemic, increased care responsibilities were disproportionately shouldered by mothers (Collins et al. 2021; Craig and Churchill 2021; Kreyenfeld and Zinn 2021; Yaish et al. 2021). Household responsibilities are performed predominantly by women in the Philippines (Bayudan-Dacuycuy and Dacuycuy 2017).

Informal employment and gender are closely intertwined subjects in developing countries: in Sub-Saharan Africa, 89.7% of employed women are informal, and in Latin America, Southeast Asia, and South Asia, informal employment rates are higher for women (ILO 2018). Rodin et al. (2012) find that the majority of informally employed women are involuntary and constrained by family responsibilities—they would have

preferred a more stable formal job had there been an opportunity. In the Philippines, Gaerlan (2011) finds that the flexible nature of informal work for female garment workers allows them to combine multiple sources of income since one job alone cannot sufficiently pay for the entirety of expenses.

Since informal employment has less barriers to entry, informal workers would likely want to continue working at the height of the lockdowns, especially in the absence of social protection mechanisms and with grimmer opportunities in the formal sector. Ogando et al. (2021) find that in twelve cities with large rates of informality, the paid and unpaid work that informally employed women did during the pandemic served as “shock absorbers” and coping mechanisms. Broadly, this *survivalist* motive among informally employed women is both an “exercise of agency to improve living conditions” (Lee et al. 2020) and a manifestation of their dual role as workers and carers of first resort. During uncertain times when there are greater threats to economic survival, women with the survivalist motive in mind simultaneously perform paid and unpaid work rather than trading off one for the other.

The informalization of women’s work when the economy is dire is not unfounded. Floro and Meurs (2009) show that female informal employment rates in South Korea increased after the 1997 Asian Financial Crisis as formal job opportunities were concentrated on men. This effect is pronounced for older women since men and younger women tend to accept lower wages. A similar increase in women’s informal employment was found in Indonesia, Mexico, and Argentina during their respective episodes of financial crisis (Albertini et al. 2019; Feridhanusetyawan and Gaduh 2000; Fernández and Meza 2015). In terms of non-financial crises, weather shocks, which are common in many Asia-Pacific countries including the Philippines, resulted to reduced wages and salaries for informal workers and lower income for the informally self-employed (Bayudan-Dacuycuy and Baje 2019). The probability of being in informal employment rises in Ecuador after local earthquakes (Mendoza and Jara 2020) and in Jamaica after hurricanes (Pecha Garzón 2017).

However, informalization could also similarly be more pronounced among males when considering a traditional male-oriented breadwinner model in the Philippines (David 1994) where males assume the sole responsibility of paid work whereas females assume unpaid care work. From a compositional perspective, if the shift from formal to informal male employment is larger in lockdown than non-lockdown regions, this would also induce informalization. Thus, it cannot be presumed or taken for granted

that informalization is a women-only phenomenon, although existing prior evidence largely points to this direction. Thus, we test the hypothesis that:

H2: Extreme lockdowns increase the probability of informal employment more for women than men.

2.3 Data and Identification Strategy

2.3.1 Labour Force Survey

The Labour Force Survey (LFS) is a quarterly survey of around 40,000 households, representative at the national and regional levels in the Philippines. Absent administrative and panel data, the LFS is the most widely used data for many developing countries in labour market research. To sufficiently establish the trends in informal employment before the imposition of the extreme lockdown, the article uses quarterly LFS data since 2016 to restrict the sample to the then-incumbent administration that assumed office in June 2016.

The analysis is likewise restricted to regions that underwent a stringently imposed regional lockdown (“treatment”) and those that did not impose any lockdowns at the regional level in favour of more targeted ones (“control”). Many regions changed the lockdown status on 01 May 2020 and since the second quarter of 2020 round of the LFS was conducted from 16 April to 08 May 2020, only regions that had a consistent classification throughout the survey period were retained in the sample.⁴

It is important to establish that the “treatment” (lockdown) is similar and comparable across all regions that were under regional lockdowns. Similarly, in the counterfactual analysis, the control group (no lockdown) does not imply the absence of any mobility restrictions but rather the absence of a regionally-imposed extreme lockdown. Since the COVID-19 response in the Philippines at the onset is centralized, the design and implementation of these regulations are uniform across all lockdown

⁴This restriction in the analysis arises due to the absence of data on the date of the exact interview of the respondent. All respondents who were in full/no lockdown regions in April but not in May were dropped from the sample to fully observe the effect of the lockdown as a treatment. In Appendix Figure B.1 I show the timeline of the lockdowns in the Philippines by region. After 15 May 2020, all regions underwent more granular semi-lockdowns—these were less restrictive and targeted lockdowns in smaller administrative units (i.e. cities) within each region. Therefore, periods after the second quarter of 2020 are excluded since lockdowns were no longer regionally imposed and the LFS data has larger margins of error for sub-regional analyses.

regions. Table 1 summarizes the policy differences that were in effect at the onset of the pandemic (April-May 2020).

Table 2.1: Comparison of measures in lockdown vs non-lockdown regions

	Lockdown Regions	Non-Lockdown Regions
Coverage	National Capital Region, Regions 3 and 4B	Regions 7, 8, 10, 11, and ARMM
Scope	Regionally-imposed	Targeted in smaller administrative sub-units
Work Restrictions	Full suspension of onsite work unless essential service	Onsite work allowed under limited capacity
Non-essential businesses	Full closure	Allowed to operate at 50-100% capacity
Sporting and recreational activities	Not allowed	Allowed unless under targeted lockdowns
Public Transportation	All forms of mass transport stopped	Allowed at reduced capacity unless under targeted lockdowns
Tourism and hospitality	Fully restricted for local and international tourists	Allowed at reduced capacity unless under targeted lockdowns
Mass gatherings	Not allowed outdoors + Indoor gatherings only for people of the same household	Allowed at reduced capacity unless under targeted lockdowns
Strictly-enforced curfew	8 PM to 5 AM	None unless under targeted lockdowns

Source: Pronouncements of the Inter-Agency Task Force for COVID-19 (IATF) and online news reports

One important distinction between the lockdown and non-lockdown regions is that police and military enforcement is heightened and concentrated in lockdown regions. Hapal (2021) argues that the government’s approach at the onset of the pandemic is comparable to “waging a war” but since “COVID is an unseen enemy”, policing was targeted towards violators of COVID restrictions. To show that this translated to different mobility changes between lockdown and non-lockdown regions, we verify, using Google Community Mobility Reports Google (2021), that there was a larger average mobility reduction in lockdown regions during the conduct of the April 2020 round of the LFS. Figure B.2 also shows that the gap between the lockdown and non-lockdown regions dissipated over time as restrictions became spatially granular and targeted.

2.3.2 Measurement and Analytical Sample

Given that the LFS is typically limited in capturing informal employment, the article relies on technical guidance from the ILO on measuring informal employment from LFS microdata (Simons and Lake 2006). From a gendered perspective, household surveys are shown to disproportionately underestimate women’s labour market participation, as they tend to undercount intermittent, home-based, and uncompensated work (Langsten and Salen 2008). Despite these limitations, LFS remains a widely used data to analyse labour market dynamics, including that of informal employment (Cabegin 2008; Comola and De Mello 2011).

Consistent with ILO’s guidelines, informal employment in this article encompasses non-professional own-account workers and contributing family workers (in their own

or another household) outside the agricultural sector and public sector. The “non-professional” exclusion restriction prevents self-practicing or freelancing professionals such as lawyers and accountants from being included in the definition. This type of self-employment, where invoices are typically issued and thus covered by taxation, is not informal by definition and thus makes the exclusion of professionals a reasonable restriction.

The agriculture sector is likewise peculiar because there is no consensus on what informality in this sector looks like, as confirmed by the International Conference of Labour Statisticians (ICLS) (Husmanns 2004). For instance, an own-account worker who is an employee of a trading agricultural cooperative is considered in informal employment if the said cooperative is an informal enterprise, which is loosely defined in the agriculture sector (ASEAN and Ducanes 2019; Vanek et al. 2014). Finally, the public sector is also excluded because the nature of employment here is largely formal. These three exclusions are consistent with ILO recommendations and practices.

Beyond these formal measures, what does informal employment actually look like in the middle of the pandemic? In the definition we adopt, non-professional self-employed individuals are classified as informally employed. This includes street vendors, non-registered online sellers, and domestic care workers, among others, some of who are able to keep their jobs during the lockdown. Those who were laid off may have also done informal online selling as an income-generating means of survival.⁵

Consistent with official definitions, the sample is likewise restricted to individuals 15 years old and above, also considering that there is no retirement age in the informal sector and informal old-age employment is prevalent in the Philippines at 52 percent (Racelis et al. 2012). The sample is also restricted to employment with positive working hours since at the height of the pandemic, some employed individuals whose basis of pay is daily or weekly, were not asked to report to work and had no income. To the extent that employed workers don’t work and have no income, employment status alone, without regard for actual work hours, becomes a poor indicator of labour market activity. The final analytical sample consists of 525,488 individuals over 16 quarters and supplementary analyses demonstrate the robustness of results when these restrictions are relaxed.

⁵We illustrate these typical labour market transitions in Appendix Figure B.3

2.3.3 Identification Strategy

The presence of a uniform widespread lockdown in select regions and its absence in others make a differences-in-differences (DD) strategy suitable in determining the causal effect of extreme lockdowns on informal employment. Given non-random treatment assignment, DD can also be applied to repeated cross-sections assuming that compositional changes do not affect the variables of interest Cunningham (2021). The key identifying assumption is that of parallel trends—the outcome variables for both lockdown and non-lockdown regions must have been moving in a parallel manner before the lockdown. The empirical challenge arises from the inability to observe what informal employment in a region would have been had it not undergone a lockdown. Empirically, placebo regressions showing whether a “lockdown effect” was present prior to the lockdown itself can be used to demonstrate parallel trends.

Using individual-level data from the LFS, the effect of extreme lockdowns on informal employment is estimated using a linear probability model, parsimoniously specified as follows:

$$Y_{igt} = \alpha + \beta LA_g + \gamma LP_t + \delta_{DD}(LA_g * LP_t) + \sum_{j=1}^n \theta_j X_{jigt} + \epsilon_{igt}, \quad (2.1)$$

where Y denotes informal employment, and LA and LP are binary variables indicating if the person was surveyed in a Lockdown Area and during the Lockdown Period, respectively. The coefficient of interest is δ_{DD} which captures the causal effect of extreme lockdowns on the probability of informal employment. In the full specification, X is a vector of individual-level j controls such as female dummy, age, and educational level. To further analyse gendered differences, DD estimates for the male- and female-only sample are presented. A limitation of employing the canonical 2x2 DD set-up is that there are multiple pre-treatment periods and multiple regions within each lockdown classification. More specifically, there are 15 pre-treatment periods, 3 lockdown, and 5 non-lockdown regions. Since there are only a few regions that count as either treated or control, clustering standard errors at the regional level introduces the problem of “too few clusters” and, consequently, small-sample bias (Cameron and Miller 2015). Aggregation techniques that correct such limitations, such as aggregating pre-treatment periods into one, have relatively low statistical power (Bertrand et al. 2004). Despite this limitation, the specification below accounts for both unobserved between-region heterogeneity and region-invariant period effects by including regional

and period (quarter) fixed effects:

$$Y_{igt} = \alpha + \beta LA_g + \gamma LP_t + \delta_{DD}(LA_g * LP_t) + \sum_{j=1}^n \theta_j X_{jigt} + \sigma_g + \tau_t + \epsilon_{igt}, \quad (2.2)$$

The two-way fixed effects estimator is used in DD setups to account for unobservable time-invariant heterogeneity as well as a group-invariant time effect (de Chaisemartin and D’Haultfœuille 2020; Pajaron and Vasquez 2023; Wooldridge 2021). This is likely present in the setup—where unobserved sociodemographic, economic, and political variables may vary between lockdown and non-lockdown regions.

2.4 Results and Discussion

2.4.1 Compositional Changes in the Labour Market

Recognizing that the lockdown has affected the labour market at large, this section contextualizes the changes within the employed population. During the lockdown quarter, the Philippines experienced a 21 percent drop in overall paid employment compared to the same quarter in the previous year, and these employment losses are most pronounced in the entertainment (-50%), hospitality (-36%), and energy (-36%) sectors (Ducanes et al. 2021). Disaggregated changes in total employment based on lockdown status and gender show that, from a macro perspective, compositional changes in the labour market partly explain informal employment dynamics as what was theoretically established in Section 2.2.

Figure 3.2 presents employment trends and compositional changes by lockdown status and gender and shows that total employment decreased during the lockdown period and the decline is more pronounced in lockdown areas. Disaggregating by gender, total employment had a steeper decline for workers in lockdown regions, with males being slightly more affected than females. This descriptive figure alone shows the stark and gendered impact of lockdowns on employment.

However, the decrease in total employment is only one part of explaining compositional labour market changes. In Figure 4.1, compositional informalization occurs when the decrease in formal outweighs the decrease in informally employed

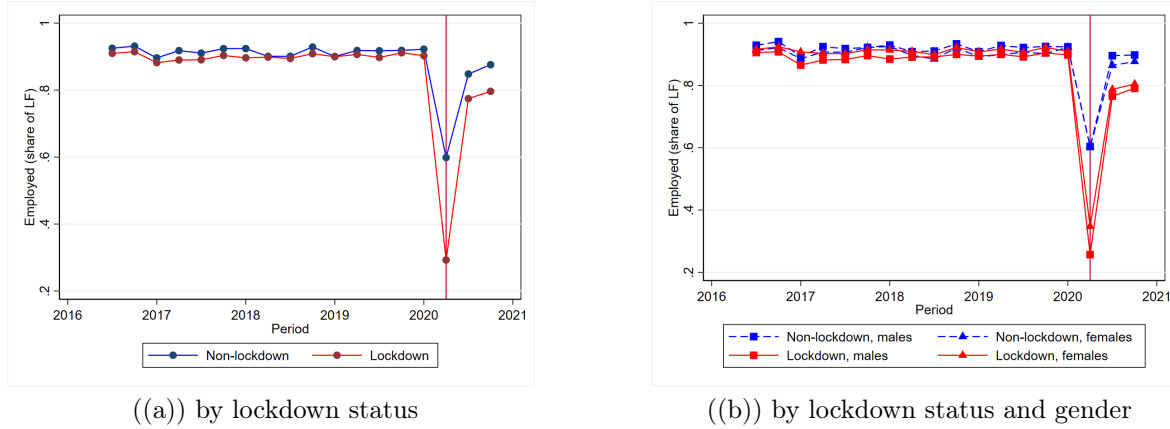


Figure 2.2: Quarterly Employment Rates, as a share of the labor force, 2016Q3-2020Q2

Note: Employed refers to people who have worked for at least an hour in the past week, including those with a job but did not work due to vacation or illnesses. By convention, these are reported as the percentage share of the labor force.

workers. Disaggregating employment changes by type and lockdown status, Table 2.2 shows two prominent findings: (1) indeed, the overall decrease in formal outweighs that of informal employment; and (2) the decrease in formal outweighs the decrease in informal employment in non-lockdown areas.

These compositional changes are not entirely surprising. Since lockdown areas are characterized by more severe mobility restrictions, their decline in total employment is inevitably larger. Regarding gendered differences, the larger decline of informally than formally employed females reflects the fact that traditionally female-dominated informal sectors, such as retail and other service work had been more restricted during the lockdown (Ducanes and Ramos 2023). Therefore, any gendered differences in informal employment are at least in part due to the gendered differences on who left and who stayed in the labour market during the lockdown.⁶

Focusing specifically on the employed population, Figure 3.3 shows estimates of informal employment, as a share of total employment, over time. From mid-2016 to January 2020, informal employment rates for lockdown and non-lockdown regions moved in similar trends and, for the most part, in parallel. Informal employment rates steeply increased in lockdown areas during the imposition of the extreme lockdown. Disaggregating by gender, the trends confirm a common finding in the literature that

⁶Appendix Table A.2 shows the percent change in employment by type, lockdown status, and gender, and the basis for these estimates in the table.

Table 2.2: Counts and percent change of employment by type and lockdown status

	Count ('000)			Percent Change	
	ΣT_{L-16}	T_{L-1}	T_L	(1)-(3)	(2)-(3)
ALL SAMPLE					
All Areas					
Formal	31,469.02	33,055.71	15,303.25	-51.37%	-53.70%
Informal	6,745.76	6,960.48	3,418.05	-49.33%	-50.89%
Lockdown Areas					
Formal	20,295.09	21,011.07	10,004.55	-50.70%	-52.38%
Informal	3,038.92	3,144.37	1,196.21	-60.64%	-61.96%
Non-lockdown Areas					
Formal	11,173.93	12,044.64	5,298.70	-52.58%	-56.01%
Informal	3,706.84	3,816.11	2,221.84	-40.06%	-41.78%
MALE SAMPLE					
Lockdown Areas					
Formal	9,655.25	10,275.31	3,016.42	-68.76%	-70.64%
Informal	1,602.50	1,649.15	438.34	-72.65%	-73.42%
Non-lockdown Areas					
Formal	5,916.62	6,509.74	2,106.90	-64.39%	-67.63%
Informal	1,863.19	1,880.20	943.94	-49.34%	-49.80%
FEMALE SAMPLE					
Lockdown Areas					
Formal	10,639.84	10,735.76	6,988.14	-34.32%	-34.91%
Informal	1,436.42	1,495.22	757.87	-47.24%	-49.31%
Non-lockdown Areas					
Formal	5,257.31	5,534.90	3,191.80	-39.29%	-42.33%
Informal	1,843.64	1,935.92	1,277.90	-30.69%	-33.99%

Note: These are percent changes in the weighted estimates of formal and informally employed workers for 2020 Q2 (lockdown period) relative to the preceding quarter (column 2) and relative to the average of all preceding quarters from 2016 Q3 to 2020 Q1 (column 1).

in most developing countries, women have a higher share in informal employment (Cucagna and Romero 2021; Floro and Meurs 2009; Horn 2010; Gaerlan et al. 2011). During the imposition of the extreme lockdown, it was women in lockdown regions who experienced the steepest increase in informal employment rates, providing first-pass descriptive evidence of a gendered informalization.

Figure 3.3 shows that the trends of lockdown and non-lockdown regions moved in parallel for the most part. While this is a “first pass” test of whether the parallel trends assumption of DD is met, additional placebo regressions are also conducted to check

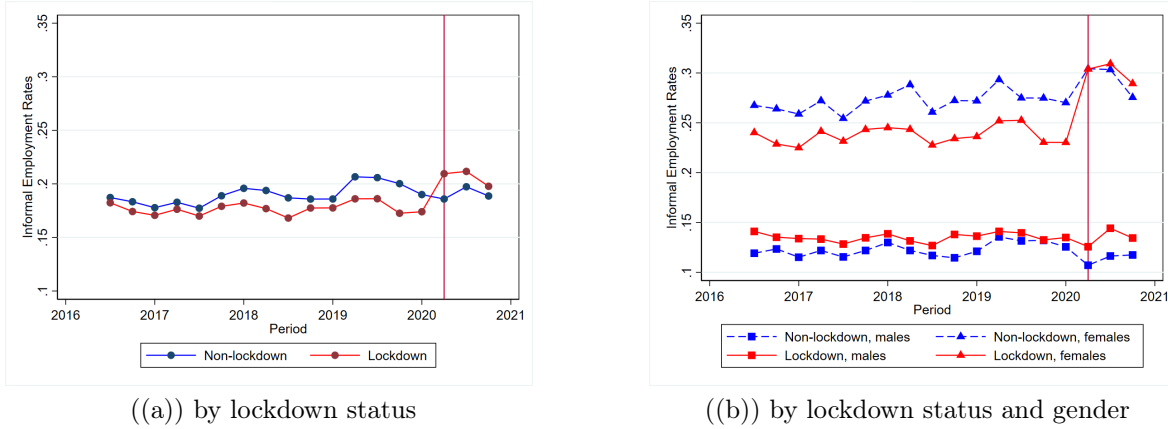


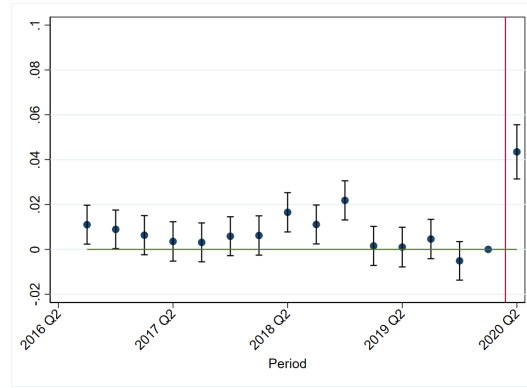
Figure 2.3: Quarterly Informal Employment, as a share of the employed population, 2016Q3-2020Q2

whether a “lockdown effect” existed before the lockdown itself, which would invalidate our key identifying assumption. These placebo regressions are extensively used as sensitivity analysis and additional evidence for parallel trends in other DD studies (Clarke and Schythe 2021; Correia 2017; Lechner 2010). Results of these analyses in Figure 3.4 show that pre-lockdown treatment effects are significantly close to zero for the most part, except for three quarters in 2018. Similar patterns also hold true by gender. These regressions include the relevant covariates in equation (2), including region and period FEs.

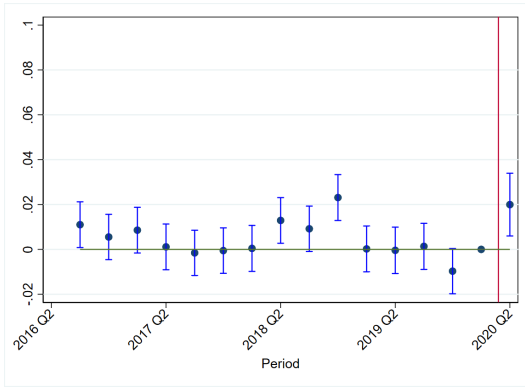
Broadly, other policy-related and structural factors not accounted for by the regional and period-fixed effects in the placebo regressions do not necessarily threaten the parallel trends assumption. Employment policies, including informal employment, are nationally determined by the Department of Labour and Employment (DOLE). While local governments sometimes perform “crackdowns” on illegal vendors along major thoroughfares and central areas, these crackdowns simply displace vendors to other locations rather than shifting them towards formal employment (Kusaka 2017).

2.4.2 Regression Results

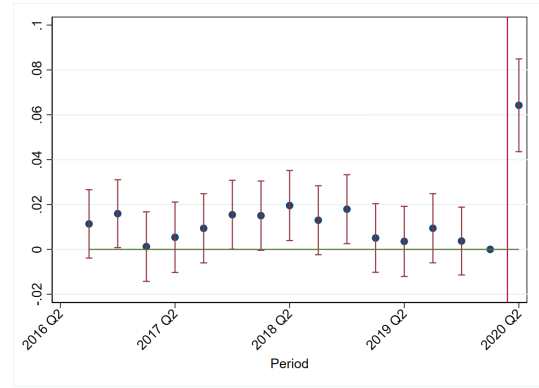
Extreme lockdowns have a positive and significant effect on the probability of being informally employed and this finding is robust to the inclusion of individual-level controls and regional and period fixed effects. In the full specification, lockdowns raise the probability of being informally employed by 1.7 pp. Disaggregating by gender,



((a)) all sample



((b)) males



((c)) females

Figure 2.4: Event Study Regression Plots

Note: To establish that there were no “lockdown” or treatment effects prior to the lockdown period, this figure presents event study regressions accounting for dynamic leads and lags prior to the treatment period. Moving the treatment period to $t-1$, parallel trends assumption is supported with treatment effects being close to zero. This specification includes all relevant covariates, including period and regional fixed effects. To account for multiple levels of fixed effects, the `reghdfe` Stata command developed by Correia (2017) was used.

the DD coefficient is robustly positive and significant only for females, implying that the effect seen in the full model is largely driven by the increase in the probability of informal employment among females of about 2.2 pp. Consistent with earlier findings in the literature of a gendered informalization of work during periods of economic uncertainty, results in Table 3.3 suggest that among women employed during the lockdown, informality increased.

One potential explanation for the overall informalization of the employed during the lockdown is the survivalist movies earlier established. Where social assistance

mechanisms and sufficient precautionary household savings are absent, the poor simply cannot afford unemployment. Thus, they face high pressure to look for employment opportunities immediately after job loss episodes (Turnham and Eröcal 1990; Udall and Sinclair 1982). Given that informal employment has lower entry barriers and more flexible conditions than formal employment, this seems to be a preferred outcome, instead of unemployment, by people who lose a job.

To establish support for the survivalist motive, various alternative labour market outcomes of the entire labour force are considered—formally employed, informally employed, unemployed, and inactive. Using multinomial logistic regression, results presented in Appendix A.8 suggest that extreme lockdowns significantly increase the likelihood of being in informal employment relative to unemployment for males, consistent with the survivalist motive. As further evidence, estimates using a 2020 World Bank survey suggest that there is a slightly higher proportion of households in lockdown regions that relied on selling assets/properties, engaging in additional income-generating work and online selling, and requesting advance payment from employers as coping mechanisms than in non-lockdown regions (see Appendix Table A.10) (Cho and Johnson 2022).

Certainly, survivalist motives are not the only mechanism at play. Compositional changes, as discussed in Section 4.1, also suggest that changes in gender-specific trajectories of formal and informal employment shares partly explain this difference. To visually demonstrate this difference, Figure 2.5 shows that compositional changes in lockdown areas are larger in general, and are steepest among informally employed females. Whether this is a household coping mechanism or a consequence of insufficient social support mechanisms should be looked into in subsequent work with more information on time use and state-to-state transition, as was done by Maurizio et al. (2023). However, to the extent that the amplifying effect of lockdowns is demonstrated, net of the adverse social and economic effects that all areas have experienced due to COVID, these results are notable.

Table 2.3: DiD Regression Results: Extreme Lockdowns and Informal Employment

VARIABLES	(1)	All (2)	(3)	(4)	Males (5)	(6)	(7)	Females (8)	(9)
Lockdown area	-0.001 (0.001)	-0.005*** (0.001)	-0.052*** (0.002)	0.012*** (0.001)	0.004*** (0.001)	-0.031*** (0.002)	-0.034*** (0.002)	-0.019*** (0.002)	-0.108*** (0.005)
Lockdown period	0.003 (0.004)	-0.003 (0.004)	0.000 (0.004)	-0.016*** (0.004)	-0.018*** (0.004)	-0.017*** (0.004)	0.033*** (0.007)	0.020** (0.007)	0.029*** (0.008)
Lockdown area*period	0.030*** (0.006)	0.019** (0.006)	0.017** (0.006)	0.007 (0.007)	0.006 (0.007)	0.003 (0.007)	0.034** (0.011)	0.022* (0.011)	0.022* (0.011)
Age		0.004*** (0.000)	0.004*** (0.000)		0.002*** (0.000)	0.003*** (0.000)		0.006*** (0.000)	0.006*** (0.000)
Basic education		0.019*** (0.001)	0.010*** (0.002)		-0.023*** (0.002)	-0.032*** (0.002)		0.058*** (0.003)	0.050*** (0.003)
Some post-secondary		0.063*** (0.002)	0.067*** (0.002)		0.011*** (0.002)	0.018*** (0.002)		0.121*** (0.003)	0.119*** (0.003)
Secondary education		0.084*** (0.001)	0.080*** (0.001)		0.030*** (0.001)	0.024*** (0.002)		0.143*** (0.002)	0.141*** (0.002)
Female		0.129*** (0.001)	0.130*** (0.001)						
Region FE			Yes			Yes			Yes
Period FE			Yes			Yes			Yes
Constant	0.178*** (0.001)	-0.065*** (0.002)	-0.036*** (0.003)	0.123*** (0.001)	0.025*** (0.002)	0.047*** (0.004)	0.272*** (0.001)	-0.038*** (0.003)	0.026*** (0.007)
Observations	591,943	579,258	579,258	361,211	353,121	353,121	230,732	226,137	226,137
R-squared	0.000	0.055	0.057	0.000	0.012	0.014	0.002	0.061	0.066

Robust standard errors in parentheses

*** p<0.001, ** p<0.01, * p<0.05, † p<0.1

Note: Dependent variable is a binary variable, indicating 1 for informal employment with positive work hours. Sample is restricted to lockdown and non-lockdown regions, from July 2016 to April 2020 (the lockdown quarter), individuals aged 15 and above.

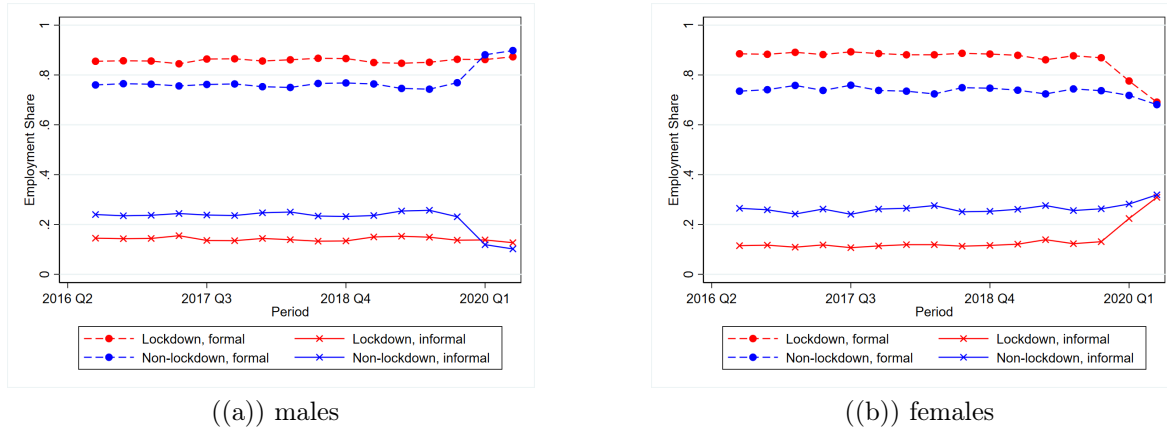


Figure 2.5: Gender-specific trajectories of formal and informal employment

2.4.3 Heterogeneous Effects and Robustness Analysis

The finding that widespread lockdowns had a positive and significant effect on the informalization of employment for women, but not for men, begs the question—does this hold for all women and is this robust to alternative model specifications and informal employment definitions?

Heterogeneous Effects Among the Women's Sample

The findings from the preceding section show that the positive effect of lockdowns on the informalization of employment is present only among women. However, this result might differ based on the marital and parental status of women. Table 2.4 presents the results of the regression on different sub-samples—single women, married or cohabiting women with no minor children, and married or cohabiting women with minor children. Informalization is pronounced among women with co-resident children under the age of 15 (minor children), who arguably have more childcare responsibilities than women without or with older or non-dependent children. Oxfam (2021) reports that women on average perform 6.5 hours of primary care work per day during COVID, more than double the 2.4 hours done by men on average. In Germany, despite men's increased involvement in household work during COVID, women also worked more and still did the bulk of the tasks (Kreyenfeld and Zinn 2021). Given the unequal distribution of care work and the flexibility that informal employment offers compared to formal

employment, it is not surprising to find that extreme lockdowns informalize work more for mothers with minor children than others.

This finding resolves the paid work versus care work dilemma that was potentially exacerbated by the lockdown. If a mother experiences a job loss during the lockdown, she can disengage in job search and paid employment to focus on care work. However, the survivalist motive posits a different outcome for developing countries—despite the increased demand for care work, individuals cannot afford unemployment, and inactivity by extension, since they have little to no access to safety nets. This confirms an earlier finding of Bayudan-Dacuycuy and Dacuycuy (2018) that among females, a wage increase has a statistically insignificant effect on housework hours—implying that women try to balance carefully the need to participate in the labor market and care responsibilities. Results in Table 2.4 show that during extreme lockdowns, mothers of minor children have a higher probability of being informally employed, which despite its instability, offers more flexibility.

These results likewise echo the findings in Brazil and Peru that, all else held constant, women with minor children have a higher probability of being in paid employment with positive hours (Aragao and Villanueva 2021). There is also little reason to believe that being informally employed is voluntary or driven by an outright preference for informality, as prior literature demonstrates that mothers in involuntary informal employment are simply unable to look for formal opportunities due to family responsibilities (Rodin et al. 2012). Overall, these results confirm prior expectations that at the height of these lockdowns, women perform the dual role as workers and carers of first resort.

Robustness to Alternative Sample Restrictions and Definitions of Informal Employment

In the main model, the pre-treatment period was limited from the third quarter of 2016-onwards. Considering that labour market policy is within the ambit of the national government, only the period of the incumbent administration, which assumed office in June 2016, was included. Estimating the models on alternative period restrictions generates the same implications. The lockdown effect on informal employment is robustly positive and significant when one considers shorter alternative periods such as only second quarter surveys and only the three preceding quarters. Note that including periods covering earlier administrations (before 2016 Q3) introduces

Table 2.4: FE Regression Results, Women Sample

VARIABLES	(1) All	(2) Single	(3) Married/Cohabiting (no minor children)	(4) Married/Cohabiting (with minor children)
Lockdown area	-0.108*** (0.005)	-0.174*** (0.015)	-0.021† (0.012)	-0.069*** (0.008)
Lockdown period	0.029*** (0.008)	0.039† (0.020)	0.026† (0.015)	0.025* (0.013)
Lockdown area*period	0.022* (0.011)	-0.042† (0.023)	0.016 (0.021)	0.080*** (0.019)
Age	0.006*** (0.000)	0.005*** (0.000)	0.006*** (0.000)	0.004*** (0.000)
Basic education	0.050*** (0.003)	0.051*** (0.009)	0.067*** (0.006)	0.038*** (0.005)
Some post-secondary ed	0.119*** (0.003)	0.108*** (0.006)	0.122*** (0.007)	0.103*** (0.007)
Secondary education	0.141*** (0.002)	0.076*** (0.004)	0.155*** (0.004)	0.153*** (0.004)
Region FE	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes
Constant	0.026*** (0.007)	0.079*** (0.017)	-0.059*** (0.015)	0.138*** (0.012)
Observations	226,137	43,548	64,090	75,185
R-squared	0.066	0.077	0.057	0.029

Robust standard errors in parentheses

*** p<0.001, ** p<0.01, * p<0.05, † p<0.1

Note: The outcome variable is a binary variable indicating informal employment. A minor child is defined as an individual younger than 15 who is living within the same household as the parent. We qualify that the child is minor because of the prevalence of co-residence of non-minor adult children (typically single) with their parents in the Philippines. While children above 15 do usually remain with the parents, they arguably require less care work and parental supervision than younger children and, therefore, are less likely to affect the labor market status of the mother. Thus, a married woman without a child or with a non-minor/adult child who lives in the same household falls under column (3) while a married woman with a minor child is classified under column (4).

more heterogeneity, because labour market policy and enforcement differ across administrations. Similarly, considering a shorter period restriction, despite yielding stronger results, cannot be conceptually justified especially if data availability allows for longer coverage that would sufficiently establish parallel trends.

In terms of age restrictions, the main model limits the sample to age 15 and above, consistent with the definition of the working-age population by the Philippine Statistics Authority and by most statistical offices (ILO nd). It is also uncommon to place an upper bound restriction on the working age for developing countries, considering the prevalence of old age employment. Racelis et al. (2012) shows that over 50 percent of the elderly report being in paid employment, even after the compulsory retirement age in the formal sector. An overwhelming majority of this work is informal. Nevertheless, results hold even under the narrowest sample restriction based on age. By including an upper bound at 64 years old and by increasing the lower bound to 18 years old, results are consistent in direction and, to a large extent, magnitude. Appendix Tables A.3 and A.4 show the results of robustness analysis for alternative timeline and age restrictions, respectively.

Finally, it should be noted that there exist various definitions of informal employment in the literature. In some definitions, informal employment has an insurance-based definition (whether one is a contributor to the social security system) (Madero-Cabib and Cabello-Hutt 2022; Villanueva and Lin 2020). Within mainland Southeast Asia, there are operational differences in the definition of informal employment with Thailand incorporating the social security-based definition, whereas Myanmar and Lao PDR incorporating both social security-based and worker-based definitions (Ducanes et al. 2023). In the Philippines, a 2008 Informal Sector Survey allowed for both an enterprise-based definition of employment in the informal sector and a job-based definition of informal employment (Cabegin 2008). This does not mean that one approach, more so one article, is superior or necessarily correct. On the contrary, highlighting and scrutinizing alternative, context-based definitions is critical in advancing research on informal employment. In this analysis, the preferred definition is both class- and sector-based with an explicit preference for an underestimate, such that the estimates presented are likely lower bounds and the true lockdown effect might even be larger.

In Tables B.4 and B.5 of the Supplementary Material, results show that across all relevant and expanded definitions of informal employment, the lockdown effect is positive and strongly significant and the gendered informalization finding holds. Results also hold even when merely using a class-based definition of informal employment and controlling for or omitting agriculture and government sectors instead (see Table A.7).

2.5 Conclusion

The Philippines imposed what many considered one of the world’s longest and strictest military-enforced lockdowns in select regions at the onset of COVID. This quasi-experimental set-up allows the article to uniquely contribute to the literature by isolating and analyzing the “lockdown effect” on the informalization of employment and distinguishing between compositional and survivalist explanations underlying this process. Using a two-way fixed effects difference-in-differences design on pooled LFS data, results show a positive and significant effect of extreme lockdowns on the probability of being in informal employment. This finding is pronounced among women, particularly for working mothers with minor children, and is robust to alternative age and period sample restrictions and definitions of informal employment.

The article explores compositional factors and survivalist motives that help explain these findings. While compositional changes in the employed population played a role, survivalist motives also suggest that some workers, males in particular, are more likely to informally work rather than be unemployed. Around 44% of households in lockdown regions also engaged in additional income generating work.⁷ Meanwhile, gendered differences are also explained by women retaining or taking on paid informal employment due to increased uncertainty brought about by the lockdowns, on top of their increased caregiving responsibilities.

One caveat when interpreting these results is that LFS data consistently undercounts informal employment in developing countries. More specifically, the LFS might exclude from the definition of informally employed those wage and salary workers employed in unregistered businesses since the LFS does not contain information about the employers’ legal status. While results are demonstrated to be robust across possible alternative definitions, the data is inherently limited in that there is no information about social security contributions or the legal status of the employer. Still, this limitation underestimates rather than obscures the causal effect of extreme lockdowns on the probability of informal employment, so the true effect might actually be somewhat larger than the estimates presented.

Similarly, further research could look into sector-specific differences in informal employment given its prevalence in some occupations (e.g. babysitting, house cleaners, etc.). Absent panel data in the Philippines that allows analysis of individual trajectories, qualitative work that focuses on individual coping mechanisms regarding

⁷See own estimates in Appendix Table A.10 using World Bank (2022) survey data.

employment might likewise be insightful.

Limitations aside, these results should be read not only as an analysis of the gendered effects of COVID, but also as part and parcel of a rich body of literature demonstrating how informally employed women bear the brunt during periods of crises and how state responses, or the lack thereof, facilitate informalization (Hammer and Ness 2021). In light of growing uncertainties, mobility restrictions are becoming either a consequence (e.g. during natural disasters) or a response mechanism (e.g. during pandemics). Either way, informal employment rises succeeding these crises (Feridhanusetyawan and Gaduh 2000; Floro and Dymski 2000; Mendoza and Jara 2020; Pecha Garzón 2017), although some evidence suggests that informal employment does not always play its usual countercyclical role (Maurizio et al. 2023).

Whereas most of the recent literature has interchangeably treated the "COVID-19 pandemic" and "lockdowns" in both language and substance, this article offers evidence that stringently enforced and widespread lockdowns, *per se*, caused an informalization of employment for women. This highlights the need to reconsider the design and targeting of safety nets for vulnerable segments in the labour market but, more broadly, challenges any notion or misbelief that employment responses during the pandemic were only male-driven. On the contrary, the lockdown effect on informalization is pronounced for women with minor children who also arguably perform more caregiving responsibilities. Women, therefore, played a dual role of being both a worker and carer of first resort. Expanding gender-sensitive and targeted active labour market policies including job search assistance, training, and public employment programs is likely to proactively establish additional "layers" of defence against crises-driven informalization. While some argue that lockdowns were the "great equalizer" that affected everyone equally, the article offers evidence that in the case of informal employment, such was not the case.

Chapter 3

Initial Employment Conditions and Subsequent Fertility Behavior

ABSTRACT

The conditions upon which people enter the labor market have been demonstrated to affect a variety of later life outcomes such as family formation, employment, and wealth accumulation. A growing strand of the literature that emerged after the 2008-09 Global Financial Crisis has shown that employment uncertainty leads to postponed childbearing and higher childlessness. Yet it is not only individual job conditions that matter, broader macroeconomic conditions upon entry also matter. Indeed, the “scarring” literature has likewise demonstrated how cohort-specific outcomes (e.g. wages, fertility) of recession entrants are lower when compared to non-recession entrants. Using detailed employment and birth histories of labor market entrants in Germany, this paper examines the effects of initial conditions, operationalized using fixed-term employment and recession year entry, on subsequent fertility behavior. Partly addressing endogenous selection into initial conditions, we employ a two-step identification strategy combining a non-parametric optimal full matching step and a parametric event history modeling step using the matched data. Results suggest that entering the labor market with a fixed-term contract has persistent negative effects on first births up to a decade after entry and the results are pronounced only for females, whereas entering during a recession has persistent negative effects only for males.

Keywords: *fertility, labor market entry, fixed-term employment, recession entry*

3.1 Introduction

Do initial employment conditions persistently shape later life outcomes? A rich body of the literature overwhelmingly suggests that the conditions upon which people enter the labor market have been shown to affect subsequent employment (e.g. wages and career trajectories), family formation (e.g. marriage and fertility), and wealth accumulation (e.g. home ownership, savings) outcomes (von Wachter 2020b,a; Huckfeldt 2022; Choi et al. 2020; Blossfeld 1987; Scherer 2004; Witteveen 2017). As such, understanding the nuances of and factors underlying these patterns has long been of interest in the social sciences literature, especially during and after periods of economic uncertainty. Among many relevant outcomes, fertility is of key interest among developed economies with aging societies such as Germany, given that its steady decline in births has considerable implications on its pension and healthcare systems as well as workforce composition.

Prior literature demonstrates that entering the labor market with some form of uncertain employment conditions typically leads to either postponed childbearing, lower probabilities of parenthood, or both (Alderotti et al. 2021; Auer and Danzer 2016; Hofmann et al. 2017; Schmitt 2021; Vignoli et al. 2020b). Indeed, among young females in Germany, the probability of childbearing for those with fixed-term contracts is 20 percentage points lower than those with a permanent contract and these differences persist even after a decade since labor market entry and are also evident in the number of children (Auer and Danzer 2016). Fixed-term contracts have risen in popularity at least in part due to calls for flexibilization in labor markets that on the one hand reduces lower turnover costs and increases optimal search matches, but on the other hand, increases inequalities in job security (Gebel 2009). In Germany, fixed-term contracts are particularly prominent among labor market entrants, who are more willing to take these positions as “stepping stones” before pursuing more stable permanent careers later in life. Indeed, Gebel (2009) estimates using panel data from 1989 to 2006 that about 31% of employed labor market entrants had a fixed-term contract. The main difference between permanent and fixed-term contracts is the tenure duration, with the former being open-ended and the latter being time-bound without any guarantee for renewal or eventual regularization.

In Italy, Vignoli et al. (2020b) estimate, using propensity score matching methods, the causal effect of first job uncertainty (e.g. fixed-term employment) on first childbearing within five years and finds that 7 percent of first-birth postponement of women, and 5 percent of that of men, is attributable to these precarious initial

conditions. In other words, individuals who had precarious first jobs would have had a first child, or had one earlier, had they secured a more stable job.

Yet it is not only individual working conditions upon labor market entry that matter for fertility choices, but how they interact with broader macroeconomic conditions. Indeed, prior work on “scarring effects” has likewise demonstrated how labor market entry during a recession negatively affects the same life-course outcomes earlier mentioned, when compared to the outcomes of the non-recession cohorts. In particular, there has been consistent empirical evidence that those who enter during a recession experience wage and career trajectory penalties that are persistent and detectable even after a decade (Cockx 2016; Oreopoulos et al. 2012; Schwandt and von Wachter 2020). In terms of family formation, labor market entry during a recession affects not only fertility behavior but also partnership formation and home ownership (Choi et al. 2020).

The mechanisms through which fixed-term entry and recession entry affects fertility are somewhat similar. On the one hand, the (objective) “income” effect captures actual wage differentials—the fact that young workers who just entered the labor market with a fixed-term contract earn lower on average and thus have narrower opportunities to achieve financial security and accumulate wealth (Kalleberg 2000; Keller and Seifert 2013; Gebel 2009). Those who entered the labor market during a recession likewise have lower wages and flatter career trajectories, *ceteris paribus* (Oreopoulos et al. 2012; Rothstein 2021). On the other hand, the subjective “uncertainty” effect pertains to differences in perceptions of economic security and overall well-being—with those in uncertain employment conditions and those who entered during recessions experiencing more worries and longer job search spells, among others (Gebel 2009; Gebel and Giesecke 2009; Van Wijk et al. 2021).

Common in the literature is separating the analysis of these individual and macro-level manifestations of employment conditions on fertility outcomes. Speaking to these two broad strands, this paper operationalizes initial conditions in two ways—fixed-term employment entry and recession entry and differentially examines their effects on subsequent fertility behavior. We look at the case of Germany, a country that has experienced an incremental but consistent increase in fixed-term employment since the reunification, of which 30 to 40 percent are underemployed.¹ Using fertility and

¹Underemployment is defined as being in involuntary nonstandard employment—those who are in fixed-term contracts but want a permanent one. The figure came from the public report of the Statistisches Bundesamt as of June 2022 based on Labor Force Survey estimates.

employment histories from the German Socio-Economic Panel (GSOEP), we employ a two-step identification strategy combining a non-parametric optimal full matching step and a parametric event history modeling step using the matched data. Doing so has a couple of advantages to relying either solely on propensity scores-based matching and weighting techniques or solely on event history models. First, optimal full matching makes use of the entire sample and creates “subsets” where the distance between the treated and control subjects is minimized. Second, Austin and Stuart (2015) argue that its combination with analyzing survival outcomes has rarely been used, even in epidemiological papers where time-to-event is a common outcome of interest. This two-step identification strategy of optimal full matching based on propensity scores and survival analysis has not been done in the fertility literature, to the best of our knowledge.

In each step of the identification strategy, we analyze different fertility outcomes. After the first step (matching), we estimate the annual Average Treatment Effects on the Treated (ATTs) on a tempo measure (e.g. annual childbearing probabilities up to 10 years after labor market entry) and a quasi-quantum measure (e.g. total number of children after 5 to 20 years). After the second step (Cox Proportional Hazards model), we estimate hazard ratios in the full and gender-restricted samples to test for aggregate and gendered differences.

The paper contributes to the literature in two ways. First, it builds on the growing strand of “causal modeling” in fertility research (Hofmann et al. 2017; Kreyenfeld 2021; Vignoli et al. 2020b) by employing a combination of matching and event history methods to find a meaningful comparison group among initial labor market entrants. Intuitively, one compares the fertility outcomes of individuals with a similar propensity to be “treated” (e.g. work fixed-term or enter during a recession) based on their socio-demographic characteristics, where one group is treated and the other is not. Thus so, this paper extends the literature that methodologically employs either pure propensity scores-based matching and weighting techniques (Hofmann et al. 2017; Vignoli et al. 2020b; Auer and Danzer 2016; Di Nallo and Lipps 2023) or pure event history models (Schmitt 2012; Kreyenfeld 2015; Schmitt 2021; Van Wijk et al. 2021; Laß 2020). This is a recent methodological advancement that has been used in epidemiology and public health where matching is used as a pre-processing step before survival analysis to partly address endogenous selection bias (Austin 2011, 2014; Austin et al. 2020a; Austin and Stuart 2015).

Our second contribution is conceptually and empirically differentiating between individual and macro-level conditions as a first step before examining any potential differences of the individual conditions across the business cycle as has been found in prior Germany-focused studies on fertility (Di Nallo and Lipps 2023; Hofmann et al. 2017). In addition, we test whether fixed-term entry and recession entry have a differential and joint effect on subsequent fertility outcomes not both the entire-sample and gender-restricted samples.

The rest of the paper is organized as follows: the succeeding section reviews the takeaways from the employment-fertility and scarring literature and subsequently develops testable hypotheses. Section 3 presents the data and methods, while Sections 4 and 5 discuss the results and concludes, respectively.

3.2 Brief Review of the Literature and Hypotheses

3.2.1 Fertility Implications of Labor Market Conditions

Labor market conditions, broadly defined, is an umbrella term referring to either individual working arrangements or the macroeconomic conditions under which one is working. At the individual level, some working conditions such as part-time employment, fixed-term contracts, and solo self-employment are more likely to be insecure, temporary, or precarious. Its relevance has arguably grown in recent decades, at least in part due to the increased demand for labor market flexibility (Allmendinger et al. 2013; Kalleberg 2000). Among scholars of fertility behavior, the operationalization of employment conditions has been varied and context-dependent, thus so providing us with various perspectives of how different types of uncertainty (subjective, objective, macro, micro, etc.) affect childbearing. For instance, defining “employment uncertainty” using both objective and subjective indicators, Schmitt (2021) and Laß (2020) find that women experiencing some form of employment uncertainty in Germany have a significantly lower transition to first birth relative to those with no reported uncertainty, especially for women. However, among women with lower levels of education, uncertainty accelerates this transition (Kreyenfeld 2010). This heterogeneity across educational levels does not seem to be universal—Pailhé and Solaz (2012) find no such distinctions in the case of France.

This paper operationalizes uncertain initial employment conditions in terms of fixed-term employment which is relevant in Germany where workers generally strive

for stability and tenure security and fixed-term workers are typically worse-off (Giesecke 2009). Fixed-term contracts are more prevalent among labor market entrants in Germany, covering about 31% on average, from 1989 to 2006 (Gebel 2009). For the whole population, only 7.8% of workers are in fixed-term employment as of 2022, around 90% of whom are in contracts with a duration of less than 3 years.² Thus, the fixed-term employment seems to be a bigger concern among labor market entrants, and whether having this type of contract as a stepping stone has long-term effects.

Along these lines, previous Germany-focused studies have shown that being in either fixed-term contracts or marginal (part-time) employment reduces the transition rate to first birth. Except for Gebel and Giesecke (2009) who find no evidence of a relationship between fixed-term employment and fertility decisions, other papers find that such conditions have a negative effect on overall childbearing (Auer and Danzer 2016; Schmitt 2012) and a postponement effect on parenthood transitions (Schmitt 2021). These different results could be explained by differences in the context, period, analytical samples, and identification strategies employed by these papers, some of which may not be able to fully address endogenous selection into employment conditions.

Of late, the literature largely points to the negative effect of fixed-term employment on fertility in many contexts (Alderotti et al. 2021) including for women in Australia (Laß 2020) and Southern Europe (Adsera 2011), men in Japan (Piotrowski et al. 2018), and both sexes in the Netherlands, Germany, and Italy (Schmitt 2021; Van Wijk et al. 2021; Vignoli et al. 2020b). The majority of these studies rely on present-time indicators of fixed-term employment, emphasizing the idea that current economic and employment conditions matter for fertility behavior. Building on this literature, this paper focuses on initial employment conditions, particularly entering the labor market with a fixed-term contract, for two substantive reasons. First, current employment conditions are path-dependent outcomes which makes earlier conditions similarly relevant in analyzing fertility. Second, the temporal framing of uncertainty has received renewed attention in the literature with some papers arguing the understated relevance of future narratives (Lappegård et al. 2022; Vignoli et al. 2020a). This motivates increased scrutiny not only of uncertainties of the future but also of those of the past and present to fully understand both the nuances and points of convergence between and among these temporal frames. Consequently, the first hypothesis addresses these recurrent findings

²Latest estimate from DESTATIS, available online here.

such that:

- *Hypothesis 1:* Entering the labor market with a fixed-term as opposed to a permanent contract reduces annual first birth probabilities, number of children, and first birth hazard rates within the first 10 years of entry.

In terms of the gendered differences, the literature on employment conditions and fertility in Germany leans towards demonstrating a negative effect of fixed-term employment among women (Kreyenfeld 2010; Auer and Danzer 2016; Schmitt 2021). Note that in Germany, while workers on fixed-term contracts are also entitled to paid parental leaves, taking it does not extend the period or duration of the contract. While this may affect the fertility decision-making process, it is unlikely to do so for labor market entrants because of the temporal spacing between labor market entry (occurring at around early-mid 20s) and first birth (occurring at around late 20s-early 30s). Regarding gendered differences, there are two competing mechanisms. On the one hand, under a breadwinner model where men serve as the main economic providers of the family, uncertainty in male careers should be more detrimental to fertility than women's. On the other hand, as women are the ones who take more extended parental leaves, it is more important for them to acquire a permanent contract prior to childbearing to avoid the disruption of their careers post-childbirth. As such, we tangentially test the gendered differences in H1 such that:

- *Hypothesis 1a:* The effects of fixed-term entry on subsequent fertility behavior are more pronounced for females than males.

3.2.2 On the Victims of "Scarring"

While the preceding section unpacks the role of micro-level employment conditions, the literature likewise tells us that macroeconomic conditions also matter for various life course outcomes. Recessionary periods that are characterized by high inflation and high unemployment, among others, have been demonstrated in the literature to affect employment and fertility outcomes persistently (Currie et al. 2014; Oreopoulos et al. 2012; Choi et al. 2020; Matysiak et al. 2021b). The "scarring" phenomenon pertains to workers whose subsequent life-course outcomes are persistently and negatively impacted by their experiences in the labor market. The literature on scarring can be grouped into two depending on the "victims"—one strand pertains to workers who

suffer from an involuntary job loss or reduction in work hours (“displaced”), while the other strand encompasses first-time labor market entrants (“entrants”).

In the scarring literature, displaced individuals consequently experience a reduction in total income in the years of and succeeding a crisis. While some of those who retained their jobs experienced income losses, Huckfeldt (2022) demonstrates that this observed aggregate pattern is driven by “occupation switchers”—those who transitioned to another occupation after being unemployed. Thus, scarring in this case pertains to the pronounced and persistent income losses experienced by those displaced. In terms of fertility outcomes, the effect of job displacement during a recession varies greatly by population subgroup. Male unemployment has a negative effect on fatherhood transitions in Germany, Denmark, and England, but the effects of female unemployment are less clear and consistent (Aksoy 2016; Andersen and Özcan 2021; Kreyenfeld and Andersson 2014; Özcan et al. 2010). Recognizing that the effects of job displacement on fertility have been extensively studied in the case of Germany (Hofmann et al. 2017; Di Nallo and Lipps 2023), this paper solely focuses on the entrants.

Labor market entrants are “victims” in the sense that starting to work during “bad times” is associated with persistent income losses through lower wages and income trajectories. Mostly studied among advanced economies (for example, South Korea, Germany, and the US), the consensus is that individuals who entered during and after recessions have lower income trajectories (Cockx 2016; Rothstein 2021; Umkehrer 2019; von Wachter 2020a). Interestingly, the literature is somewhat less clear in terms of family formation outcomes. While women who enter the labor market during a recession year accelerate the transition to parenthood (Choi et al. 2020; Hofmann and Hohmeyer 2016), the total number of children of “unlucky entrants” is found to be higher (Choi et al. 2020), lower (Schwandt and von Wachter 2020), and not significantly different (Hofmann and Hohmeyer 2016) from those who enter the labor market in better times. For men, evidence from South Korea points to recession entry having a negative effect on subsequent marriage and childbearing (Choi et al. 2020). These studies have of course focused on different contexts and unpacked the nuances of these aggregate findings.

In this paper, we focus on entrants rather than displaced individuals. Considering the empirically ambiguous relationship between recession entry and fertility outcomes, the second hypothesis weighs in on this debate and tests whether a scarring (e.g.

negative) effect is observed:

- *Hypothesis 2:* Entering the labor market in a recession year as opposed to a non-recession year reduces annual first birth probabilities, number of children, and first birth hazard rates within the first 10 years of entry.

Similarly, a tangential interest in this paper is to understand whether the effects of recession entry on fertility behavior are different by gender, although prior literature has been ambiguous in this regard, as discussed above. On the one hand, young adult males who enter the labor market during a recession and consequently begin with an uncertain or precarious economic position might be perceived as less desirable in dating and matching markets and, when coupled with a heightened sense of subjective economic insecurity, further delay partnership formation and fertility (Currie et al. 2014). On the other hand, young adult females who enter during a recession might postpone births even later than non-recession entrants as they take a longer time to stabilize their occupational positions. This extended postponement might ultimately lead to higher childlessness. This is exactly what was found in the US. Currie et al. (2014) demonstrate that a one percent increase in the unemployment rate in the year of labor market entry reduces the completed fertility of that cohort by about 14.2 per 1000 conceptions. As such, while we have no strong priors regarding the direction and significance of the gendered differences in H2, we consistently adopt the phrasing of H1a, where women are presumed to value employment stability before childbearing, and test the hypothesis that:

- *Hypothesis 2a:* The effects of recession entry on subsequent fertility behavior are more pronounced for females than males.

Finally, we likewise extend the analysis to test whether a “double scarring” phenomenon exists for workers who entered on a fixed-term contract during a recession. In particular, the negative effect on subsequent fertility experienced by fixed-term entrants might be more pronounced if they entered during a recession year. For one, their subjective economic worries are more likely two-pronged— they are worried about both the temporary nature of their employment on one hand and turbulent macroeconomic conditions which will shape their future employment opportunities. As such, we formally test the hypothesis that:

- *Hypothesis 3:* Entering the German labor market with a fixed-term contract during a recession as opposed to during a non-recession year has a more pronounced reduction in annual first birth probabilities, number of children, and first birth hazard rates within the first 10 years of entry.

3.3 Methodology

3.3.1 Data and Analytical Sample

This paper uses individual employment and fertility data from the German Socio-Economic Panel (GSOEP) from 1990 until 2022. The main analytical sample consists of childless individuals at the time of LM entry, with available information on their subsequent childbearing behavior within 10 years post-entry. We dropped individuals whose reported first employment status was unemployed since there won't be any information about their first job characteristics and it likely means that they started working at some point in the future but not at the year of reported LM entry. This also allows us to focus our analysis between the types of contract (fixed-term vs. permanent), *conditional on* being employed. GSOEP contains two variables indicating the year of LM entry and the results shown in this paper use the measure that has a slightly higher sample size.³ Both measures define LM entry as the year of or after leaving school or graduation, excluding any military service or apprenticeships. Appendix Table B.1 reports the step-by-step derivation of the analytical sample and summary statistics. In our sample, we find that about 38% of labor market entrants have a fixed-term contract, a slightly higher estimate than the 31.4% observed by Gebel (2009), which uses the same dataset up to 2006. Similarly, about 11% of entrants are recession cohorts.

The main independent variables of interest are indicators of LM entry conditions including fixed-term employment and labor market entry during a recession year. Particularly, the focus is on the first job succeeding school- or university-leaving or end of vocational training, excluding apprenticeships (which despite being typically fixed-

³One measure is based on retrospective personal biographies (PBIO) and the other is based on monthly activity calendars from education to employment (ARTKALEN). While these two measures yield fairly similar sample sizes and employment status shares, all results shown in this paper use the latter measure based on monthly calendars due to the slightly higher sample size of 5,500 compared to the former with 4,795. Appendix Figure B.1 illustrates the sample selection process.

term are part of one’s vocational training)⁴ and military service. Recession years, on the other hand, are the four recession years in Germany identified by the World Bank’s World Development Indicators and Federal Reserve databases (1993, 2002, 2003, and 2009). Finally, the main dependent variable is the date of first childbirth, backdated by 9 months to capture the approximate period of childbearing decision.

3.3.2 Identification Strategy

Event history analysis is commonly used in many studies analyzing parenthood transitions. However, one limitation of using this approach to analyze the impact of initial LM conditions is the likely selection into fixed-term employment and the year of LM entry. For instance, some highly educated young adults might want to take a gap year or delay their university completion because of poor employment prospects during a recession. Similarly, university graduates and those with vocational training are more likely to enter a fixed-term contract in Germany (Gebel 2009). Broadly, whether an individual enters the LM with a fixed-term contract or during a recession year is likely non-random since certain socio-demographic groups may have a higher propensity of doing so. Partially addressing this issue, we follow and calibrate matching techniques to establish a covariate balance between permanent and fixed-term entrants and between recession and non-recession entrants (Vignoli et al. 2020b; Gebel 2009). Specifically, we employ a **two-step identification strategy** that combines optimal full matching using propensity scores to construct a balanced sample and estimate year-specific Average Treatment Effect on the Treated (ATT) and Cox proportional hazards model to obtain the Hazard Ratio (HR) between the treated and control groups. Austin and Stuart (2015) and Austin et al. (2020b) suggest that the combination of optimal full matching and covariate-adjusted survival analyses performs better than nearest neighbor and caliper matching techniques due to its flexibility and full use of the entire sample. Simply put, optimal full matching differs such that it creates matched subsets that could vary in size. A treated observation can be matched with one or more controls, or vice versa, based on what yields the smallest overall difference across all covariates within matched sets.

In the first step, we employ propensity score matching using pre-entry socio-demographic information including sex, educational attainment, age at entry, an

⁴An apprenticeship or *ausbildung* is not considered as “labor market entry”. For young workers who go through this track, their year of entry is when they have their first full-time job.

indicator variable for whether the labor market entrant had a first job in East Germany, and migration background which yields the smallest overall difference across these variables within the matched sets. Succeeding this matching procedure, we present two outcomes of interest— ATTs for first birth probabilities in each year up to a decade after entry as a *tempo* measure and the ATTs for the total number of children within a decade after entry as a *quasi-quantum* measure.⁵ The ATT represents the difference in outcomes between fixed-term and permanent first jobs among those with a fixed-term job and between recession and non-recession entrants among recession entrants.

Matching methods have gained popularity in fertility research recently, and an overview of these studies thus far, as well as others that employ comparable causal modeling techniques, has been reported by Kreyenfeld (2021). Similar to Vignoli et al. (2020b), Hofmann et al. (2017) employ inverse probability weighting methods to account for differences between women who were displaced (treated) and were not displaced (control) from the job. These weights were obtained using logistic regression estimates, similar to how propensity scores were estimated by Vignoli et al. (2020b). The approach of this paper also builds on Auer and Danzer (2016) by using matching as a non-parametric pre-processing stage to reduce model dependence instead of fully relying on a regression adjustment.

One limitation of matching based only on individual pre-treatment characteristics such as age and educational levels is that compositional factors might partly explain the treatment effects. For instance, women who entered fixed-term employment during an economic recession might have different characteristics from women who entered part-time employment during an economic boom. In a quasi-experimental set-up, inverse probability weighting accounts for this, as was done by Hofmann et al. (2017). A straightforward alternative that accounts for macroeconomic conditions is to obtain the ATT of fixed-term first jobs during recession vs. non-recession years. In effect, the paper likewise distinguishes between the effects of individual first-job conditions on fertility across the business cycle.

The paper proceeds with the analysis in two phases. First, we obtain the ATT, as was done by Vignoli et al. (2020b), which straightforwardly compares the outcomes of interest between the “treated” and “control” individuals. We present the ATTs for each year up to a decade after LM entry which could be interpreted as the difference

⁵Quasi-quantum in this case pertains to the incomplete cohort fertility of LM entrants. It is incomplete in the sense that some (female) LM entrants in the sample, particularly those from recent survey years, have not yet reached the end of biological childbearing ages.

in first birth probabilities within 1-10 years after entry. Second, we use the weights and subclasses of the matched sample to estimate first birth Hazard Ratios (HR) using a Cox Proportional Hazards (PH) model and show adjusted survival probabilities over time Austin (2011, 2014); Austin and Stuart (2015). We estimate the model, parsimoniously specified as:

$$h_p(t) = h_0(t) \exp(\beta_p X_p + \beta_q X_q),$$

where $h_0(t)$ is the baseline hazard, X_p is a treatment dummy for fixed-term or recession entrants, and X_q is a vector of time-constant socio-demographic and temporal controls. Note that despite the matching procedure creating comparable groups based on observed covariates, there might still be some residual imbalance or variability within the matched sets. Greifer (2024b) suggests that it's ideal to include covariates, although it's unnecessary when "excellent balance" is achieved through matching.

3.4 Results and Discussion

3.4.1 ATTs from Matching Analysis

Key Finding 1. *Persistent negative effect of fixed-term entry of up to -5pp on first birth probabilities within 5-10 years after entry and on the number of children within 5 and 10 years.*

Figure 3.1 presents the ATTs suggesting a persistently negative effect of fixed-term first jobs on first birth probabilities beginning the 5th year after entry, lending partial support to the first hypothesis. As seen in the left panel, the absence of a detectable treatment effect in the first few years is not surprising, given the typical temporal gap between labor market entry (around early to mid-20s on average) and first childbirth (around early 30s on average as of 2023). At its strongest, the effect size hovers around a 5pp reduction in first birth probability for fixed-term compared to permanent entrants. Meanwhile, the negative effect on the number of children is barely significant only within the first 5 and 10 years of labor market entry (~6pp), and negative but insignificant within the first 15 and 20 years. These results lend support to the idea that unstable initial LM conditions have negative fertility implications on both tempo and quasi-quantum measures.

Key Finding 2. *Null effect of recession entry on first birth probabilities within the*

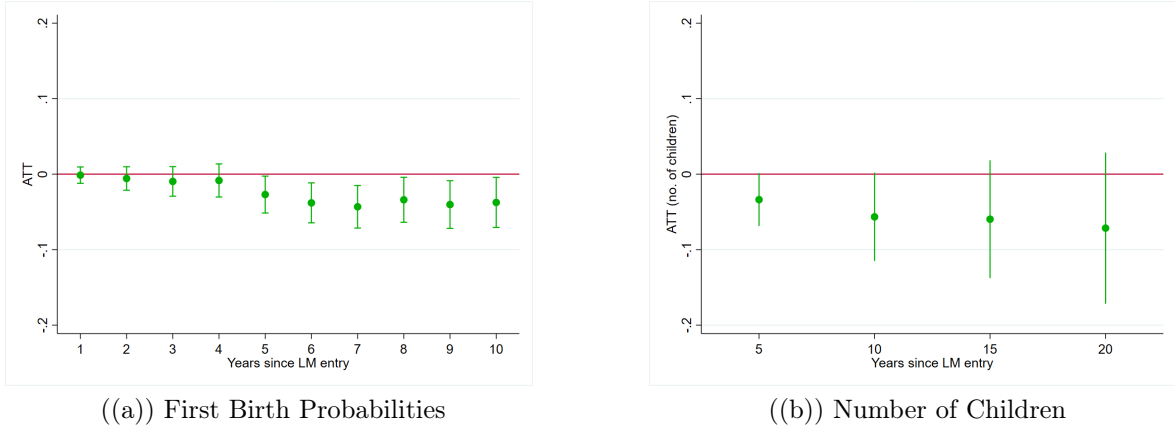


Figure 3.1: Fixed-Term First Jobs and Subsequent Fertility Outcomes

Note: The matching variables used in these models are socio-demographic pre-treatment characteristics including sex, educational attainment at entry, East Germany, age at entry, and migration background. Appendix Figure B.2 shows that the optimal full matching procedure achieves overall covariate balance.

first 10 years after entry and insignificant effect on the number of children.

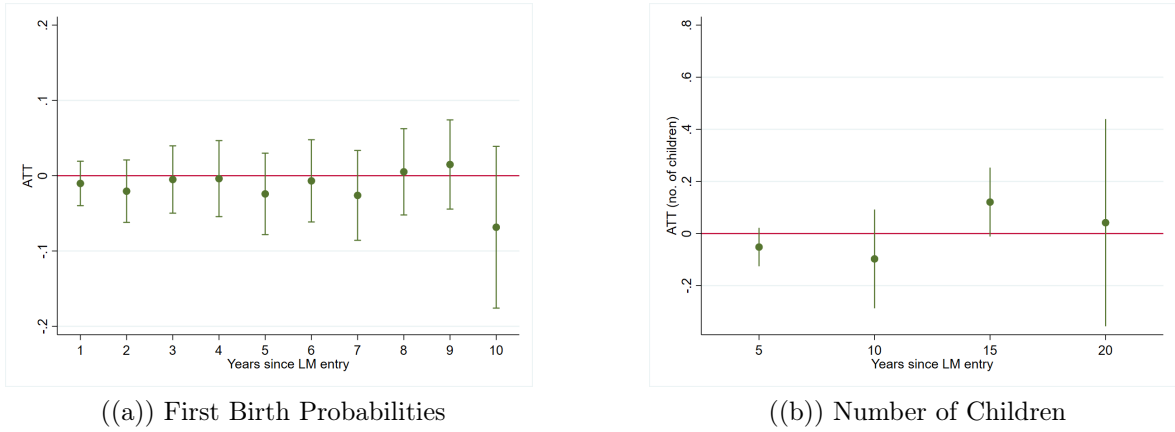


Figure 3.2: Recession Entry and Subsequent Fertility Outcomes

Note: The matching variables used in these models include the state-level (BuLa) unemployment rate and pre-treatment socio-demographic and temporal characteristics including sex, educational attainment at entry, and age at entry. Appendix Figure B.2 shows that the optimal full matching procedure achieves overall covariate balance.

On the other hand, results do not suggest any effects of recession entry on subsequent fertility outcomes, contrary to findings in South Korea (Choi et al. 2020)

and the US (Schwandt and von Wachter 2020). These results lend support to earlier findings in Germany of a non-significant difference between recession and non-recession entrants in terms of the number of children (Hofmann and Hohmeyer 2016) but not to the “acceleration” phenomenon observed among female recession entrants. Figure 3.2 shows that across all years post-entry, the ATTs for first birth probabilities are not statistically different from zero. In comparison to the analysis of fixed-term entry, it is notable that these estimates have considerably larger confidence intervals due to lower cases of recession entrants (i.e. there are only four official recession years).

3.4.2 Hazard Ratios using the Matched Data

Following Austin (2011, 2014) and Austin and Stuart (2015), matching was conducted as a pre-processing step before survival analysis. While a simple comparison of the outcome between the treated and control units in the matched data yields the ATT presented above, the matching procedure likewise produces a set of weights and subclasses that can be used for a standard event history model. In this section, we use the weights and subclasses from an optimal full matching procedure using propensity scores to estimate hazard ratios for the timing of first birth and plot the adjusted survival probabilities within the first 10 years of entry.⁶ As an additional test of heterogeneity by gender, we present estimates using the full, male-restricted, and female-restricted samples.

Key Finding 3. *Significant difference in survival probabilities by fixed-term entry, but the effect is pronounced only for females with a 19% lower first birth hazard*

In our preferred specification in Table 3.1, results suggest that fixed-term entrants have a 12 percent reduced first-birth hazard compared to permanent counterparts. Disaggregating the analysis by gender, this difference is only significant among females whose fixed-term entrants have a 19 percent lower hazard. To visualize the effect of fixed-term entry on cumulative survival probabilities over the first 10 years of entry, we show adjusted survival curves obtained using estimates from the Cox model. Thus, these survival curves control for the potentially confounding impacts of other variables, namely sex, educational attainment at entry, East Germany indicator, age at entry, and migration background, on survival. Results in Figure 3.3 confirm earlier findings that

⁶Note that the coefficient of the treatment variable obtained from estimating a Cox proportional hazards model with covariates is the conditional effect on first birth timing. For a detailed discussion on this, see (Greifer 2024b).

Table 3.1: Cox PH Results on Matched Sample: Fixed-Term Entry

	<i>Dependent variable:</i>			
	TIME			
	(Base)	(Full)	(Males)	(Females)
Fixed-Term Entrant	0.906 (0.057)	0.879* (0.058)	1.043 (0.094)	0.813** (0.073)
Female		1.825*** (0.057)		
Education-Mid		0.987 (0.097)	1.076 (0.157)	0.919 (0.123)
Education-High		1.475** (0.118)	2.283*** (0.190)	1.123 (0.153)
East Germany		1.368*** (0.068)	1.155 (0.121)	1.504*** (0.082)
LM Entry Age 21-25		1.275** (0.086)	1.189 (0.143)	1.354** (0.109)
LM Entry Age 26-		1.689*** (0.111)	1.554** (0.175)	1.799*** (0.145)
LM Entry Full-Time		1.178** (0.062)	1.456** (0.108)	1.045 (0.076)
Migration Background		1.419*** (0.068)	1.536*** (0.111)	1.357*** (0.086)
Observations	5,500	5,500	2,666	2,834
R ²	0.001	0.051	0.045	0.028

Note:

*p<0.1; **p<0.05; ***p<0.01

secure and permanent employment facilitates higher first-birth survival probabilities and that the small but significant difference between curves is apparent for the female sample. Further analyses using interaction models do not support any detectable difference of the treatment effect between East and West Germany (see Appendix Table B.4). These results lend support to our hypotheses 1 and 1a and the prior literature in Germany showing the pronounced impact of female employment conditions on fertility (Kreyenfeld 2010; Auer and Danzer 2016; Schmitt 2021).

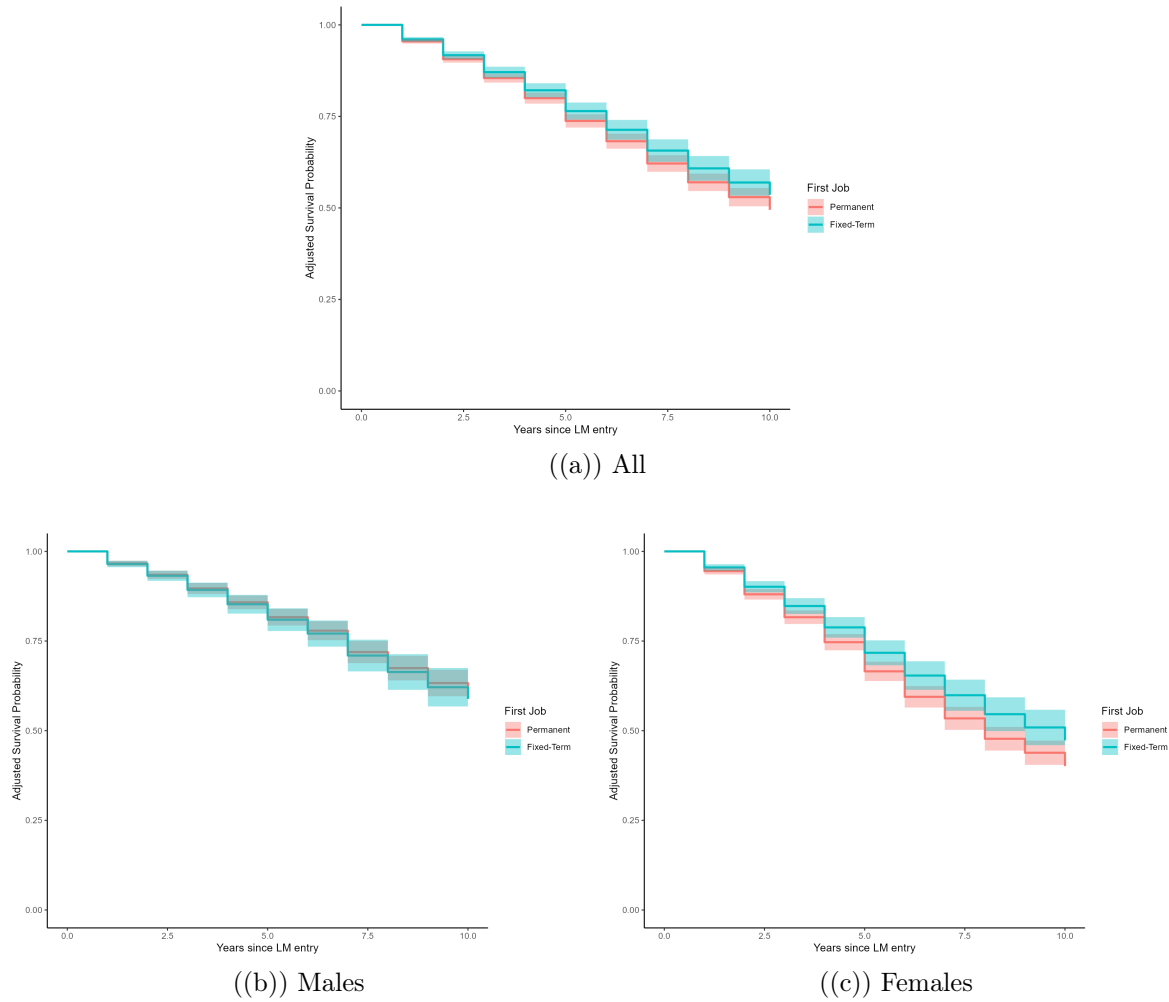


Figure 3.3: Adjusted Survival Curves for Fixed-Term Entry using Matched Sample

Note: These survival curves are estimated with a maximum exit time capped at 10 years after LM entry. Panels (b) and (c) are group-specific estimates for males and females, respectively. Appendix Figure B.3 shows that the difference in survival probabilities over time is barely significant for the full sample and significant for the female sample. These plots were generated using the R `adjustedCurves` package (Denz 2023).

Key Finding 4. *No significant difference in hazard ratios and cumulative adjusted survival probabilities by recession entry, except in the male-restricted sample.*

Table 3.2: Cox PH Results on Matched Sample: Recession Entry

	<i>Dependent variable:</i>			
	TIME			
	(Base)	(Full)	(Males)	(Females)
Recession Entrant	1.016 (0.080)	1.038 (0.080)	0.765* (0.140)	1.193 (0.098)
Female		1.248* (0.055)		
Education-Mid		1.056 (0.098)	1.038 (0.157)	1.043 (0.126)
Education-High		1.799** (0.118)	2.034 (0.186)	1.648 (0.153)
LM Entry Age 21-25		1.241 (0.090)	0.873 (0.143)	1.549** (0.117)
LM Entry Age 26-		1.346 (0.113)	1.580 (0.166)	1.125 (0.155)
State UR at Entry		1.033*** (0.006)	1.024 (0.010)	1.036** (0.008)
Observations	4,814	4,814	2,291	2,523
R ²	0.00001	0.035	0.076	0.025

Note:

*p<0.1; **p<0.05; ***p<0.01

On the other hand, results from our preferred specification in Table 3.2 suggest that while there is no statistically significant difference in first-birth hazard of recession entrants compared to non-recession counterparts in the full and female sample, male recession entrants have a 23 percent reduced hazard. These estimates account for the potential confounding effect of state-level variation in unemployment rates upon entry as well as other socio-demographic information such as sex, educational attainment at

entry, and age at entry. Compared to the Cox model on fixed-term entry, migration background is not included as a control variable here as there has been no empirical evidence of its confounding effect on recession entry. Meanwhile, instead of an East Germany indicator, we opted for a more disaggregated measure of state-level unemployment rates.

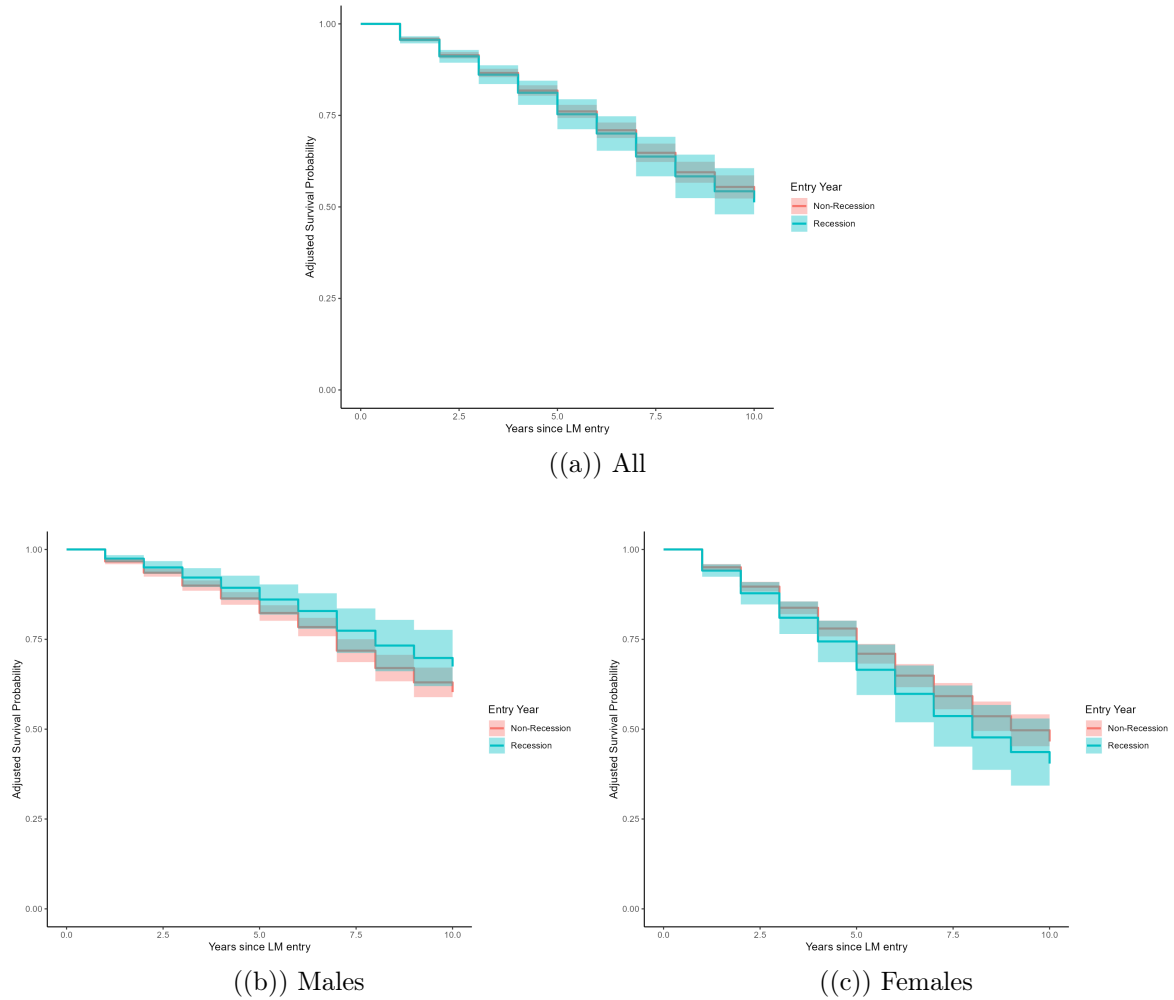


Figure 3.4: Adjusted Survival Curves for Recession Entry using Matched Sample

Note: These survival curves are estimated with a maximum exit time capped at 10 years after LM entry. Panels (b) and (c) are group-specific estimates for males and females, respectively. Appendix Figure B.4 shows that the difference in survival probabilities over time is significant for the male sample. These plots were generated using the R `adjustedCurves` package (Denz 2023).

Further, adjusted survival curves in Figure 3.4 show that male non-recession entrants have higher first-birth survival probabilities. Thus, the null results of recession

entry as captured in Figures 3.2 and panel (a) of Figure 3.4 do not hold when zeroing in on the male population. While prior findings in Germany show that entering during a recession significantly accelerates first birth transitions for females (Hofmann and Hohmeyer 2016), our results suggest a similarity in direction (e.g. acceleration) but not in significance. Similarly, while (Hofmann and Hohmeyer 2016) find a negative but statistically insignificant effect for males, our results confirm the direction but not the (in)significance. These results thus warrant a rejection of hypothesis 2a and echo the South Korean experience during and after the Asian Financial Crisis where male entrants had persistent negative effects on subsequent marriage and childbearing (Choi et al. 2020).

Analyzing the gendered differences of recession entrants, more broadly, extends the findings of Currie et al. (2014) who find that female recession entrants in the US have lower completed fertility. There are a few non-exhaustive and non-mutually exclusive mechanisms that explain this gendered effect—past episodes of recessions might have had a stronger effect on young males which resulted in income losses and lower feelings of economic security. In addition, these objective and subjective worries might have put them at a disadvantage in dating or matching markets (Currie et al. 2014).

3.4.3 “Double Scarring” Effect During Recessions

As an additional test for heterogeneity of the effects of fixed-term entry on fertility across the business cycle, we estimate Cox proportional hazards models by recession and non-recession years as a first-pass test. Fixed-term entrants during recession years might experience a “double scarring” phenomenon where the negative effect on subsequent fertility might be more pronounced compared to fixed-term entrants during non-recession years. By comparing sub-group estimates, “double scarring” can be observed when the hazard ratios of fixed-term entrants are significantly lower for recession compared to non-recession entrants. If so, these individuals would be experiencing a scarring effect from fixed-term employment and an amplifying scarring effect from a recession entry. Results in Table 3.3 suggest that after accounting for socio-demographic information and state-level variation in the unemployment rate at entry, the hazard ratios of fixed-term entry are both insignificant for recession and non-recession years with a comparable magnitude. Further, adjusted survival curves were obtained from interaction models of fixed-term and recession entrants likewise showing no significant difference between group-specific survival probabilities, as shown

in Appendix Figure B.5. Thus, results do not lend support to a “double scarring” effect of fixed-term entry during recessions, contrary to expectations from Hypothesis 3.

3.5 Conclusion

Conditions upon labor market entry have persistent effects on subsequent life course outcomes. Prior literature has demonstrated that individual-level conditions at entry affect subsequent demographic behavior including fertility and marriage (Choi et al. 2020; Hofmann and Hohmeyer 2016; Schmitt 2021; Auer and Danzer 2016). Similarly, macro-level (or cohort-specific) conditions at entry, for instance, recession years, have likewise been demonstrated to affect these outcomes (Schwandt and von Wachter 2020; Scherer 2004; Rothstein 2021; Witteveen 2017).

In this paper, we build on the literature on initial conditions and subsequent fertility behavior in two respects. Methodologically, we employ optimal full matching based on propensity scores as a non-parametric pre-processing step to estimate treatment effects from survival analysis. Common in epidemiological studies, this combination partly addresses endogenous selection bias while maintaining the sample size and properties (Austin et al. 2020b; Austin and Stuart 2015). In the fertility literature, this paper builds on the existing strand that relies either solely on propensity scores-based matching and weighting techniques (Hofmann et al. 2017; Vignoli et al. 2020b; Auer and Danzer 2016; Di Nallo and Lipps 2023) or solely on event history models (Schmitt 2012; Kreyenfeld 2015; Schmitt 2021; Van Wijk et al. 2021; Laß 2020) by using a combination of both approaches. Our second contribution is to make a distinction between individual conditions (e.g. fixed-term vs. permanent) and macro-level conditions (e.g. recession vs. non-recession) at entry and extend the analysis to detect any potential interactive effects. In doing so, we test whether these conditions have a differential and joint effect on subsequent fertility outcomes not both the entire-sample and gender-restricted samples.

Four notable findings emerge from the analyses. First, results suggest that there are persistent negative effects of fixed-term entry on first-birth probabilities and the number of children within 10 years post-entry. This shows that, in general, individual uncertainties upon labor market entry in the form of fixed-term employment have an impact on subsequent fertility behavior, lending support to prior findings in Germany (Schmitt 2021; Auer and Danzer 2016). Second, survival analyses using the matched

Table 3.3: Cox PH Results on Matched Sample: Fixed-Term by Recession Entry

	<i>Dependent variable:</i>			
	TIME			
	Full	with SUR	RY	Non-RY
Fixed-Term Entrant	0.879* (0.058)	0.902 (0.062)	0.944 (0.159)	0.894 (0.068)
Recession Entrant	1.112 (0.080)	1.148 (0.081)		
Female	1.820*** (0.058)	1.824*** (0.063)	2.015*** (0.165)	1.799*** (0.068)
Education-Mid	0.982 (0.097)	1.074 (0.118)	0.905 (0.297)	1.093 (0.129)
Education-High	1.467** (0.118)	1.648*** (0.136)	1.500 (0.340)	1.673*** (0.149)
East Germany	1.369*** (0.068)	1.250* (0.101)	2.035** (0.312)	1.176 (0.109)
LM Entry Age 21-25	1.272** (0.086)	1.489*** (0.107)	1.772* (0.253)	1.470*** (0.119)
LM Entry Age 26-	1.692*** (0.111)	1.957*** (0.128)	2.234** (0.322)	1.958*** (0.142)
LM Entry Full-Time	1.174** (0.062)	1.100 (0.066)	0.680* (0.169)	1.196* (0.072)
Migration Background	1.425*** (0.068)	1.423*** (0.076)	1.939*** (0.206)	1.366*** (0.082)
State UR at Entry		1.020* (0.009)	1.014 (0.029)	1.018 (0.010)
Observations	5,500	4,814	583	4,231
R ²	0.052	0.058	0.110	0.054

Note:

*p<0.1; **p<0.05; ***p<0.01

data suggest that the effect is pronounced only for the female-restricted sample, lending support to prior findings in Germany on the impact of female employment conditions on first-birth transitions (Kreyenfeld 2010; Laß 2020).

Third, as it relates to entry during a recession year, results suggest no significant effect on overall fertility behavior, at least within the first 10 years post-entry. This differs from the US context where recession entrants have persistently lower childbearing on average (Schwandt and von Wachter 2020). It can be argued, however, that the magnitudes of these recessions were different—unemployment rates during the official recession years in Germany pre-COVID were generally lower than the US unemployment rates in their recession in the early 80s which was what Schwandt and von Wachter (2020) focused on. However, our fourth finding demonstrates that when analyzing by gender, male recession entrants have a lower hazard of first birth within the first 10 years by about 23% compared to non-recession entrants. This lends support to findings in South Korea where male entrants during the 1997-98 Asian Financial Crisis had persistent negative effects on subsequent marriage and childbearing (Choi et al. 2020).

Additional robustness analyses using an alternative but comparable measure of labor market entry are shown in Appendix B.3. While the paper contributes methodologically in addressing selection into initial labor market conditions using a two-step identification strategy combining a non-parametric matching step and a parametric Cox Proportional Hazards modeling step using the matched data, we note a couple of limitations. First, we limit our analysis to 10 years post-entry to minimize sample attrition in the estimation of annual ATTs post-entry. Although the size of our base analytical sample from the SOEP (5,500) is slightly larger and comparable to other longitudinal data-based studies of labor market entrants using the National Educational Panel Survey with 2,474 individuals (Hofmann and Hohmeyer 2016) and the Korea Labor and Income Panel Study with 4,794 individuals (Choi et al. 2020), this sample size is still relatively small, compared to what register data can provide. Our second limitation is that although we detect effects in the gender-restricted samples, we are unable to proffer insights as to the underlying mechanisms. While initial labor market conditions likely affect a variety of later life economic and non-economic outcomes (e.g. wage growth, marriage markets, asset formation), we do not empirically tease out the underlying mediators in this nexus.

Moving forward, future research can address these gaps by conducting extensive

mediation analyses using both objective indicators of subsequent employment and life course outcomes (e.g. partnership status, income, etc.) and subjective indicators of uncertainty (e.g. self-reported worries) using register data as was done in the case of the Netherlands (Van Wijk et al. 2021). Further, as labor market entrants during the COVID-19 pandemic move further along the life course, it would be interesting to replicate this analysis later to further understand the long-run demographic implications of the pandemic.

Initial labor market conditions are multi-dimensional. These could be proxied by individual employment conditions such as having a fixed-term, part-time, informal, or low-waged position. However, these could also be operationalized by understanding the macroeconomic conditions upon entry such as recession years or sub-national variations in unemployment and/or output growth. Regardless of measurement and identification, the idea that subsequent life course outcomes are shaped by a set of initial conditions is a recurrent and well-tested one. Indeed, policy responses aimed at improving the enabling conditions for parenthood should also take into consideration the experiences and conditions in young adulthood, including labor market entry.

Chapter 4

Second Birth Fertility in Germany: Social Class, Gender, and the Role of Economic Uncertainty

co-authored with Michaela Kreyenfeld, Dirk Konietzka, and
Philippe Lambert

ABSTRACT

Building on a thick strand of the literature on the determinants of higher-order births, this study uses a gender and class perspective to analyze second-birth progression rates in Germany. Using data from the German Socio-Economic Panel from 1990 to 2020, individuals are classified based on their occupation into upper service, lower service, skilled manual/higher-grade routine nonmanual, and semi-/unskilled manual/lower-grade routine nonmanual classes. Results highlight the “economic advantage” of men and women in service classes who experience strongly elevated second birth rates. Finally, we demonstrate that upward career mobility post-first birth is associated with higher second birth rates, particularly among men.

Keywords: *Fertility, Germany, Uncertainty, Social class, Employment*

4.1 Introduction

Classical demography has devoted substantial attention to the issue of class differences in marriage and fertility behavior. Malthus (1998) is unquestionably the most foundational scholar in this context. A general premise of his work is that there is a strong negative class–fertility gradient. He argued that the higher social classes, which at that time were composed of landlords and members of the aristocracy, would limit their number of children out of a “fear of lowering their condition in life” (ibid.: 6). He also assumed that although the lower social classes often lacked the necessary wealth and economic security to support a large family, their unrestrained sexual behavior would result in high fertility. Malthus’ writings certainly reflect a striking degree of presumptuousness and a strong bias towards believing that the behavior of his own social class was rational and conscientious (Petersen 1990; Pullen 2019). Nevertheless, his framework generated clear and testable hypotheses regarding the association between class, economic security, and fertility behavior. Notestein (1936, p. 29) later elaborated on this perspective by asserting that class differences would “narrow or perhaps even reverse” if the fertility of the lower classes could be brought “more completely under control”.

In contrast to early classical demographic research (Brentano 1910; Malthus 1998; Sallume and Notestein 1932; Notestein 1936), contemporary demography has devoted relatively little attention to the role of social class differences in fertility behavior. Fertility researchers only rarely refer to this concept and instead tend to focus on differences in birth dynamics by education (e.g. Bartus et al. 2013; Nitsche et al. 2018; Nisén et al. 2021), earnings (e.g. Andersson et al. 2014; Heckman and Walker 1990), or employment (Hofmann et al. 2017; Matysiak and Vignoli 2008). Further, the role of economic uncertainty in fertility has garnered substantial attention among fertility researchers (Vignoli et al. 2020a), particularly in the context of the global financial crisis (e.g. Goldstein et al. 2013; Schneider 2015, 2017) and the recent COVID-19 pandemic (e.g. Guetto et al. 2021).

Given the increase in levels of labor market uncertainty, analyses of differences in behavior across social classes may help to shed new light on contemporary fertility behavior. Following the Weberian distinction between “class” (*Klasse*) and “status” (*Stand*), one’s “status” is rooted in social recognition and esteem (prestige). Social class is defined through people’s labor market positions, which are at the root of their long-term life chances, their economic vulnerabilities, and their employment risks (Erikson

and Goldthorpe 1992; Goldthorpe 2007, 2010; Grusky and Sørensen 1998). Within this framework, social classes represent firmly theorized and validated occupation-based categories that reflect economic uncertainties.

The main goal of this paper is to elaborate on the concept of social class in the analysis of contemporary fertility behavior. Moreover, we provide empirical evidence on the relationship between social class and second birth rates in post-reunification Germany. Studying a single parity may be criticized as a "piecemeal approach" (Heckman and Walker 1990, p. 1416). However, there are benefits related to such a "piecemeal approach" when the interest is in the relation between social class and fertility. Assuming that people typically achieve a certain class position by the time they have their first child, the advantage of focusing on the second birth is that it enables use to examine how people's class mobility after the first birth affects their subsequent fertility behavior. The analysis relies on a proportional hazard model in which we use a piecewise constant specification for the underlying process. Some concerns have been raised that proportional hazard models may conflate timing and quantum effects (Bartus et al. 2013; Kreyenfeld 2002). Thus, we address these using a cure fraction model that disentangles the two components.

4.2 Theoretical Considerations and Prior Research

4.2.1 Prior Research on Uncertainty and Fertility

The Great Recession of 2008 has led to renewed scholarly interest in the role of economic uncertainty in fertility behavior. This broad strand of the literature relies on objective measures of macroeconomic and labor market conditions, such as unemployment and fixed-term employment, as proxies for uncertainty (Alderotti et al. 2021). Recognizing that uncertainty is high when economic conditions are dire, these studies generally agree that fertility rates are procyclical: i.e. they decrease during business cycle troughs, and increase during peaks (Adsera 2005, 2011; Cazzola et al. 2016; Currie et al. 2014; Goldstein et al. 2013; Gozgor et al. 2021; Karaman Örsal and Goldstein 2018; Sobotka et al. 2011). On the one hand, these short-run declines in period fertility may eventually translate into a "true" decline in completed cohort fertility, which implies a decrease in the total number of children that women of a certain cohort will have. On the other hand, these short-run declines in period fertility may be explained in part by postponement. For instance, Adsera (2011) found that first and second births occurred

later in European countries that experienced high and persistent unemployment in the 1980s. However, one challenge that most of these studies encounter is that isolating fertility postponement (tempo effect) from a permanent decline in fertility (quantum effect) can be difficult (Sobotka et al. 2011).

In addition to unemployment rates, an important strand of the literature has also considered other measures of economic uncertainty, including GDP (Luci-Greulich and Thévenon 2014; Matysiak et al. 2021b), consumer confidence (Comolli 2017; Schneider 2015), and press coverage of economic developments (Gozgor et al. 2021; Guetto et al. 2023b; Schneider 2015). These studies have also provided support for the claim that adverse economic conditions are negatively correlated with fertility. Particularly during the global financial crisis of 2007–08, which was characterised by sudden and steep increases in unemployment, firm closures, and, more broadly, negative reports on the state of the economy, (Schneider 2015) found that states in the USA that were hit hardest by the recession also had the largest decreases in general fertility rates. The fertility declines in these states at the height of the recession were attributed not just to the overall increase in uncertainty and economic hardship in these areas, but also to the increase in contraceptive use among selected population subgroups, particularly among unmarried women and women from lower-income backgrounds (Schneider 2015, 2017).

The findings mentioned above are complemented by an equally thick strand of the literature that has used subjective measures of economic uncertainty as determinants of fertility behavior. These indicators are usually constructed from items in individual and household surveys that ask respondents whether they are worried about their own finances or the general state of the economy. Studies that used subjective measures of economic uncertainty have found that its effect on fertility is more nuanced; that is, economic uncertainty seems to affect only select population subgroups. Kreyenfeld (2010) and Hofmann and Hohmeyer (2013) have both reported that there is little support for the claim that financial worries translate into first birth postponement. However, studies that have taken levels of education into account have shown that economic uncertainty accelerates the transition to the first birth among less educated women (Kreyenfeld 2010, 2015). Indeed, there is strong evidence of differences in fertility behavior in response to economic uncertainty by gender, population subgroup, and birth order.

In addition, a large body of research has examined how financial worries affect

not just fertility *behavior*, but fertility *intentions*. It has, for example, been shown that subjective economic uncertainty negatively affects birth intentions and that this relationship is more pronounced among men, given that men are often expected to take on a "primary provided role" (Busetta et al. 2019; Fahlén and Oláh 2018; Kuhnt et al. 2021). Finally, a relatively recent body of research has also pointed to the role of future narratives of uncertainty as a determinant of fertility intentions (Brauner-Otto and Geist 2018; Gatta et al. 2021; Vignoli et al. 2020a).

4.2.2 Social Class Position and Economic Uncertainty

Many of the above-mentioned studies have grappled with the question of how a valid operational definition of economic uncertainty can be found. Having children is a long-term and binding commitment. Thus, it is not only people's current economic conditions, but also their future employment prospects that influence their decisions about whether and, if so, when to have children. In this context, the concept of "social class", which is well established in research on social stratification and mobility, provides a potentially useful link. The theoretical backbone of contemporary class concepts is that in capitalist societies, individual life chances are essentially shaped by labor market, occupational, and employment conditions. Thus, social class is not interchangeable with education or income. Instead, it is a well-defined and "parsimonious indicator of the social positions of individuals" that helps us to "better understand fundamental forms of social relations and inequalities to which income is merely epiphenomenal" (Connelly et al. 2016, p. 3). Class researchers typically aggregate similar occupations into broader socio-economic class categories (Erikson et al. 1979b; Goldthorpe 2007; Oesch 2006; Wright 1985). Although class concepts differ with respect to their theoretical underpinnings and the basic mechanisms that are assumed to define and to distinguish classes, the prevalent class schemes, as developed by Erikson et al. (1979b), Goldthorpe (2007), Wright (1985), Oesch (2006), are aligned in terms of their basic occupational distinctions. For this study, the class schema proposed by Goldthorpe (2007) is particularly useful, as it suggests that occupational classes are inherently defined through employment relations. Accordingly, it is assumed that members of the same social classes have similar overall life chances, and are also exposed to similar degrees of economic vulnerability and uncertainty.

Goldthorpe (2007, p. 110-118) differentiated occupations based on whether the related *tasks are difficult to monitor*, and by whether the *human assets* required for

the occupations are specific. At the one extreme are occupations in which the tasks are difficult to monitor. People in these occupations usually have highly specific human assets (knowledge and expertise). At the other extreme are occupations in which the tasks are easy to supervise, and the quantity of work output is easy to measure. Furthermore, the human assets needed in these occupations are not specific. According to Goldthorpe (2007), the "nature of the tasks" and the "specificity of the human assets" determine the employment relationship. Based on this premise, he distinguishes nine categories, as described below (for details, see also the "Data, variables, and analytical" section).¹

Unskilled and semi-skilled manual workers [VIIa] and *lower-grade routine nonmanual employees* [IIIb] are often employed under short-term contracts. This implies that the jobs these categories take generally do not involve a long-term commitment from either the employer or the employee (Erikson and Goldthorpe 1992, p. 41). Thus, the workers in these classes are subject to considerable economic uncertainty. The typical occupations in these classes include waiter, cleaner, shop assistant, housekeeper, taxi driver, and truck or van driver. In contrast to them, occupations in the *upper and lower service classes* [I, II] are mostly embedded in larger organizations, and "involve a longer term and generally more diffuse exchange" (ibid.: p. 103). Most importantly, the rewards associated with these occupations typically include "prospective elements", such as employment security and "well-defined career opportunities" (ibid.: p. 103). Although the service classes have also been affected by the rise of fixed-term contracts, members of these classes generally enjoy greater employment stability than unskilled and semi-skilled laborers or routine nonmanual employees. These occupations include lawyer, scientist, engineer, higher-grade manager, in the German case also secondary school teacher (upper service class), as well as nurse, kindergarten teacher, technician, and lower-grade manager (lower service class). The *skilled manual workers and supervisors* [V, VI] as well as *the higher*

¹Goldthorpe (2007, p. 104) distinguishes the following nine classes:

- I professional and managers, higher grade;
- II professional and managers, lower grade;
- IIIa routine nonmanual employees, higher grade;
- IIIb routine nonmanual employees, lower grade;
- IVabc small proprietors, self-employed;
- V technicians, lower-grade supervisors of manual workers;
- VI skilled manual workers;
- VIIa semi- and unskilled manual workers (other than in agriculture); and
- VIIb agricultural workers.

routine nonmanual employees [IIIa] hold an intermediate position. These occupations involve mixed forms of employment relationships. The type of work done is either more difficult to monitor than un-/semi-skilled work, or it requires medium levels of specific human assets/human capital. The typical occupations in this category include machine operator, plumber, and electrician. Small employers/self-employed, farmers, and agricultural workers occupy separate classes [IVabc, VIIb]. These categories will not be considered in this investigation as they are small and heterogeneous. Although class concepts do not necessarily entail a hierarchical ordering (Connelly et al. 2016), social classes can be ranked by their degree of employment risk, with levels of economic vulnerability and uncertainty being highest among the un- and semi-skilled workers and lower-grade routine nonmanual employees, and lowest among the upper service class.

4.2.3 Prior Research on Occupational Class and Fertility

While social class is a well-established concept in research on social inequality, only a relatively thin strand of recent literature has focused on the relationship between social class and fertility. Moreover, they have often adopted different strategies for classifying occupations. Some of the early studies, which were published when a large fraction of the population was still working in the agricultural sector, were particularly concerned with the elevated fertility of people working as farmers or farm laborers. An example is the study by Dinkel (1952), who argued that people in different occupations have different “ways of life” in terms of the practices and values that affect fertility. He showed that in the early twentieth century, farm owners and laborers had fertility rates that were 40–72% higher than those of professionals, depending on the region of residence in the USA. He attributed this gap in part to the labor needs of farming households (Dinkel 1952; Maloney et al. 2014). Similar patterns have also been observed in Sweden in the mid-1900s, where farmers were shown to have the highest fertility rates among all occupational groups (Dribe and Scalone 2014). More recent work has challenged these findings. For example, Köppen et al. (2017) found a drastic increase in childlessness among male farmers in France starting with the 1960s cohorts.

More recent research has also emphasized the importance of incorporating a gender perspective into explanations of relationships between social class and fertility (Szreter 2015). It has been reported that since the 1990s in Sweden, women’s occupational class has had a U-shaped relationship with the transition to parenthood, with women in low-skilled and high-skilled occupations having higher birth risks than women in medium-

skilled occupations (Dribe and Smith 2021). Research in Austria has found that women whose educational levels typically lead them to have lower-class occupations are less likely to remain childless than women whose educational levels lead them to have higher-class occupations (Neyer et al. 2017). Begall and Mills (2013) used data from the Netherlands to study the birth behaviour of women of the 1940–1985 cohort by groups of occupations and found that women in teaching-related occupations transitioned relatively quickly to first birth. Their results also indicated that women in communicative jobs (healthcare, teaching) transitioned relatively rapidly to higher-order fertility, while women in technology-related occupations had comparatively low higher-order birth risks.²

The studies that come closest to using the established sociological concepts of social class and analyzing its relation to fertility generally find a positive relationship between being in a higher social class with second-order births (Baizan 2020, 2021; Ekert-Jaffé et al. 2002). Using event history and simultaneous equation models on longitudinal data reveal elevated risks of women in the higher professional class in the transition to second birth in Spain (Baizan 2020), as well as Austria, France, Norway, and the UK (Baizan 2021). Gender also matters in this nexus—(Ekert-Jaffé et al. 2002) find no strong variation in women’s fertility depending on their social class. However, they observed that the second birth rates of women with a spouse in a higher managerial position were well above average. Building on this strand of the literature, we likewise incorporate in our analysis the role of gender and economic uncertainty.

4.2.4 Hypotheses

As argued Goldthorpe (2007), social classes are based on people’s occupations and employment positions, which provide them with differing levels of socioeconomic resources, including varying degrees of employment security. Employment security is rooted in the nature of the job-related tasks the employee is expected to perform and in the kind of job contract deemed necessary to incentivize the employee to perform the tasks. Accordingly, class positions differ with respect to employee-employer commitment levels and trust relationships, and in terms of the long-term character of employment contracts. Assuming that fertility choices are long-term, binding

²In the Latin American context, scholars have also examined the class and fertility nexus (Castro Torres 2021). However, instead of relying on occupation-based class concepts, they measured social class using a large battery of variables, including electricity and water supply.

biographical decisions that require some degree of economic certainty, it can also be assumed that fertility behavior differs by social class. Given the more advantaged positions of the upper service class, individuals in this class should have the highest second birth rates, while the semi-/unskilled workers and the lower-grade routine nonmanual employees should have the lowest second birth rates (*hypothesis 1*).

Compared to their income and earnings, peoples' class positions are rather stable traits that mirror their long-term employment and lifetime chances. However, the childbearing years coincide with a period in people's lives in which they are typically seeking to advance in their professional careers or are participating in education or vocational training. In Germany, as in most other European countries, the age at first birth has risen to about age 30 for women and to about age 32 for men. Although the scholarly literature often assumes that class positions are rather fixed beyond age 30, upward mobility—and, to a lesser extent, downward mobility—may occur beyond that age. As having a higher-class position is linked to greater economic security, we assume that upward mobility will lead to higher second birth rates (*hypothesis 2*).

Traditional concepts have understood social class as a household concept, where the social class of an individual in a household is defined over the social class of the (male) prime earner (Goldthorpe 1983). Feminist scholars have challenged that view, calling for a gender and individual perspective on social class (see, for example Baxter 1994; Bonney 2007, p. 146). We define social class as an individual trait defined over one's own occupation. Still, labour market options are strongly gendered in most societies, also in the case of reunited Germany, which is the focus of this investigation. Important family policy reforms were enacted in recent years in Germany, most notably the expansion of childcare in 2005 and the reform of parental leave benefits in 2007. Scholars have argued that these reforms have been consequential, as they represent a sharp departure from Germany's previously well-established path of providing policy support for a conservative family model centered on the male breadwinner (Fleckenstein 2011). While the full-time employment rates of mothers have increased in recent years, employment patterns after the first birth are still strongly gendered, particularly in West Germany. Against this background, we assume that social class is a stronger predictor of men's than of women's fertility transitions (*hypothesis 3*).

Finally, people's class positions reflect their long-term employment chances and their levels of economic security and vulnerability. Among the benefits of the German Socio-Economic Panel (GSOEP) dataset, which we will use in our investigation, is

that it includes not only measures of social class but also items that estimate levels of economic uncertainty and vulnerability, such as the subjective feeling of having financial worries. This information allows us to study whether and, if so, how this measure correlates with the respondents' social class positions. It also enables us to explore whether the effect of social class is robust to the inclusion of more direct measures of uncertainty. Generally, we expect to find that having financial worries may explain some of the class differences. Thus, we assume that the effect of social class becomes weaker after controlling for other measures of uncertainty (*hypothesis 4*).

Beyond these four guiding research hypotheses, the analyses will take into account the possibility that social class has a distinct influence *on the timing and the quantum* of second birth fertility. Because they tend to be older when they have their first child, members of the service class are likely to face a “time squeeze” that leads them to progress more rapidly to the second birth than, for example, members of the unskilled and semi-skilled and the routine nonmanual classes, who often have their first child at a younger age. We will use a cure fraction model to check whether a more fine-grained modeling approach that differentiates between timing and quantum effects generates the same results as standard event history models.

4.3 Data, Variables, and Analytical Strategy

4.3.1 Data and Analytical Sample

Data for this investigation come from the German Socio-Economic Panel (GSOEP) release 37 (Panel 2022). The GSOEP is a yearly household panel that was launched in 1984. The original sample includes West German respondents and an oversample of migrants from the former labour recruitment countries. Since its inception, various subsamples have been added to this dataset, most prominently an East German subsample in 1990. For this investigation, we use data from the years 1990 to 2020.³ Thus, the investigation covers post-reunification Germany. We limited the sample to respondents who had a first child from 1990 onwards and provided valid information on each individual's social class in the year of the first birth. We omitted respondents who were self-employed or farmers when they had their first child, as this group was

³The COVID-19 crisis may have affected fertility in Germany. However, most interviews are conducted at the beginning of the year. As 2020 is the last year when interviews were conducted, only a few 2020 births are included in the analysis.

rather small and heterogeneous. Finally, we censored the cases 12 years after the first birth and restricted the sample to episodes in which the respondents were aged 18–55. The final analytical sample includes 2282 men who fathered 1161 children and 2819 women who contributed 1371 births to the study population.

4.3.2 Variables

Dependent Variable

The dependent variable is the transition to the second child, with the process time being measured in months from the birth of the first child to the start of the second pregnancy (i.e. the date of childbirth backdated by nine months). In some cases, the month of childbirth was missing from the data. We imputed the missing information using a random number generator. While we had information on the month of childbirth, information on the time-varying covariates (such as financial worries and subsequent class positions) was available only at the time of the interview. We assumed in these cases that the respondents' characteristics were fixed until the next interview. Figure C.1 in the appendix plots the survival curves for the second child by social class and gender. The figure suggests that when the first child was age 10, the probability of having a second child was around 63% which is a bit lower than recent official estimates (Bundesamt 2019).⁴

Social Class

The key variable of interest is the *class position in the year of first childbirth*, which we treat as a time-constant covariate in the investigation. We have chosen to focus on the class position at first birth in part because it allows us to examine how downward or upward mobility relates to fertility behavior (see below). We have operationalized the class position using the Erikson Goldthorpe Portocarero (EGP) class schema which has been adopted in the GSOEP (Erikson et al. 1979b). We have removed from the sample the self-employed and agricultural labourers because they were a very heterogeneous group, and because they comprised only a very small fraction of the population. We also grouped routine service workers and unskilled manual workers into a single category because the shares of men employed in routine service jobs and the shares of women

⁴The parity progression ratio that was calculated based on German vital statistics for women ages 45–49 in the year 2018 was about 68% (Bundesamt 2019).

employed in unskilled manual jobs were extremely low. If a person was not employed in the year when s/he had a child, we used the person's class position in the previous year. However, some respondents, particularly women, were not working both in the year when they had a child and in the previous year. For these cases, we built a separate category that indicates that the person was not employed (including a small fraction of individuals who were in education). In total, we distinguished the following five class positions:

- Upper service class [I],
- Lower service class [II],
- Skilled (manual and higher-grade routine nonmanual) classes [IIIb, V, VI],
- Semi- and unskilled (manual and lower-grade routine nonmanual) classes [IIIa, VIIa],
- Not employed (including individuals in education)

Among men, the largest group is composed of persons who were in the skilled (manual and higher routine nonmanual) class, while the lower service class is the most common class for women at first birth (see Table C.1 in Appendix). A larger fraction of men than women were in the upper service class at the time of their first birth. For both women and men, typical occupations in the upper service class are medical doctors or public service administrative professionals. There is a higher fraction of computer systems designers and engineers among men, while architects and sales, marketing, and department managers are more prevalent among women in the upper service class. A striking difference between women and men furthermore is that more women than men were not employed in the year when their first child was born (or in the prior year).

Partner's Social Class

We also account for the partner's social class at first birth in some parts of the investigation. Unfortunately, there is a large fraction of the respondents for whom we do not have any information on the partner's social class (see Table C.1). For those respondents for which information is available, we observe strong class homogamy on the couple level (see Tables C.3 and C.4). For example, 48% of the women who were employed in the upper service class around the time of first birth had partners who were also in the upper service class.

Social Mobility

To capture the effects of *social mobility*, we have generated a time-varying variable that combines the class position in the year of first childbirth and the class position in the years thereafter. We distinguish between individuals who remained in the position they had in the year of their first birth and individuals who experienced upward or downward mobility (compared to the class position in the year of first birth). Here, we assume the following hierarchy of class positions: upper service class > lower service class > skilled manual/higher-grade routine nonmanual classes > unskilled and semi-skilled manual/lower-grade routine nonmanual classes. A separate category includes the respondents who were not employed either at the first childbirth or in the period that followed. Thus, we do not consider the move from nonemployment at the first birth to participation in the labor market as upward mobility. Tables 4.1 and 4.2 report the sample statistics for social mobility by class and gender. Within the given class framework (disregarding nonemployment), it was not possible for members of the upper class to move up, or for members of the semi-skilled and unskilled classes to move down. Thus, upward mobility was possible for the lower and medium social classes only, while the semi-skilled and unskilled classes did not contribute any exposure time to the downward moves. These floor and ceiling effects need to be considered when interpreting the model results (see also Breen and Müller 2020). The table provides some relevant insights into the class-specific mobility patterns. While upward moves were rare for women after the first birth, a significant fraction of the men—particularly those who were previously in unskilled or semi-skilled occupations—experienced upward mobility after the first birth.

Economic Uncertainty

The GSOEP includes various measures of economic uncertainty, which allows us to examine how these measures are correlated with class positions. In our analysis, we use the subjective feeling of having *financial worries*. The respondents' financial worries were operationalized with a survey question that asked them whether they were very worried, somewhat worried, or not worried about their personal financial situation. In line with our theoretical expectations, we observe that the respondents' financial worries varied considerably depending on their class position. For example, individuals who were in semi-skilled or unskilled positions were three times more likely than members of the upper service class to report being very worried about their

Table 4.1: Financial worries by social class, distribution by person-years, men, column %

	Men's social class in the year of the first birth				
	Upper service	Lower service	Skilled	Semi-/unskilled	Not employed
<i>Social mobility</i>					
Stable	86	73	71	66	-
Upward	-	11	8	19	-
Downward	11	11	9	-	-
Not employed	3	5	11	14	100
<i>Financial worries</i>					
Very worried	9	13	24	27	34
Somewhat worried	50	51	54	55	49
Not worried	41	36	22	18	17
<i>Age at first birth</i>					
Mean	34.1	33.5	29.9	29.3	27.8
N (person-months)	17,687	21,142	40,722	22,389	6,301

Source: SOEP, v37, 1990-2020, unweighted estimates

Table 4.2: Financial worries by social class, distribution by person-years, women, column %

	Women's social class in the year of the first birth				
	Upper service	Lower service	Skilled	Semi-/unskilled	Not employed
<i>Social mobility</i>					
Stable	52	49	42	40	-
Upward	-	3	6	10	-
Downward	13	10	6	-	-
Not employed	35	38	45	50	100
<i>Financial worries</i>					
Very worried	11	16	22	30	29
Somewhat worried	54	57	60	55	52
Not worried	35	27	19	15	19
<i>Age at first birth</i>					
Mean	32.1	30.3	28.5	27.0	25.0
N (person-months)	11,292	34,791	34,451	33,721	20,093

Note: The sample includes persons at risk of a second birth. For the distribution of financial worries, persons with missing information for that particular variable were excluded

Source: SOEP, v37, 1990-2020, unweighted estimates

economic situation (Table 4.1 and 4.2). Moreover, individuals who were not working at first birth also indicated that they were very concerned about their economic situation. The patterns were similar for both women and men, but the association was slightly stronger for the women than for the men (Spearman's rank correlation coefficient was -0.23 for women and -0.17 for men).

Socio-Demographic Covariates

The regression analysis also controls for the standard socio-demographic variables. (The table with the complete summary statistics is included in the appendix; see Tables C.1 and C.2.) We account for *the duration since the previous birth* (baseline hazard). Region is included in the models by distinguishing between West and East Germany (including West Berlin). We also control for migration background and differentiate between natives and individuals with a *migration background*. Education was distinguished between low (lower than vocational degree), medium (vocational degree), and higher education (university degree). *Age at first childbearing* is another important predictor for subsequent births. This variable may be strongly correlated with social status, and it may have a distinct influence on the timing and the quantum of second-birth fertility. Table 4.1 and 4.2 provide support for the assumption that there was a strong age gradient in first childbearing by social class. The average age at first childbearing was much higher among the respondents in the upper service class than among the respondents in the manual and routine occupations.

4.3.3 Analytical Strategy

Main Analysis

Our analytical strategy consists of two steps. In the first step, we use a piecewise constant model to estimate the transition rates to the second child. The piecewise constant model is particularly suitable for studying vital events, such as birth dynamics (Hoem 1987; Hoem and Hoem 1989). Like the Cox model, it belongs to the large group of proportional hazard models. However, unlike the Cox model, it provides parameter estimates for the baseline hazard, which allows the researcher to better grasp the underlying process. The baseline hazard in our analysis is the duration since the last birth. The baseline $h_0(t)$ is partitioned into different pieces, while the hazard is assumed to be constant within the respective segments. In our model, time t is

partitioned into four segments: i.e. into 0–23, 24–47, 48–71, and 72–143 months after the first childbirth. The hazard at time t , given a set of X covariates, is defined as:

$$h(t|X) = h_0(t) \times \exp(\beta x) \quad (4.1)$$

While the baseline hazard is defined as follows:

$$h_0(t) = \begin{cases} h_{1,t}(0, \tau_1), \\ h_{2,t}(\tau_1, \tau_2), \\ h_{3,t}(\tau_2, \tau_3), \\ h_{4,t}(\tau_3, \tau_4), \end{cases} \quad (4.2)$$

The main covariates in the first model (M1 for men and W1 for women) are the variables for social class position and class mobility. In a second model (M2 for men and W2 for women), we also account for the subjective feeling of uncertainty. In a final model (M3 for men and W3 for women), we furthermore account for the partner’s social class.

Robustness Checks

The descriptive statistics (see Tables 4.1 and 4.2) show that the age at first birth varied systematically by social class. This finding may have implications for the model’s assumptions. Individuals who were older when they had their first child may have faced a “time squeeze” that led them to progress rapidly to having a second child. Thus, the elevated birth rates may indicate that individuals who were older when they had their first child were less likely to have a second child, but spaced their first and second births closer together. As the age at first childbearing and social class are closely related, the failure to properly model the influence of the age at first birth on second birth rates may lead to biased results.

We employ a cure fraction model to separate the timing and the quantum effects. There are two broad families of cure fraction models. First, the mixture cure model (Berkson and Gage 1952) relies on a survival function written as a mixture of two components: one corresponding to the proportion of “immune” subjects in the population (those who never have a second child) and a second one corresponding to the survival function of the population who will experience the event (see for example, Beaujouan and Solaz 2013, in a fertility context). Second, the promotion time model,

also named the bounded cumulative hazard model (Tsodikov and Yakovlev 1996), which has recently been employed in fertility research as well (Bremhorst et al. 2016, e.g.). This approach explicitly acknowledges that the population survival function $S_p(t)$ converges (when $t > T$) to a nonzero value π corresponding to the "immune fraction. In our context, T would denote the minimum number of months after the first birth after which it is reasonable to conclude that subjects with one child will not have a second, π denoting the expected proportion of subjects in this situation. It implies that the cumulative hazard function $H_p(t)$ is bounded by $\theta = -\log(\pi)$ and can be written as $H_p(t) = \theta F(t)$ where $F(t)$ is a cumulative distribution function such that $F(0) = 0$ and $F(t) = 1.0$ when $t > T$. The associated density function $f(t)$ can be viewed as a normalized form of the population hazard function $H_p(t) = \theta \times f(t)$ governing the dynamics of events. A flexible spline-based form of $f(t)$ will be considered. Covariates x can enter the specification of θ and $F(t)$ using $\theta(x) = \exp(\beta_0 + \beta'x)$ and $S(t|x) = 1 - F(t|x) = S_0(t)^{\exp(\gamma'x)}$.⁵ The promotion time model has been formalized for models with time-constant covariates, but the inclusion of time-varying covariates is still an emerging topic in the statistical literature (see Lambert and Bremhorst 2020; Lambert and Kreyenfeld 2023, for recent proposals). For this reason, we have fixed the values found for education and region as the first birth. As this is not meaningful for social mobility, we omit this variable from this part of the investigation. We also omit financial worries. One could fix this variable at first birth; but the meaning of this variable as a fixed and time-varying covariate would differ.⁶

4.4 Results

4.4.1 Social Class, Social Mobility, and Second Birth Fertility

Table 4.3 reports the results from a set of event history models that estimated the transition rate to the second child. All models report a consistent pattern with respect

⁵Note that identification issues can arise if the follow-up is not sufficiently long to observe the "plateau" in the population survival function (see Bremhorst et al. 2016; Lambert and Bremhorst 2019).

⁶We have conducted further robustness checks and estimated models excluded age at first childbearing and education (see Figure C.2 in the Appendix). The class gradient is less pronounced for the male sample when age at first birth is removed from the model. A model that does not control for education leads to a less pronounced social class gradient for both genders. Note that it is currently debated whether education should be included in models that control for social class (see, for example, Grätz 2022)

to the control variables: i.e. the second birth rates are highest two to three years after the first childbirth. Furthermore, there is a strong negative correlation between the age at first childbirth and the progression rates to the next childbirth. Moreover, the second birth rates are roughly 40% lower in East than in West Germany. This finding is very much in line with prior research on the East-West differences in higher-order childbearing behavior (Becker et al. 2010). We find no relevant differences between native and migrant populations. This may be surprising, as it is often assumed that the migrant population has higher fertility than the native population. It should be noted that this study focuses on second-order births, for which native migrant differences tend to be less pronounced. Furthermore, apart from migrants of Turkish origin, many of the more recent migrants in Germany come from Central and Eastern European countries that are characterized by low second birth rates.

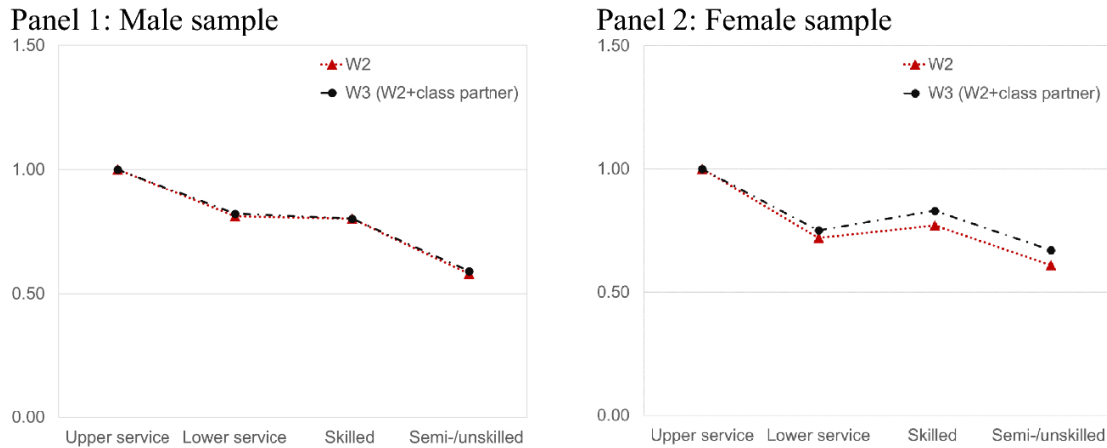
The analysis shows that the men's social class is strongly related to their second birth behavior (Model M1): i.e. the lower the social class, the lower the second birth rate. The groups that stand out are the semi-skilled and unskilled classes, as their second birth rate is 42% lower than that of the reference group (upper service class positions). The pattern for women is similar, but the differences are attenuated (Model W1). Class mobility also plays out differently for women and men. Among men, upward mobility is associated with an increase of roughly 25% in the second birth rate, while downward mobility and nonemployment are unrelated to the second birth rate. The parameter for upward mobility for women is in the same direction and of similar magnitude as for men, but it is only weakly significantly different from the reference category (stable class position). This weak significance may not come as a surprise, given the small fraction of women who experienced social upward mobility after their first birth (see Table C.2 in the appendix).

Models M2 and W2 display the results from the models that include the additional measures of economic uncertainty (subjective financial worries). While the men's financial worries do not seem to influence the second birth rates (Model M2), the women's financial worries are associated with a postponement of second childbearing (Model W2). The second birth rates of women who reported being very worried are 20% lower than those of women who reported being not worried. Women who said they are somewhat worried have a birth rate that is 17% lower than those without worries. The effect size seems strong, and the inclusion of this variable also increased the model fit (log-likelihood ratio test was conducted, $p < 0,05$). However, it does not greatly

affect the class pattern. Thus, among women, there seems to be an independent effect of subjective worries that is not captured by their own class position.⁷

4.4.2 Partner’s Social Class and Second Birth Fertility

Figure 4.1: Piecewise constant event history model. Relative second birth risks (hazard ratios)



Note: The analysis was conducted on the sample with respondents with valid information on partner’s class position ($n = 81,669$ person-months for male sample; $n = 83,654$ person-months for female sample). Further variables in the model are the age of the first child, age at first birth, education, region, migration status, own social class, social mobility, and financial worries. The hazard ratio for “not employed” is not displayed in the figures due to the small sample size in the male sample.

Source: SOEP, v37, 1990–2020. Own unweighted estimates

Figure 4.1 displays the results from the models which additionally controls also for the partner’s class position. The figures include the hazard ratios from the model that includes the partner’s class (M3 and W3) as well as results from a model that does not include it. It contains furthermore the same covariates as in the prior analysis (M2 and W2). Note, however, that this part of the analysis was restricted to respondents

⁷For illustrative purposes, we have also estimated a median duration time to second birth by social class from Model M1 and W1 (Table 4.2). Such an approach requires assumptions on the covariate constellations. We estimated values for the following constellation: A German citizen in West Germany with medium education who had the first child between ages 24 and 28, and who did not experience upward mobility. For the male model (M1), the estimated median duration to second birth for the upper service class is 3 years, while it is roughly 4.5 years for the semi-/unskilled class. For the female model (W1), the estimated median duration to second birth for the upper service class is about 3.2 years, while it is about 5 years for the semi-/unskilled class.

with valid information on partner's class characteristics so that the parameters slightly deviate from the previous investigation. As can be depicted from the figure, patterns for the male sample remain unchanged regardless of whether the partner's social class is controlled or not. However, patterns are attenuated for the female sample, suggesting that men's social class has a more positive impact on second-birth transitions than women's in the German context. Still, the overall pattern remains unchanged: The higher the social class of both women and men, the higher the second birth rate.

4.4.3 Timing and Quantum Effects

Table 4.4 reports the results from the cure fraction models. The model results for the male sample show that the previously reported "class effects" are mainly quantum effects. Thus, we can ascertain that there is a positive class and fertility nexus. The members of the upper service class are the most likely to progress to the second birth, and the semi/unskilled workers and lower-grade routine nonmanual employees are the least likely to have a second child. We also find that the age at first fatherhood has a distinct influence on the timing and the quantum of male second-birth fertility. While a late age at first fatherhood leads to the first and second child being more closely spaced, it also lowers the quantum of fertility. We find that a migration background increases quantum, but the parameter is only significant in the male sample. Further, the reduced hazard rate that we found for East Germany in the previous investigation seems to be related to both timing and quantum effects. Further, men's higher levels of education seem to affect second birth quantum, but not timing.

In many respects, the results for the female sample concur with the results for the male sample. Most of the class differences can be attributed to quantum effects. A pronounced pattern is found for women who were not employed in the year of the first childbirth: i.e. they are rather unlikely to have a second child, but if they have a second child, they often have it at short durations after the first birth. The age at first childbearing has the same effect in the female sample as in the male sample. An early age at childbearing increases the quantum, but it also increases the birth interval. Late childbearing has the opposite effect, as it lowers the quantum, but it shortens the birth interval.

4.5 Conclusion

While classical demography had a strong interest in the relationship between social class and fertility, contemporary fertility research rarely uses the class concept to investigate birth behaviour. Instead, scholars mostly focus on income, education, and employment when examining how labor market conditions are related to fertility behavior. However, the global financial crisis, the COVID-19 pandemic, and, more recently, the recession that is expected to follow the Russian war of aggression in Ukraine have led to increasing scholarly interest in the uncertainty and fertility nexus. Nonetheless, there is still considerable ambivalence about how to properly operationalize economic uncertainty and long-term employment chances. In this context, it is conspicuous that most demographers have failed to take into consideration the large body of sociological work on the relationship between social class, economic vulnerability, and life chances. We argued in this paper that social class is a well-theorized concept with firmly validated categories that have been effectively employed in sociological labour market research and that can also prove useful in demographic investigations.

The empirical part of this investigation focused on second-birth fertility in post reunification Germany (1990–2020). We chose to look at second childbearing in order to explore how class mobility after the first childbirth affected birth behavior. The results of the descriptive investigation indicated that women were less likely than men to experience upward mobility. Furthermore, while we found that moving up the social ladder increased men’s second-birth risks significantly, we only observed a statistically weak association between women’s mobility and their second-birth fertility. We also found that the association between social class and second-birth fertility was stronger in the male than in the female sample. Nevertheless, the overall pattern was similar for both genders, with members of the upper service classes having the highest birth rates and members of the unskilled manual/ lower routine nonmanual classes having the lowest birth rates. We also examined the question of whether subjective feelings of uncertainty explained the differences by class. Our findings indicated that while having financial worries was associated with lower second birth rates, particularly among the female sample, the inclusion of this variable did not change the class patterns. An important methodological question we considered was whether the model results would be robust if the timing and quantum effects were differentiated. To this end, we employed cure fraction models. The cure fraction model showed that the age at first childbearing had a very different impact on the timing and the quantum of fertility:

i.e. a later age at childbirth reduced the quantum, but it led to a closer spacing of the first and the second child. The model also showed that the class differences were mostly quantum effects.

While our investigation generated novel results on the class–fertility nexus, there are important limitations to this investigation that should be mentioned. *First*, we focused on second births. As the transition to the first birth usually coincides with the phase of life when people are getting established in the labor market, first-birth analyses would have required additional considerations. Moreover, as higher-order births are rare in Germany, we would not have sufficient case numbers to study third- or higher-order births. Thus, while our focus on second births may be justified, the analysis of a single transition may still be characterized as a “piecemeal approach” (Heckman and Walker 1990, p. 1416). We cannot rule out the possibility that the patterns for other birth parties are different from the patterns we found for second births. Furthermore, there may be selection into the study population based on social class characteristics. Earlier studies have revealed that highly educated women in Germany are more likely to remain childless than less educated. As social class and education are correlated, one may conclude that women who belong to the upper service class and are at risk of second birth are a selective population. It may be that it is their particular characteristics that have selected them into the pool of mothers also drive their higher progression to the next child (see, for example, Bartus et al. 2013; Kreyenfeld 2002). *Second*, we assumed that social class is a solid and firmly validated indicator of economic uncertainty, economic vulnerability, and long-term life chances. The GSOEP offers various additional variables that indicate different facets of economic uncertainty and economic standing (e.g., labor market earnings, term-limited working contracts, and worries about global economic development). Among the many variables that this dataset offers, we picked having financial insecurity. We included this variable in our model, but it did not ultimately explain much of the class differences. The “stepwise procedure” we used may be criticized for failing to sufficiently account for other measures of uncertainty that may affect the relationship between social class and fertility behavior. Possibly, more elaborated mediation analysis that accounts for the complex interplay of social class and various dimensions of uncertainty may be a way forward here (Kuha et al. 2021).

Table 4.3: Piecewise constant event history model. Relative second birth risks (hazard ratios)

	Men		Women	
	M1	M2	W1	W2
<i>Age of first child</i>				
Age first child 0-1	Ref.	Ref.	Ref.	Ref.
Age first child 2-3	1.66***	1.65***	1.58***	1.59***
Age first child 4-5	0.82**	0.81**	0.85	0.86
Age first child 6-11	0.32***	0.32***	0.34***	0.34***
<i>Age at first birth</i>				
Age 18-23	1.04	1.04	1.13	1.15
Age 24-28	1.15*	1.15*	1.04	1.05
Age 29-32	Ref.	Ref.	Ref.	Ref.
Age 33-55	0.71***	0.71***	0.48***	0.48***
<i>Region</i>				
West Germany	Ref.	Ref.	Ref.	Ref.
East Germany	0.60***	0.60***	0.66***	0.68***
<i>Migration Background</i>				
Native	Ref.	Ref.	Ref.	Ref.
Migration Background	1.02	1.03	1.09	1.10
<i>Social class at first birth</i>				
Upper service	Ref	Ref	Ref	Ref
Lower service	0.79**	0.79***	0.75***	0.76***
Skilled	0.78**	0.78***	0.81*	0.83*
Semi-/unskilled	0.58***	0.58***	0.62***	0.64***
Not employed	0.86	0.86	0.64***	0.64***
<i>Education</i>				
Low (no degree)	Ref.	Ref.	Ref.	Ref.
Medium (vocation)	1.09	1.08	1.22**	1.22**
High (university)	1.33**	1.34**	1.39***	1.37***
<i>Social mobility</i>				
Upward	1.26**	1.27**	1.32*	1.31
Stable	Ref.	Ref.	Ref.	Ref.
Downward	0.88	0.88	1.00	1.01
Not employed	0.89	0.90	1.17**	1.18**
<i>Financial worries</i>				
Very worried		1.02		0.80***
Somewhat worried		1.15**		0.87**
Not worried		Ref.		Ref.
<i>Sample size</i>				
Person-months	108,241	108,241	132,348	132,348
Events	1161	1161	1371	1371

Note: Further variables in the model for "other" social mobility and "missing" for financial worries and education. * p<0.05; ** p<0.01; *** p<0.001

Source: SOEP, v37, 1990-2020, unweighted estimates

Table 4.4: Cure fraction model. Relative second birth risks (hazard ratios)

	Men		Women	
	Quantum	Timing	Quantum	Timing
<i>Age at first birth</i>				
Age 18-23	1.16	0.77	1.49***	0.60***
Age 24-28	1.24***	0.84*	1.19**	0.74***
Age 29-32	Ref.	Ref.	Ref.	Ref.
Age 33-55	0.61***	1.45***	0.44***	1.45***
<i>Region</i>				
West Germany	Ref.	Ref.	Ref.	Ref.
East Germany	0.62***	0.85	0.73***	0.77***
<i>Migration Background</i>				
Native	Ref.	Ref.	Ref.	Ref.
Migration Background	1.13*	0.82**	1.08	1.02
<i>Education</i>				
Low	Ref.	Ref.	Ref.	Ref.
Medium	1.10	1.18*	1.28***	0.98
High	1.38**	1.10	1.39***	1.06
<i>Social class at first birth</i>				
Upper service	Ref.	Ref.	Ref.	Ref.
Lower service	0.85*	0.86	0.72***	1.13
Skilled	0.90	0.78**	0.77**	1.27**
Semi-/unskilled	0.64***	0.90	0.61***	1.16
Not employed	0.78	1.12	0.61***	1.42**
<i>Log-Likelihood</i>				
Null model (no covariates)	-6268		-7446	
Final model	-6190		-7345	
<i>Sample size</i>				
Person-months	127,701		158,161	
Events	1171		1381	

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Person-months and events are slightly different from Table 4.2 as the analysis was not censored after 12 years.

Source: SOEP, v37, 1990-2020, unweighted estimates

Chapter 5

Ascribed Fertility Intentions Under Future Caregiving Responsibilities and Employment Uncertainties

co-authored with Michaela Kreyenfeld, Julie O’Sullivan, Jan Paul Heisig, Enrique Alonso-Perez, and Paul Gellert

ABSTRACT

The impact of uncertainty on fertility has been widely explored in the literature, with a strong focus on past and present experiences of both objective and subjective employment uncertainty. There has been limited research on the effects of caregiving responsibilities toward a sick parent on fertility decisions. This paper explores how various employment and caregiving scenarios impact ascribed fertility intentions (AFI). Using a vignette experiment from the nationally representative German Socio-Economic Panel Innovation Sample (SOEP-IS) (n=1,750), respondents were randomly exposed to hypothetical scenarios of couples facing different levels of future caregiving responsibilities and employment uncertainties who subsequently assessed the likelihood that the couple would have a child within the next three years. Results show that having no caregiving responsibilities and no employment uncertainty for both partners in the foreseeable future increases fertility intentions by 2.8 and 1.9 units, respectively, on a 0-10 scale, relative to when future caregiving responsibilities and employment

uncertainties are high. Further testing whether these effects vary by respondents' characteristics, results lend support to a gendered difference in the effect of caregiving responsibilities, whereas neither objective nor subjective measures of respondents' employment markers moderate the effect of future employment uncertainty. This paper highlights individual perceptions of what scenarios and conditions enable the transition to parenthood and, more broadly, how caregiving responsibilities and employment instability translate to concerns about family formation in aging societies.

Keywords: *caregiving responsibilities, employment uncertainty, ascribed fertility intentions, factorial survey experiment, Germany*

5.1 Introduction

Families in contemporary societies around the world are confronted with multiple uncertainties and challenges including economic and employment instability, caregiving responsibilities, and growing individual concerns about various issues. The consequences of these uncertainties for social and economic phenomena such as family formation and dynamics, wages and labor supply, or political behavior are a pervasive interest among social scientists, with prior work covering uncertainties in multiple domains and on various levels—micro and macro, subjective and objective, and short-term as well as long-term (Currie et al. 2014; Gatta et al. 2021; Goldstein et al. 2013; Schneider 2015; van Wijk et al. 2022).

Regarding family formation and dynamics, there are strong theoretical arguments suggesting that uncertainties and anticipated responsibilities in the work and family domains can lead to fertility postponement and reduced total fertility. However, empirical evidence is not fully conclusive, as results vary not only by how uncertainty is operationalized (Alderotti et al. 2021; Buh 2023; Gatta et al. 2021) but also by whether first birth timing, higher-order births, or fertility intentions are examined (Kreyenfeld et al. 2023a). Results also differ widely across country contexts. Some studies show that the association between employment uncertainty and fertility is positive in the US (Schwandt and von Wachter 2020), ambiguous in Germany (Hofmann and Hohmeyer 2016), and negative in South Korea (Choi et al. 2020). Finally, the literature has also pointed to heterogeneity across population subgroups, showing that the direction and strength of the uncertainty-fertility nexus vary, for example, according to gender and educational attainment (Alderotti et al. 2021; Kreyenfeld and Andersson 2014).

In terms of the operationalization of economic uncertainties, researchers have mostly focused on either *past* or *current* experiences including involuntary job loss, graduation during a recession, or certain forms of employment arrangements (e.g. fixed-term contracts). Commonly present in surveys and official national statistics, objective indicators of unemployment and fixed-term employment are typically used in operationalizing labor market uncertainty or precarity (Schmitt 2021; Kreyenfeld 2010; Auer and Danzer 2016). Rarely do survey data consistently include items capturing future-oriented economic uncertainties that could shed light on people’s assessments and perceptions of future world states and how they relate to fertility behavior and intentions.

Vignoli et al. (2020a) argue that narratives and perceptions of the future are important yet relatively understudied determinants of present-time fertility behavior and intentions. This framework was pivotal in expanding the uncertainty-fertility nexus to incorporate the role of future-oriented perceptions. In subsequent empirical work testing this framework, Lappegård et al. (2022) demonstrate that a positively portrayed future macroeconomic situation has a positive effect on present fertility intentions. Further, Guetto et al. (2023b) demonstrate that negative news cycles about the economy are negatively associated with fertility, suggesting that media plays a role in shaping individual perceptions of the future and, consequently, fertility decisions.

In this paper, we contribute to the growing literature on the enabling conditions of family and employment for parenthood transitions (Brauner-Otto and Geist 2018; Gatta et al. 2021; Vignoli et al. 2020a; Zimmerman et al. 2022), with a focus on two domains: (1) caregiving responsibilities and (2) employment uncertainty, both with an explicit orientation towards the future. This is one substantive contribution of our study to focus on the impact of caregiving responsibilities in addition to economic uncertainty, whereas most prior work has only considered the latter. This is especially relevant in aging societies, such as Germany, with an increasing share of “sandwiched” caregivers (Alburez-Gutierrez et al. 2021) who have dual care responsibilities both towards aging or sick parents and young children.

Recent demographic trends not only show a decline in birth rates in Europe but also that women are postponing first births, resulting in overlapping periods of child and elderly care (Roustaei et al. 2019; Ester Lazzari and Beaujouan 2024). This scenario can pose challenges for individuals and families in terms of time, material resources, and mental and emotional capacities to fulfill caregiving responsibilities. The idea of

having competing caregiving responsibilities may affect family formation as couples seek to avoid compounding the cognitive, emotional, and physical strains of care work. To date, however, there has been little empirical research on the effects of present and future caregiving responsibilities on fertility and fertility intentions.

Our study is conducted in Germany which is experiencing persistent increases not only in the share of elderly in the population but also in childbearing ages. As of 2022, the median age at first childbearing for women is age 30, and for men age 33 and no trend reversal seems in sight (DESTATIS 2023). Although healthy life expectancy in Germany has generally risen to a similar extent as in other countries (EUROSTAT 2023), the postponement of childbearing has increased the risk of people having to care for their dependent children and their elderly relatives in close succession or even at the same time in their life courses, placing an increased care burden on mothers in particular.

In terms of identification, we design a factorial survey experiment with random assignment of vignettes (i.e., various hypothetical situational contexts and constraints) to enable causal inference. Survey-based experiments are increasingly popular tools in the Social Sciences to study human attitudes and decision-making processes considering their advantages (Mize and Manago 2022). Through the random assignment of participants to hypothetical vignettes, researchers control for the potential confounding of unobservables (Auspurg and Hinz 2014; Sauer 2020). Survey experiments also provide an opportunity to explore sensitive topics (including own fertility intentions) and scenarios that may not be ethically or practically feasible to observe in real-world settings. Finally, experimental designs address the common issues in inferring causal claims using observational data such as confounding, response bias (e.g., social desirability), and sample selection. This paper thus contributes to a growing strand of research employing comparable experimental methods in studying fertility intentions (Karabchuk et al. 2022; Lappegård et al. 2022; Guetto et al. 2023a).

Our experiment was fielded in the nationally representative German Socio-Economic Panel Innovation Sample (SOEP-IS), a subsample from the main GSOEP respondent pool intended for the implementation of one-off surveys and experiments. Each respondent received five randomly selected vignettes that varied the future employment situation of a fictitious heterosexual couple as well as the female partner's caregiving responsibilities. Our focus on female caregiving is motivated by persistent gender inequalities, with women still performing as much as double the care work than

men do (Schäper et al. 2023). Respondents were then asked to rate the likelihood of each couple having a child in the next three years under these constraints. It is important to note that the outcome does not directly measure the respondents’ own fertility intentions but rather the *ascribed* decisions of the couple in the vignette, as evaluated by the respondents. Therefore, we refer to this outcome as **ascribed fertility intentions (AFI)** for clarity and precision, as has been used in other factorial survey experiments on fertility (see Guetto et al. (2023a) for instance).

5.2 Background and Hypotheses

This paper builds on three overlapping strands of the uncertainty-fertility nexus literature—(1) the role of employment uncertainties and caregiving responsibilities; (2) experimental analyses of the enabling conditions of parenthood; and (3) heterogeneities across population subgroups. In this section, we briefly summarize the most salient findings and recent developments in each strand and subsequently highlight our main contributions.

5.2.1 Employment Uncertainty and Caregiving Responsibilities

On the substantive front, the concept of “uncertainty” as a determinant of fertility intentions and behavior has been extensively studied. While uncertainty can manifest and be measured in different ways, the most extensively studied type is employment uncertainty, which has been analyzed through various measures (e.g., objective and subjective) and on various levels (e.g., macro and micro). After the 2008 global financial crisis, objective macro-level measures such as shocks in unemployment rates are used to demonstrate that, at least in the case of the US, states hit hardest by the recession experienced a more pronounced dip in fertility (Currie et al. 2014; Schneider 2015). Subsequent papers operationalized uncertainty beyond labor market indicators, used Google trends (Comolli and Vignoli 2021) and (traditional) media coverage (Guetto et al. 2023b) instead, and generally found that economic uncertainty negatively influences fertility. When uncertainty is operationalized through objective individual-level markers of unemployment or fixed-term work contracts, more evidence suggests a negative effect of adverse economic conditions on fertility (Alderotti et al. 2021; Gatta et al. 2021; Hofmann et al. 2017; Modena and Sabatini 2012; van Wijk et al. 2022).

An alternative approach is to use survey data with subjective item batteries measuring feelings of financial uncertainty or worries regarding one’s own or the country’s future economic situation (Kreyenfeld, 2010). Studies using such subjective markers of economic uncertainty have provided mixed results, with some finding a negative (or postponement) effect (Brauner-Otto and Geist 2018; Hofmann and Hohmeyer 2013), while others show an acceleration in select population subgroups such as women with lower levels of education (Kreyenfeld 2015). Alderotti et al. (2021) provide a comprehensive meta-analysis and show that these mixed results are due to the variety in measurements of uncertainty, outcomes, country contexts, and specific population subgroups. By and large, uncertainties experienced by males lead to postponement of first birth and delay in second birth transitions, while the effect of female uncertainties on first birth is context- and group-specific. Results for higher-order births are mixed.

These mixed findings may reflect the theoretically competing income and substitution effects of economic conditions on fertility. The income effect posits that having limited resources, as is the case under adverse or uncertain economic conditions, disincentivizes and lowers fertility. However, limited economic (e.g., labor market) opportunities also lower the opportunity cost of childbearing, potentially inducing accelerated or increased childbearing (substitution effect).

While the operationalization of uncertainty in the uncertainty-fertility literature is predominantly economic, there has been growing interest in other domains including health worries (Zimmerman et al. 2022), environmental concerns (Bodin and Björklund 2022), and policy access worries (Zhang et al. 2022). Another less explored determinant of fertility in the literature is caregiving responsibilities, which are increasingly common in aging societies (towards a sick parent, for instance), disproportionately shouldered by women. In the case of the US from 2003 to 2012, 8.9% of 25-44-year-old women are “sandwiched” caregivers who simultaneously care for both a parent or in-law and a child, compared to 6.5% of men in this age group (Suh 2016). In Germany, 6% of respondents from 2010 to 2020 who have care responsibilities have dual elderly and childcare duties based on estimates from the German SOEP—5% for men and 7% for women (Schäper et al. 2023). This is still a relatively small share. However, with the steady increase in childbearing ages, this is anticipated to grow. In terms of care work, broadly defined, women carry the brunt by performing as much as double what men do (Schäper et al. 2023; DESTATIS 2023). Since sandwiched caregivers experience

worse health outcomes, higher care burden, and more career interruptions (Alburez-Gutierrez et al. 2021), women who are currently taking care of a parent with care needs or who expect to do so in the near future might avoid or postpone childbearing to avert compounded caregiving duties.

In this paper, we therefore substantively extend the uncertainty-fertility literature by analyzing the effects not only of future-oriented employment uncertainties but also forthcoming caregiving responsibilities. In effect, we also analyze whether the simultaneous absence of these work and family-related instabilities yield a “compounded advantage” for childbearing decisions.

5.2.2 Experimental Analysis of the Enabling Conditions of Parenthood

On the methodological front, prior research on the uncertainty-fertility nexus has almost exclusively studied past or current experiences using observational data. One example of past experiences is the effect of initial labor market conditions or employment characteristics immediately post-graduation on subsequent fertility outcomes (Auer and Danzer 2016; Choi et al. 2020; Hofmann and Hohmeyer 2013; Raymo and Shibata 2017). Using observational data, many of these studies employ quasi-experimental techniques, leveraging on inter- or intra-cohort differences as treatment-control groups (e.g., fixed-term vs. permanent, recession vs. non-recession entrants) and comparing average outcomes. On the other hand, current uncertainty is typically operationalized as present-time worries using questions such as “How worried are you about your own economic situation?”. This type of question, as it appears in the German Socio-Economic Panel, focuses on present-time concerns and at best only partly and imperfectly captures their perception of the future (Vignoli et al. 2020a). In other cases, present-time uncertainty is operationalized through the respondents’ current employment status, i.e., whether a respondent is unemployed or holding a fixed-term working contract (Kreyenfeld 2010; Schmitt 2012, 2021).

This strand of research is worth extending methodologically in two ways. First, given that there are inherent limitations of inferring causality from observational data due to possible bias arising from selection and response quality (Winship and Morgan 1999), recent studies have turned to employing quasi-experimental methods such as difference-in-differences (DiD) and matching and weighting (Hofmann et al. 2017; Comolli and Vignoli 2021). However, these methods may still not fully address selection

bias if, for example, there remain substantial residual differences between the treatment and control groups after matching or in the case of DiD designs, when the parallel trends are barely or not at all observed. Social scientists recognize this limitation and are thus increasingly using experiments (e.g., lab, field) to fully isolate and estimate the causal effect of a certain variable (Mize and Manago 2022).

Experimental approaches, particularly factorial survey experiments, have been increasingly used to study ascribed fertility intentions. Recent work has looked at the effects of policy reforms (Guetto et al. 2023a), narratives of future economic scenarios (Lappegård et al. 2022), and income and employment conditions (Karabchuk et al. 2022), and all these papers find the importance of people’s (positive) economic conditions, both measured objectively and subjectively, in forming the enabling conditions for parenthood.

The second methodological extension follows through the explicit future orientation of uncertainty. As discussed above, whereas prior measures of economic uncertainty focused heavily on past and current experiences, recent work argues that perceptions or narratives of the future likewise matter. In a seminal work, Vignoli et al. (2020a) develop a “Narrative Framework” and argue that fertility decisions are based on a narrative of the future—one’s imagined and perceived future conditions. They argue that narratives of the future are shaped on the one hand by the individuals’ social network and their interactions (e.g., peers and family), and on the other hand by external influences and framers of narratives (e.g., media).

Recent work has moved in this direction and empirically tested the effects of perceptions of the future. Lappegård et al. (2022) use an experimental design to tease out how future economic scenarios affected fertility intentions in Norway, operationalizing positive economic scenarios as a country with higher rates of employment in permanent and full-time contracts and more job opportunities in the next three years. Respondents who were exposed to these positive economic scenarios expressed higher fertility intentions than those exposed to a negative scenario and those not exposed at all (Lappegård et al. 2022).

This paper adopts these two methodological extensions and, in light of the discussion above, substantively addresses: what are the enabling (e.g., positive) conditions of parenthood as it relates to one’s caregiving and employment domains? Under which scenarios or situations are ascribed fertility intentions the highest? We thus test the hypotheses that:

- *H1a (caregiving): Ascribed fertility intentions are higher for couples without any future caregiving responsibilities.*
- *H1b (employment): Ascribed fertility intentions are higher for couples without any future employment uncertainties.*

Beyond the separate effects of caregiving and employment domains, we likewise analyze whether the effect of future employment uncertainty varies at different levels of future caregiving responsibilities and vice-versa. To analyze whether a “compounded advantage” exists for couples who face no uncertainties and responsibilities in both domains such that:

- *H2 (simultaneous): Ascribed fertility intentions are highest for couples without any future caregiving responsibilities and future employment uncertainties.*

5.2.3 Heterogeneities by Respondent’s Gender

We also explore whether the effects of caregiving responsibilities and employment uncertainties vary by the respondent’s gender. We test our priors that the positive effect of having no future caregiving responsibilities on ascribed fertility intentions is stronger for female respondents given that they traditionally shoulder these obligations and thus understand the cognitive, emotional, and physical demands of caregiving work more than men do.

- *H3 (gendered differences): The effect of future caregiving responsibilities on fertility intentions is more pronounced for females compared to male respondents.*

Gendered differences in the hypothesized positive effect of having no future employment uncertainty are more difficult to predict and, indeed, the literature presents mixed results. Using a quasi-experimental design that exploited an exogenous variation of economic concerns in Germany, Hofmann and Hohmeyer (2013) found that economic uncertainty more strongly reduced the fertility of females compared to males. Meanwhile, the presence of male breadwinner norms would suggest that men’s employment situation is more central to fertility decisions. Indeed, male employment uncertainty is associated with delayed first and second-childbearing in Germany and Denmark (Kreyenfeld and Andersson 2014). These competing predictions might point either way in terms of whether the positive effect of having no employment uncertainty

is larger for male or female respondents, although most studies point to gendered differences in both fertility and employment patterns. In this paper, we will explore the gendered differences in the effect of employment uncertainty without strong priors or expectations about its direction.

5.2.4 Heterogeneities by Respondent's Own Employment Markers

One advantage of quasi-experimental or experimental studies conducted on representative samples is the possibility of examining heterogeneities by population subgroups. For instance, fertility intentions and behavior vary across employment types and working conditions. The background for this investigation is that there are generally strong arguments supporting the notion that labor market uncertainties lead to fertility postponement. While this may be the result of rational calculations, it may also be the result of social norms that require individuals to first settle into a stable employment position before getting married and having children. Prior qualitative work has shown that these norms are particularly strong in the case of West Germany, but lesser in the case of the East (Bernardi et al. 2007). Further, the notion that a secure economic base is a prerequisite for having children may be a more 'middle-class' one, so that respondents from lower or more precarious labor market positions may perceive employment responsibilities less often as a barrier to having children than higher social classes. This necessitates an examination of whether the treatment effects are heterogeneous across respondents based on usual markers of employment position including employment status (e.g. employed vs. unemployed), contract duration (e.g. fixed-term vs. permanent), and net earnings.

Furthermore, the treatment effects may likewise vary by social class. While present-day demographic research hardly ever addresses the class-fertility nexus (see however Albertini et al. (2024); Kreyenfeld et al. (2023a)), classical demography has extensively been occupied by class differences in fertility behavior. With a strong moral imperative, this strand of research has stipulated that the fertility behavior of the lower social classes is typically less subject to careful rational consideration than that of the higher social classes. Turning the argument around, the belief that having children requires a stable career may be a historically deep-rooted 'bourgeois' approach to fertility choice that is not universal across all social classes.

Finally, it might be argued that it is not the objective markers of employment that

matter but rather the subjective experiences or worries of individuals. These measures better account for how satisfied or worried individuals are about their own employment conditions, regardless of what type of employment they have. For instance, someone on a fixed-term contract may not be worried at all about their employment prospects but someone on a permanent contract might be worried about, say, their job being prone to automation or outsourcing. Subjective measures of worries capture these nuances and have been shown to also be a predictor of fertility (Kreyenfeld 2015; Hofmann and Hohmeyer 2013). Against this backdrop, we formally test the following hypothesis:

- *H4 (heterogeneity by the respondent’s employment markers): The effect of future employment uncertainty on ascribed fertility intentions is moderated by the respondents’ own objective and subjective employment markers.*

5.3 Methodology

The German SOEP is a nationally representative longitudinal survey covering a broad set of demographic, social, and economic variables that has been conducted annually since 1984 (Goebel et al. 2019a). We designed and conducted a factorial survey experiment in the SOEP Innovation Sample (SOEP-IS), a subsample of the SOEP that was established in 2011 and is dedicated to the implementation of experimental and innovative instruments. SOEP-IS respondents only receive a subset of the regular questionnaire to make space for the additional survey content.

Factorial survey experiments are used to study judgments or decision-making in complex social situations. A popular approach, and the one we use in this study, is to expose respondents to different “vignettes” of a case of interest and elicit their assessment of each case with one or several outcome questions. Vignettes vary in one or (more typically) several domains of interest (e.g., future caregiving responsibilities and employment uncertainties). These factors are randomized and respondents are subsequently exposed to vignettes with different combinations or levels (Auspurg and Hinz 2014). The vignettes used in the present survey experiment were pre-tested in late 2021 (Alonso-Perez et al. 2021) and data was collected in the 2022 round of the SOEP-IS. An advantage of conducting a survey experiment through the SOEP-IS is that we are able to use standard variables from the SOEP about the respondents’ socio-demographic and economic characteristics including gender, age, marital status, and own employment information.

5.3.1 Experimental design

In our experiment, vignettes varied with respect to two attributes of interest, future caregiving responsibilities and future employment uncertainty, each comprising 4 levels, leading to 16 distinct vignettes. For future caregiving responsibilities, the hypothesized positive effect of having no sick parent on ascribed fertility intentions might be driven by two channels— the positive effect of grandparental support with childcare and of having no caregiving responsibilities. We limit this confounding in our vignettes by explicitly designing the scenarios to only capture and refer to the female partner’s caregiving responsibilities, without any mention of grandparental involvement in childcare. The implied idea of grandparental support as a “benefit” of having no future caregiving responsibilities did not come up in the pretests of the experiment (Alonso-Perez et al. 2021). For future employment uncertainty, the treatment levels pertain to which partner, or if both or none, experiences job loss worries. The experiment began with a brief contextualization of a fictitious heterosexual couple, Tom and Lena, with the following fixed characteristics:

- Tom (age 27/32/37) and Lena (age 25/30/35) have been living together for three years and generally agree that they may have children together;
- Both are currently in stable employment, work full-time, and wish to continue doing so;
- Tom has no siblings and his parents live further away; and
- Lena’s older sister and her mother both live close by, while her father died several years ago.

Except for age, all of these characteristics were fixed and included to provide the respondents with additional contextual information about the couple and minimize ambiguities about the employment and family status of both partners. By providing constant contextual information, we ensure that respondents think about the same situation when judging the vignettes. Age was varied across vignettes but the age difference between Tom and Lena was consistently set to 2 years. The decision to include these characteristics was based on a cognitive pre-test where vignettes of various characteristics were tested for clarity and brevity (Alonso-Perez et al. 2021).

Each respondent was presented with 5 randomly selected vignettes with varying future care and employment situations of the couple. After each vignette, the

respondent is asked to evaluate the likelihood of this couple having a child in the next 3 years on a scale from 0 (very unlikely) to 10 (very likely). To eliminate order effects (Sauer 2020), vignettes are ordered randomly—respondents are presented with 5 random vignettes from a pool of 16 possible combinations. Below are the exact wordings of the caregiving and employment situations in the experiment.

Dimension A: Future care situation	Dimension B: Future employment situation
High: Her mother has developed a severe illness, which means that Lena will soon need to take care of her alone.	Both: Tom and Lena are worried that they may lose their jobs.
Some: Her mother has developed a severe illness, which means that Lena and her older sister will soon need to take care of her.	Male: Tom is worried that he may lose his job.
Low: Her mother has developed a severe illness, which means that she will soon move in a long-term care facility.	Female: Lena is worried that she may lose her job.
None: Her mother is still very active and does not need any care.	None: Tom and Lena are not worried that they may lose their jobs.

The outcome is the respondent’s evaluation of the likelihood that the fictitious couple will have a child within the next three years, rated on a 0-10 scale. As noted earlier, this outcome is a measure of **ascribed** fertility intentions: rather than the respondents’ own intentions, our experiment captures what their intentions could be under exactly the same circumstances as the fictitious couple described in the vignette. While this outcome may not be indicative of the actual behavior, especially among biological male or female respondents outside childbearing age, our main interest is not fertility *per se* but rather the enabling conditions and underlying mechanisms that influence the fertility decision-making process. As such, our outcomes are presented as the effect of a certain treatment category (i.e. high employment uncertainty) *relative* to a reference category (i.e. no employment uncertainty).

To minimize the burden for the respondents and thus avoid bias due to fatigue, each respondent evaluated 5 vignettes that were randomly selected from the vignette universe (16 vignettes). The survey experiment was conducted with a final sample of 1,750 respondents, who were part of the SOEP-IS. In total, we have collected 8,750 vignette observations with 240 non-responses (2.75%). Each of the randomly assigned 16 vignette combinations has at least 500 responses, allowing us to make robust between-group comparisons (Auspurg and Hinz 2014)

While our survey experiment provides a powerful way to randomly “expose”

respondents to various situational contexts and constraints, it also comes with the potential limitation that survey experimental responses might not be a good proxy for respondents' actual behavior. Hainmueller et al. (2015) reassuringly find that their estimates from a survey experiment on support for immigrant naturalization are in line with respondents' actual political preferences and behavior in real-world elections, but it remains an open question if and to what extent this congruence of experimental and real-life outcomes generalizes to other settings.

5.3.2 Analytical Strategy

Equation 5.1 provides our baseline specification, estimated using Ordinary Least Squares (OLS) regressions with standard errors clustered at the individual level,

$$Y_{v,i} = \beta_1 C_v + \beta_2 E_v + \delta S_v + \gamma I_{v,i} + \epsilon_{v,i} \quad (5.1)$$

where $Y_{v,i}$ refers to respondent i 's evaluation of fertility intentions for vignette v . $C_{v,i}$ and $E_{v,i}$ are vectors of the caregiving and employment treatment level dummies. In the full model specification to improve precision, we additionally include $S_{v,i}$, a vector indicator of the other varying characteristic of the vignette—dummies of Lena's age—and $I_{v,i}$, a vector of individual characteristics including sex, age group, marital status, and own financial worries. Due to random assignment, the coefficients on the vignette attributes capture the so-called Average Marginal Component Effect (AMCE), that is, the causal effect of an attribute (or rather a specific attribute level, relative to the omitted baseline), averaged over the joint distribution of the remaining attributes (Hainmueller et al. 2014).

Following our additional hypotheses, we explore three extensions of the analysis using interaction models. The first addresses the possibility of a “compounded advantage” for couples with stable employment and no caregiving responsibilities by introducing a full set of interactions between levels of employment and caregiving scenarios (in the terminology of Hainmueller et al. (2014), the corresponding coefficient identify the Average Component Interaction Effect (ACIE)) as specified in equation 5.2.

$$Y_{v,i} = \beta C_v * E_v + \delta S_v + \gamma I_{v,i} + \epsilon_{v,i} \quad (5.2)$$

The main advantage of conducting this survey experiment on a large and population-representative sample such as the GSOEP is that it allows us to examine

the differences of the treatment effects by population subgroups (i.e., Conditional Average Treatment Effects)– an extension that is more difficult to do in smaller and/or non-random samples (Sauer 2020). To test hypotheses 3 and 4 and explore whether the treatment effects vary by the respondents’ gender (H3) and the respondents’ own employment characteristics (H4), we estimate interaction models. For H3, we separately run models with caregiving-gender and employment-gender interactions. For H4, we interact the employment uncertainty treatment with measures capturing four objective markers of respondents’ current employment situation: employment status (i.e., employed full-time, employed part-time, not employed, and others including those retired and inactive), contract type (i.e., fixed-term and permanent), occupational class (i.e., self-employed, blue-collar, white-collar, civil service, and apprentice/trainee), and net monthly earnings which divide those with reported net earnings into 5 quintiles. In the last three markers, we omit from the sample those reported as not employed to focus the comparison between different employment states in each marker among employed respondents. We likewise present interactions with two subjective markers (financial satisfaction and own job worries) that capture the role of perceptions and worries, which are more aligned with the subjective phrasing of the employment uncertainty treatment in the vignette.

5.4 Results and Discussion

5.4.1 Baseline Results

Results in Figure 5.1 show that having no caregiving responsibilities increases ascribed fertility intentions on average from 3.0 (SD=2.6) to 5.8 (SD=2.8), or by 2.8 units (on a 0-10 scale). Having fewer caregiving responsibilities, either through sharing it with a family member (“some”) or outsourcing them to a facility (“low”) also has a positive effect on AFIs relative to when the female partner fully shoulders care work. Indeed, consistent with our expectations of avoiding competing care demands from aging parents and young children, it is when the female partner has no forthcoming caregiving responsibilities that AFIs are highest.

On the other hand, couples where both partners do not worry about losing a job in the future have higher AFIs by 1.9 units relative to couples where both are highly worried. This finding lends support to the observational literature on the employment uncertainty-fertility nexus that demonstrates a negative relationship. Results also

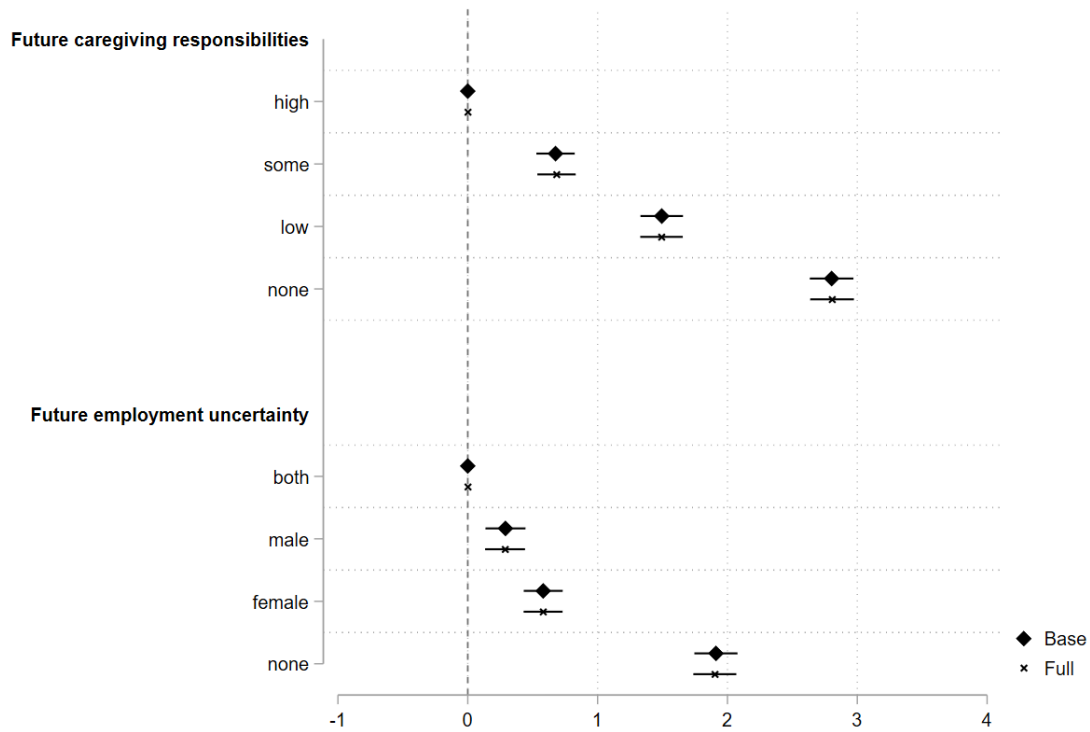


Figure 5.1: Effect of future caregiving responsibilities and employment uncertainties on fertility

Note: Coefficients are comparable to the Average Marginal Component Effects (AMCE)—the effect of a change in attribute relative to the base category on fertility. The full model controls for Lena’s age and the respondents’ age, gender, marital status, and financial worries. The command “conjoint” in Stata developed by Frith (2021) is used. Standard errors are clustered at the individual level.

suggest slightly higher fertility intentions when future employment uncertainty is experienced by the female rather than the male partner.

5.4.2 Compounded Advantage of Stable Employment and No Caregiving

With AFIs measured as the likelihood of the couple having a child (in the next three years) on a scale from 0 (very unlikely) to 10 (very likely), the vignette describing a couple with no employment uncertainty for both partners and no future caregiving responsibilities had the highest average AFIs (7.39), while the vignette reporting future

employment uncertainty for both partners and high future care responsibilities had the lowest average (2.57). It is worth noting that persons who are exposed to high caregiving responsibilities, but low employment uncertainties still report rather low AFIs (3.81). The complete summary means for each combination of caregiving and employment treatments give a sense of the range of AFI values and are presented in Appendix Figure D.1.

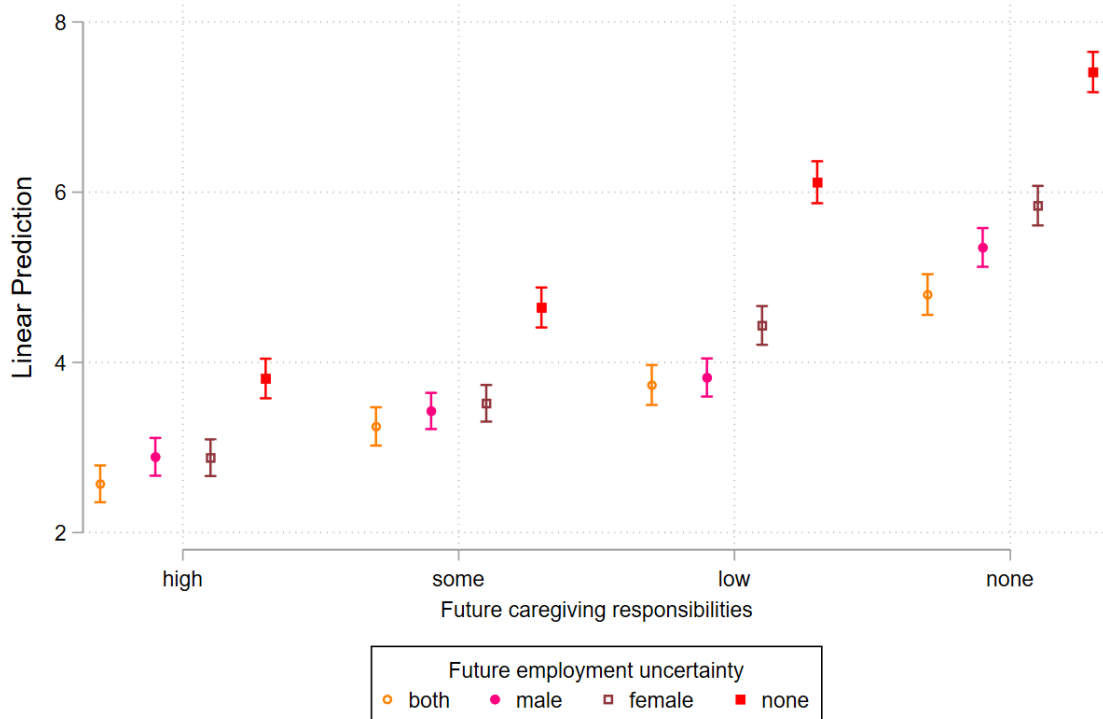


Figure 5.2: Interactive effects of future caregiving responsibilities and employment uncertainties

Note: These models show the predictive margins of the interaction between future caregiving responsibilities and employment uncertainty. These estimates control for Lena's age and the respondents' age, gender, marital status, and financial worries. Standard errors are clustered at the individual level.

In Figure 5.2, we take a closer look at the interactive effects from the full model and show that the highest fertility intentions are for couples without future employment uncertainty and caregiving responsibilities. An omnibus F test confirms that the interaction terms are jointly significant ($F(9,1712)=8.38$, $p<0.001$). Conversely, fertility intentions are lowest for couples with both partners experiencing high

employment uncertainty and where future female care responsibilities are high. The difference between these two categories is stark—around 5 units on an 11-point scale. All these taken together lend support to our second hypothesis that the absence of multiple and simultaneous uncertainties is where fertility intentions are highest.

5.4.3 Moderating Role of Respondents' Gender

In this section, we explore whether the effect of future caregiving and employment uncertainty on fertility intentions is moderated by the respondents' gender. Results in Figure 5.3 show some gendered differences in caregiving responsibilities but not in future employment uncertainty. Consistent with our expectation in Hypothesis 3, formal hypothesis tests reveal that the marginal effect of having low or no future caregiving responsibilities is significantly higher for female respondents by 0.43 ($p=0.002$) and 0.27 ($p=0.049$) units, respectively, compared to males. Indeed, as females have traditionally disproportionately shouldered caregiving responsibilities, they might better understand the cognitive, emotional, and physical demands of care work that are compounded when they are in a situation of having to care for both an aging parent and a young child. This gendered effect demonstrates how the caregiving responsibilities of young adults towards older parents are somewhat a barrier for childbearing but more for women than men.

On the contrary, we do not find any gendered difference in the positive effect of no future employment uncertainty on intentions, or any of the other employment treatment levels. Thus, contrary to our earlier expectations of seeing gendered differences in both caregiving and employment, we find this to be the case only for the former where the effects of certain levels are unambiguously stronger for female respondents.

5.4.4 Moderating Role of Respondents' Employment Markers

Finally, we address how the effect of future employment uncertainty on fertility intentions varies based on the respondents' own objective and subjective employment markers. As discussed in the preceding section and for completeness, we use all available and relevant objective employment markers including employment status, occupational class, contract type, and net monthly earnings, and subjective markers including financial satisfaction and own job worries. Formally testing this, we estimate interaction models of employment uncertainty with each employment marker.

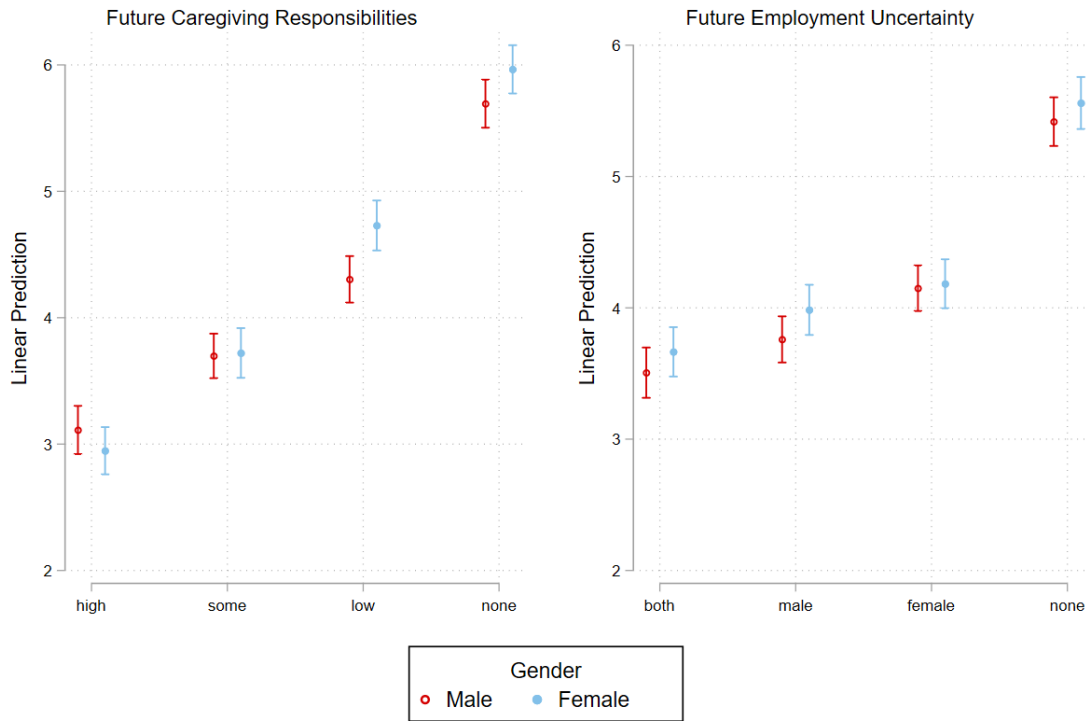


Figure 5.3: Interactive effects of caregiving and employment with gender

Note: These models show the predictive margins of the interaction between gender and future caregiving responsibilities and employment uncertainty, respectively. These estimates control for Lena's age and the respondents' age, gender, marital status, and financial worries. Standard errors are clustered at the individual level.

Results in Figure 5.4 show that the effects of future employment uncertainty are not significantly moderated by any of these respondent markers.

Respondents have notably rated the highest fertility intentions for vignettes where none of the partners have an uncertain job, regardless of their employment status, contract term, occupational class, and monthly earnings. While recent findings on the uncertainty-fertility nexus suggest fertility differentials by employment markers such as occupational class (Kreyenfeld et al. 2023a), the broader absence of a significant interaction or heterogeneities by employment characteristics may be due to the fact that respondents were evaluating fertility intentions of the couples in the vignettes with different preferences and constraints rather than their own situation.

This finding hints that respondents who are asked to make judgments about other people's intentions and behavior may be taking into account the context and

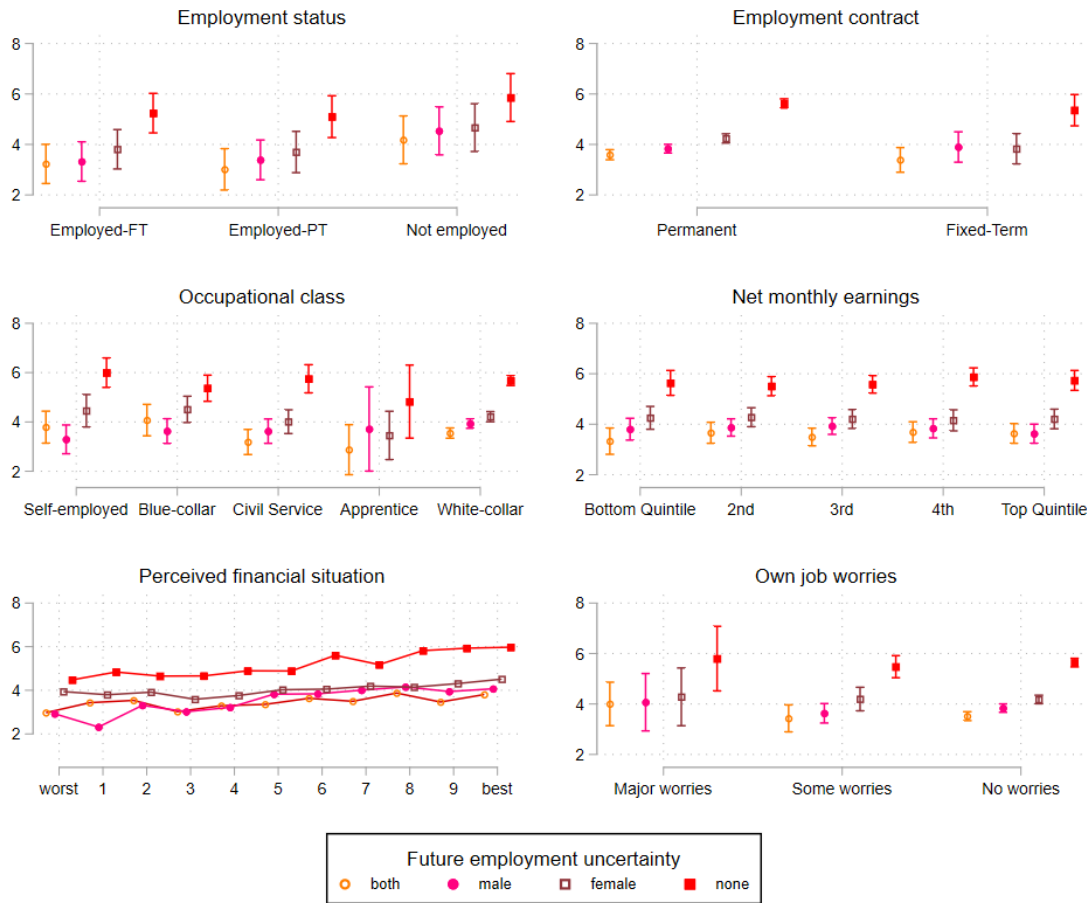


Figure 5.4: Moderating Effect of Respondent's Own Objective and Subjective Employment Markers

Note: These models test the interaction of the vignette's employment uncertainty treatment with each of the six markers of the respondent's objective and subjective employment characteristics—employment status ($n=7,846$) classified between full-time, part-time, and unemployment, employment contract ($n=4,260$), occupational class ($n=4,890$), net monthly earnings ($n=4,630$), perceived financial situation ($n=8,485$), and subjective job worries ($n=5,159$). Apart from the employment status and perceived financial situation interactions, respondents not employed are dropped in the analyses to focus the comparison among employment types, conditional on being employed. These models control for other salient vignette characteristics (Lena's age), socio-demographic variables (age group, marital status, and gender), and the remaining employment measures. Cluster robust standard errors at the individual level are used.

environment of the people they are assessing more than their own situation. Thus, this result suggests that one's ideas of how other people will behave under a certain set of constraints are less affected by one's own situation. We further note that this

analysis is descriptive and exploratory—as the covariates of own employment markers are non-randomly assigned, these are mere associations between each marker and the treatment effect and not the causal effect of moving from one group (e.g., fixed-term) to another (e.g., permanent).

5.5 Conclusion

In this paper, we analyze the effects of future caregiving responsibilities and employment uncertainties on ascribed fertility intentions in Germany. Extending the survey experiments by Lappegård et al. (2022) and Karabchuk et al. (2022) on the impact of macro-economic conditions, we conducted a factorial survey experiment that presented respondents with varying caregiving and employment scenarios faced by hypothetical couples.

Our contribution is threefold. First, avoiding key limitations of observational designs (e.g., endogenous selection, desirability, limited measures of uncertainty), our survey experiment allows us to estimate the causal effect of both caregiving responsibilities and employment uncertainties on ascribed fertility intentions by leveraging on the randomized exposure of respondents to vignettes. Second, we extend the literature on the economic uncertainty-fertility nexus by simultaneously analyzing how employment worries interact with another and relatively understudied scenario that is common and relevant for young adults in aging societies such as Germany: caregiving responsibilities toward a sick parent. Finally, building on Vignoli et al. (2020a), our vignettes are explicitly framed as responsibilities in and uncertainties about the future, thus complementing prior work, which has predominantly focused on past and present experiences of uncertainty (e.g., experiencing a job loss, graduating during a recession, etc.).

Three findings are worth highlighting. First, couples without female caregiving responsibilities and employment uncertainties for both partners in the foreseeable future have higher ascribed fertility intentions by 2.8 and 1.9 units, respectively (on a 0–10 scale), relative to when both are high. Consequently, we find that ascribed intentions are highest when there are no future caregiving responsibilities and employment uncertainty simultaneously. Second, the positive effect of having low or no future female caregiving responsibilities is stronger for female respondents, while the effect of future employment uncertainty does not differ by gender. The gendered gradient

in the effect of female caregiving responsibilities might be explained by the fact that females continue to carry a disproportionate share of parental caregiving responsibilities (Labbas and Stanfors 2023). These expectations may have socialized females to better understand the cognitive, emotional, and physical demands of caregiving work, and that these are particularly pronounced when elderly/parental care occurs simultaneously with child-rearing. Finally, the effect of employment uncertainty did not seem to be moderated by the respondents' employment status, contract type, occupational class, monthly earnings, financial satisfaction, and own job worries, suggesting that their expressed fertility intentions are shaped more by their evaluation of the couples' given constraints and preferences rather than their own.

The main limitation of our analysis is the inability of the survey experiment to capture individual fertility intentions and subsequent behavior, an inherent limitation of vignette designs and survey experiments more broadly. While at least one previous study (Hainmueller et al. 2015) found a high consistency between survey experimental and real-world decision patterns, we would caution not to interpret our results in terms of respondents' own fertility intentions. Beyond general concerns about a possible mismatch between survey experimental and real-world behavior, this is due to our outcome of interest being the intentions ascribed to fictitious couples rather than the respondents' own stated intentions. Additionally, our sample is representative of the German population, which means that a sizable portion of our sample is actually incapable of biological childbearing either due to their age or sex at birth. Ascribed fertility intentions as assessed in this study must therefore be interpreted only within the constraints set in the vignette and not as indicative or predictive of the respondents' own intentions and behavior.

Finally, we recognize the limitation that employment and caregiving constellations are much more nuanced in reality, and could be far more complex than the four scenarios included in our experiment. For instance, while our caregiving treatment levels vary the female's caregiving workload, these could also be shared between the partners. It is also possible that these caregiving responsibilities toward a sick parent occur at a much later age than the early 30s or late 20s in a low-mortality context (e.g. high life expectancy) such as Germany. Future experimental work can therefore further explore these nuances in caregiving and employment arrangements, as well as focus on other outcomes including higher-order births and intentions to remain childless, among others. Other future avenues of inquiry are possible, including differentiation between

having no caregiving responsibilities and benefitting from caregiving support, analyzing other forms of uncertainty (e.g., environmental, healthcare, etc.), and comparing the relative contributions of past experiences and future perceptions on fertility.

Our work speaks to how having competing caregiving responsibilities and high worries about future employment positions disincentivize childbearing decisions. For caregiving, our results lend evidence to the idea that the availability of institutional healthcare and elderly care facilities that could partially or fully augment the caregiving needs of elderly and sick parents may help enable couples to realize their childbearing goals. On the other hand, policies that minimize people's worries about the security of their labor market positions also seem to be beneficial. More broadly, social policies aimed at supporting families to meet their desired size should take into consideration tangential policy domains (e.g. in elderly care and labor markets) that make childbearing conducive and viable for couples.

Chapter 6

Conclusion

*“Working 9 to 5
What a way to make a living
Barely getting by
It’s all taking and no giving”*

-Dolly Parton, 9 to 5

Armand Garcia¹ from the Philippines had been working as a plant operator in an oil refinery in Saudi Arabia for over a decade until the start of the COVID-19 pandemic when he lost his job. Unemployed back in his hometown in one of the poorest neighborhoods in Metro Manila during the height of the lockdowns, he realized that he was not alone—many of his neighbors also lost their jobs with nothing to do and little to spend. With whatever little they had—scraps of cloth and old sewing machines, around 50 of them organized and collectively produced handmade bags, pillows, and tablecloths and sold them on social media platforms. To this date, their community organization is slowly but steadily growing online, sometimes participating in local bazaars and flea markets. Inspired by these familiar stories of resilience during periods of economic difficulties, the essays in this dissertation are demonstrably driven by the causes and demographic consequences of **employment uncertainty**. From these four essays, six cross-cutting takeaways are worth noting.

¹A randomly selected pseudonym

6.1 Summary and Contributions

1. **Employment uncertainty is a loaded term—operationalize with care.**

In each chapter, “employment uncertainty” as it appears in the literature seems to change meaning from one use case to another. This lack of universality in what scholars mean with the term “uncertainty” or “precarity” is embedded in the empirical literature. Dating back to seminal theories of social stratification where Weber et al. (1922) makes a distinction between class, which is determined by the economic sphere of social interaction, and status, which is shaped by the identity and honor associated with people in a society. From this, and many other contemporary scholars of stratification, we know that employment uncertainty or precarity could be constructed, understood, and operationalized differently.

This dissertation makes one such attempt. In Chapter 1, a schematic classification of *employment uncertainty* is proffered. It is a direct extension of the earlier work of Blossfeld et al. (2006) who characterize uncertainty in three dimensions— economic, temporal, and employment-based. In this expanded schema, we develop 7 main dimensions of employment uncertainty: (1) positional; (2) durational; (3) statutory; (4) recessionary; (5) class-based; (6) subjective; and (7) caregiving. Within each dimension, certain groups inherently face more uncertainty than others.

As indicated in Figure 1.7 and Table 1.7, each chapter deals with multiple types of employment uncertainty, offering a cross-cutting and transdimensional assessment of its causes and demographic consequences. In this regard, the contribution of this dissertation lies in the schematic expansion of employment uncertainty and the simultaneous operationalization of various dimensions in the empirical analysis.

2. **Employment uncertainty is a determined phenomenon.** The analysis in Chapter 2 builds on the stand of the literature demonstrating the countercyclicality of informality (Maurizio 2021), which typically increases during periods of economic downturns (Floro et al. 2009; Albertini et al. 2019; Fernández and Meza 2015). Prior work has shown that during and after past episodes of crises like the 1997 Asian Financial Crisis or the Financial Crises episodes in Latin America in the early 2000s, unemployment and informal employment rose. These findings suggest a two-pronged precarization in the labor market as a

result of crises: (1) increased joblessness and (2) increased informality among the remaining employed population.

Therefore, this dissertation suggests that uncertain employment positions, though not deemed ideal, serve as risk-coping cushions for individuals who do not want or cannot afford to be unemployed. While this article demonstrates this in the Philippine context, the cushioning role of employment uncertainty is not a developing country-only phenomenon. Among some developed countries (Germany, in particular), “short time work” or reduced working hours sharply increased during past episodes of crisis (e.g., Global Financial Crisis and COVID-19 pandemic) (Naujoks et al. 2022). While uncertain employment positions are determined, Chapter 2 also shows that this phenomenon is gendered. In the Philippines, the informalization of employment as a result of extreme lockdowns was pronounced for women more than men.

Chapter 2 makes two substantive contributions to our understanding of informality. The first is the conceptual and empirical distinction between the underlying causes of employment informalization—namely compositional factors and survivalist motives. Second, it establishes a causal link between state policies and informalization by isolating the “lockdown” effect from the baseline recessionary effects of the pandemic. Broadly, this chapter speaks to three main dimensions of employment uncertainty as illustrated in Chapter 1—extreme lockdowns have had a significant impact on the lowering of employment rates (*positional employment uncertainty*) and the increasing of informality within the working population (*statutory employment uncertainty*).

3. **Employment uncertainty is deterministic. Past, present, and future experiences matter for fertility decisions.**

A sizable portion of this dissertation is devoted to analyzing the demographic repercussions of employment uncertainty. Chapters 3 to 5 examine how actual or potential experiences of employment uncertainty at various points in the life course affect fertility intentions and behavior. This builds on a thick strand of the literature— from seminal work linking economic conditions and fertility Malthus (1798); Boserup (1981) to contemporary theoretical models of fertility choice (Becker 1960) and empirical analyses of employment uncertainty and fertility outcomes (Kreyenfeld 2010; Alderotti et al. 2021; Matysiak et al. 2021a).

Using data on initial employment conditions in Germany, Chapter 3 demonstrates that females entering the labor market with a fixed-term contract and males entering during a recession experience persistently lower first birth hazard rates within a decade after entry compared to their permanent and non-recession counterpart entrants, respectively. This article highlights how experiences of *past* employment uncertainty (e.g., labor market entry that occurs in young adulthood) affect subsequent life course outcomes.

In terms of measuring *present* experiences of uncertainty, Chapter 4 shows the “economic advantage” of males and females in service-class occupations in the transition to higher-order births, when compared to workers in low and unskilled social classes. In terms of the role of subjective worries, the article also finds that among women, there is an independent effect of subjective worries that aren’t already captured through their occupational class positions.

Finally, Chapter 5 differs from the earlier chapters in three ways— first, it explicitly frames both (subjective) employment uncertainty and caregiving responsibilities with a future orientation. Building on the “narratives of the future” framework (Vignoli et al. 2020a), it highlights the role of perceptions and imageries of the future in shaping current present-time intentions and decisions. Second, it adopts an experimental approach. Employing a vignette experiment design that randomizes respondents’ exposure to a hypothetical couple’s future employment and caregiving scenarios allows for causal identification. Finally, results in this chapter demonstrate that people experiencing simultaneous uncertainties (having both employment uncertainty and caregiving responsibilities) have the lowest fertility intentions.

4. Employment uncertainty is determined and deterministic across different country contexts.

While each chapter of the dissertation identifies *local* effects based on individual-level data in specific country contexts, Chapter 1.2 shows that employment uncertainty is a global phenomenon. Official unemployment rates are persistently higher in high-income countries, partly owing to robust unemployment assistance and other welfare support. Among developing countries where unemployment cannot be afforded and labor markets are less developed, vulnerable employment is the norm, rather than the exception. Descriptive trends in Figure 1.1 depict the

countercyclicality of unemployment. A sharp increase in the unemployment rate is notable in high-income economies during the 2008 Global Financial Crisis, most notably for North America, the epicenter of this recession. Similarly, unemployment also rose during the political and financial turmoils in the late 90s in Latin America. The observation that certain measures of employment uncertainty (i.e., unemployment rate) rise during and after economic shocks seems to travel across contexts. Udall and Sinclair 1982; Lee et al. 2020, see discussions in.

In terms of the demographic repercussions of employment uncertainty, Chapter 1.3 proffers a global reappraisal. In Table 1.3, the negative effect of a one-unit increase in the lagged unemployment rate on total fertility rate seems to be stronger for low and lower-middle-income economies, despite limited country-specific research in these contexts. Both are robustly negative, but the effect of unemployment among lower-income countries is almost double the effect size in higher-income countries. These cross-country estimates lend support to the idea that, at least at the macro level, *positional employment uncertainty* and *statutory employment uncertainty* contribute to the global phenomenon of fertility decline (Skirbekk 2008; Karaman Örsal and Goldstein 2018).

In sum, while the examination of developing country contexts in this dissertation is rather limited, we can deduce two overarching findings that apply to them. First, developing countries have an inherently different form of employment uncertainty. Whereas unemployment is the important metric in more advanced economies, measures of informality and underemployment seem to be more relevant elsewhere. Second, the negative effect of employment uncertainty on fertility is **stronger** in developing countries. Future work could emphasize whether this difference in magnitude is attributable to cultural or welfare policy-related differences, or both.

5. **Women play a dual role as workers and carers of first resort.** One cross-cutting finding from Chapters 2 and 5 is the dual role that women play as workers and carers of first resort, rather than women having to give up one role for the other. In the Philippines, the informalization of women's employment, which is most pronounced for mothers with minor children, meant that they continued to work informally even during the height of extreme lockdowns at the

onset of the pandemic. During these times, they simultaneously perform paid and unpaid work rather than trade off one for the other. In Germany, where the brunt of caregiving responsibilities are carried by women, our vignette experiment shows that lifting caregiving responsibilities towards a sick parent from working women increases fertility intentions by a significant magnitude (3 units on a 0-10 scale). Building on earlier arguments of Kotowska et al. (2008) and Doepke et al. (2023), this dissertation lends evidence to the continued blurring of the trade-off between paid employment and caregiving responsibilities among women. Circling back to Chapter 1, this dissertation argues that caregiving responsibilities, per se, constitute an uncertain labor market position as there is inherent uncertainty as to the length, amount, and intensity of caregiving required. When these caregiving needs are deemed excessive or unpredictable, these arguably affect the work domain as well.

6. **Novel methodological tools bring fresh perspectives to established ideas.** Each chapter contains a novel methodological contribution to their respective strands of the literature. In Chapter 2, the use of a two-way fixed effects difference-in-differences design leverages the regional variation in lockdown assignment to isolate the lockdown effect, while its militarized implementation that sealed regional borders automatically addresses spatial spillovers that are pervasive in area-based treatments. In Chapter 3, the two-step identification strategy combines a pre-processing non-parametric matching step with a parametric survival analysis (Cox Proportional Hazards modeling) step and addresses endogenous selection into initial employment conditions. This builds on prior fertility literature that has relied either solely on matching-based or solely on survival analysis-based designs. Chapter 4 employs cure fraction models to differentiate the fertility effect of social class between tempo (timing) and quantum (quantity), an approach that has roots also in epidemiological studies. Finally, Chapter 5 develops a survey experiment design that randomized respondents' exposure to a hypothetical couple's future employment and caregiving scenarios. This identification strategy addresses endogeneity issues arising from social desirability and sample selection that are potentially pervasive in pure survey-based designs exploring this topic.

6.2 A Subsequent Research Agenda

While the examination of the demographic consequences of employment uncertainty in this dissertation seems to outweigh the examination of its causes, a future research agenda that examines why people work in precarious positions in the first place is a promising path forward. Two extensions come to mind. First, the role of unions and the collective bargaining (CB) process in augmenting insufficient statutorily guaranteed minimums (e.g., wages, parental leaves) is well-documented in European and other welfare state contexts. Owing to a lack of complete population-representative data, this is widely understudied in alternative contexts with weaker union presence and social protection systems. Simply, we don't know what unions get out of the CB process that's more generalizable than insights from qualitative papers analyzing a very select subset of CB agreements (CBAs) in specific sectors. One key interest in this project could be how parental leave provisions are embedded in CBAs. Do CBAs augment parental leaves already guaranteed by law? In other words, if the law already guarantees X weeks of parental leave, do CBAs increase it to $X + n$? To preliminarily answer this question, we initiated the collection, systematic encoding, and analysis of the population of all CBAs filed in the Philippines from 2016 to 2021. Our initial estimates show that around 65% of CBAs contain reinforcing provisions that merely restate the statutory leave entitlements, while only 5% contain augmenting provisions that provide higher leaves. In terms of potential mechanisms, semi-structured interviews with union leaders lend support to the idea that where compliance and enforcement of family policy laws are perceived as weak, redundancy is as much of an objective as augmentation is in the collective bargaining process. This work is a natural extension of this dissertation project. Considering that none of the standalone articles or the developed schema in Chapter 1 discusses the effects of union membership and collective bargaining in reducing employment uncertainty, it is interesting to pursue a subsequent research agenda along these lines.

Relatedly, while working on these dissertation chapters, I more strongly developed the idea that employment uncertainty, regardless of how it is operationalized (see Table 1.1), is **nonrandom**. When we talk about fixed-term contracts or being unemployed or informally employed, these decisions are ultimately made by an agent who has the discretion and agency to do so. Indeed, employment conditions emanate from firm behavior. Large and powerful firms are increasingly subjecting their workers across all levels to restrictive covenants in employment contracts such as non-competes.

These provisions, which prohibit working for a competitor firm after the term of their employment, are outright curtailments of workers' mobility and freedom of occupation in favor of supposedly safeguarding corporate interests. Lately, this topic has been of interest among legal scholars and economists, evidenced by a surge of related papers in their top general and field journals in recent years. Sociologists and demographers have not explored this topic at all, despite these provisions potentially affecting outcomes of core interest to ours—employment trajectories, labor market stratification, and family formation.

Therefore, the claim that there are demographic consequences of employment uncertainty is, in my view, recurrent but perhaps incomplete. The unstable working conditions that may hamper family formation do not arise by chance or from workers' preferences but rather emanate from the actions and policies of firms, willingly or otherwise. In continental Europe, the outsized hiring and term-setting power possessed by firms, in the form of more concentrated labor markets, reduces the probability of hiring on a permanent contract and converting a temporary contract to a permanent one (Bassanini et al. 2022). However relevant it may seem, the direct link between firm behavior and demographic outcomes has been relatively **understudied**. Building on this dissertation project, a future research agenda that would highlight the **demographic and life course responses to firm behavior** using experimental (and quasi-experimental) methods.

Firms have the bandwidth and discretion to set working hours, establish flexibility in working time and locations, provide additional leaves where necessary, and even subsidize childcare. These are *de facto* and *direct* firm decisions that may improve the balance between work and family life. However, other *de jure* firm policies might have an *indirect* effect on demographic decisions. For instance, if the firm restricts one's future career opportunities, this might indirectly affect future demographic behavior as well.

Along these lines, a novel subsequent research agenda could look into the effects of firms' imposition of **restrictive covenants in employment contracts**—provisions that explicitly restrict workers from working or engaging with competitor firms within a certain period succeeding their employment with that firm. Recent work shows that the use of these provisions disproportionately restricts workers' future career opportunities, increases job search duration, and generates unnecessary turnover costs because of the need for retraining (Starr 2019; Starr et al. 2020). Especially worrying are recent

estimates that among private sector employees, 16% (Italy), 19% (Netherlands, UK), and 20% (US, Australia) are bound by some type of restrictive covenant. These outright curtailments of and exceptionalities in workers' mobility and freedom of occupation potentially have accompanying adverse effects on family formation intentions, mediated by a flatter career trajectory and reduced future employment opportunities.

This could also be tested using a survey experiment that randomizes respondents' exposure to contracts without and with various types of restrictive covenants through vignettes and subsequently measures family formation intentions (e.g., fertility, marriage). Through the conduct of an experiment, one could directly identify and estimate individual-level responses without encountering the problems of data incompleteness and recall bias that might be pervasive when analyzing this topic using survey-based designs. Vignette experiments have also been used to demonstrate the life course consequences of precarious working conditions (Auspurg and Gundert 2015) and more recently, the effects of uncertainty on fertility (see Chapter 5) and marriage (Gong and Wang 2022). Post-exposure, the main outcomes of interest are short- and long-term marital and fertility intentions and the expressed ideal age for these events. Additional questions on subsequent job search intentions and career expectations would allow for a formal mediation analysis that may unpack the mechanisms between firm-imposed contractual restrictions and demographic behavior.

6.3 Policy Implications and Limitations

In terms of policy implications, this dissertation, as a collective, can only speak in broad directional strokes. Notably, governments recognize that employment uncertainty is a pressing concern. From specific legislative packages like the 2022 CHIPS and Science Act in the United States to country-specific medium-term plans like the 2022-2028 Philippine Development Plan and even internationally agreed targets like the Sustainable Development Goals, improvements in employment conditions (and a reduction in uncertainty, more broadly) are deeply incorporated. Therefore, there doesn't seem to be a lack of *direction* and *consensus* on this issue—policymakers explicitly strive to foster employment stability. A multi-pronged approach is needed to achieve such aims.

First, in many contexts, strengthening **social protection systems** and improving targeting towards the most vulnerable groups in the labor market is necessary. In

the Philippine case, as shown in Chapter 2, among households with a non-working head, only 0.07% and 0.13% have received job search assistance and livelihood support, respectively. To reiterate and put it in context, this is less than 1/10th of 1 percent of all households in the Philippines with a non-working household head. To say that social protection systems are barely existent, if at all, in some contexts is not an understatement. Reducing employment uncertainty thus necessitates continuously fine-tuning social safety nets.

Second, Chapter 4 highlights the role that upward social mobility (e.g., from semi/unskilled occupation to a service class occupation, for instance) can play in increasing higher-order births, especially among men. Governments can play a pivotal role in this regard by investing in skills development and lifelong learning initiatives tailored to the demands of the labor market. Some cases of successful upskilling programs include that of Singapore. Their SkillsFuture initiative includes the provision of subsidies for training courses and grants for skills acquisition, thereby enabling upward class mobility and cushioning their local labor market from shocks and other technological disruptions.

Finally, extending the application of labor market regulations to nonstandard employment categories also ensures that rules aimed at worker protection do not discriminate based on type. For instance, in the EU, the recently adopted Platform Workers Directive aims to clarify the classification of platform workers from self-employed or independent contractors to direct employees if their actual working conditions meet certain criteria. This law, therefore, reduces the uncertainty experienced by platform workers by making them eligible for the same protection and benefits received by regular and permanent employees. In the Philippines, the 2013 Domestic Workers Act aimed at improving the minimum standards of remuneration and protection for domestic workers and caregivers, including, most notably, the need for their employers to contribute to their social insurance premiums. Therefore, this law removed part of the uncertainty of being informally employed by covering them under the country's national social protection system.

While each dissertation chapter has identified its respective limitations, three overarching ones are worth noting and reflecting upon. First, each chapter presents estimates that are arguably local and context-specific. As the literature review section has shown, for instance, there are certain peculiarities in each country's context that do not apply elsewhere. For example, Chapters 3 to 5 are all about Germany— a

country that has a marked East/West distinction in social and economic life that has not yet fully converged even after three decades since reunification (Goldstein and Kreyenfeld 2011). While these regional disparities have been “controlled for” in each of these articles, the coefficient sizes and significance for East Germany indicators in the regression models in Chapters 3 and 4 reveal that it’s still a relevant component in explaining fertility dynamics in Germany. Therefore, beyond the macro-level analysis in Chapter 1, one should not infer that the local and context-specific effects detected in the stand-alone chapters are automatically true in other country contexts.

Further, the dissertation has only tangentially touched on the underlying theories of change that affect the relationship between the respective “treatments” and outcomes. In Chapter 2, while the effects are shown to be driven by both compositional and survivalist factors, this is relatively underdeveloped since the data used (repeated cross-sections) is not suitable to track individual transitions to different labor market states. Similarly, in Chapter 3, while alternative specifications robustly demonstrate the effects of initial conditions on subsequent fertility, there is little discussion as to the underlying channels (e.g., what happened between entry and fertility?) owing in part to the relatively small and unbalanced sample in the study. Similarly, in Chapter 5, while the effect of future caregiving and employment uncertainties on fertility intentions are tested for heterogeneities by gender and employment characteristics, there may be other personal psychological factors that might also explain the relationship. Moving forward, subsequent work that extends these stand-alone chapters and looks into specific mechanisms and more tests for heterogeneity is likely to be insightful.

Finally, the operationalization of some of the variables used in the chapters might benefit from fine-tuning or comparison with alternative measures. For instance, in Chapter 3, the recession treatment is a dummy variable that denotes whether the World Bank and FRED databases classify a certain year as a recession year. However, Gandrud and Hallerberg (2015) make a point that these dummy indicators do not capture the severity of crises nor do they capture instances when early signs of economic stress were avoided or managed. Future work that analyzes how workers respond to brewing economic stress is a promising way forward. Among this novel strand of the literature is recent work showing that the quantity and emphasis of real-time media coverage of a country’s (negative) economic situation are negatively associated with fertility behavior (Guetto et al. 2023b).

In Chapter 5, on the other hand, the caregiving uncertainty treatment is highly

gendered—fully allocating the care responsibilities unto the female partner whereas they can be shared in reality between the couple. Thus, further survey experiments can better map out various caregiving constellations that reflect actual divisions of paid and unpaid work within families.

In sum, this dissertation demonstrates that employment uncertainty is multi-dimensional and simultaneously determined and deterministic. People's labor market positions cannot be described in a black-and-white manner, nor do they solely predict life course outcomes by themselves. However, they are a focal component of one's identity, perhaps even central for some. This necessitates our continued attention to and study of these topics in light of changing times and recurrent crises.

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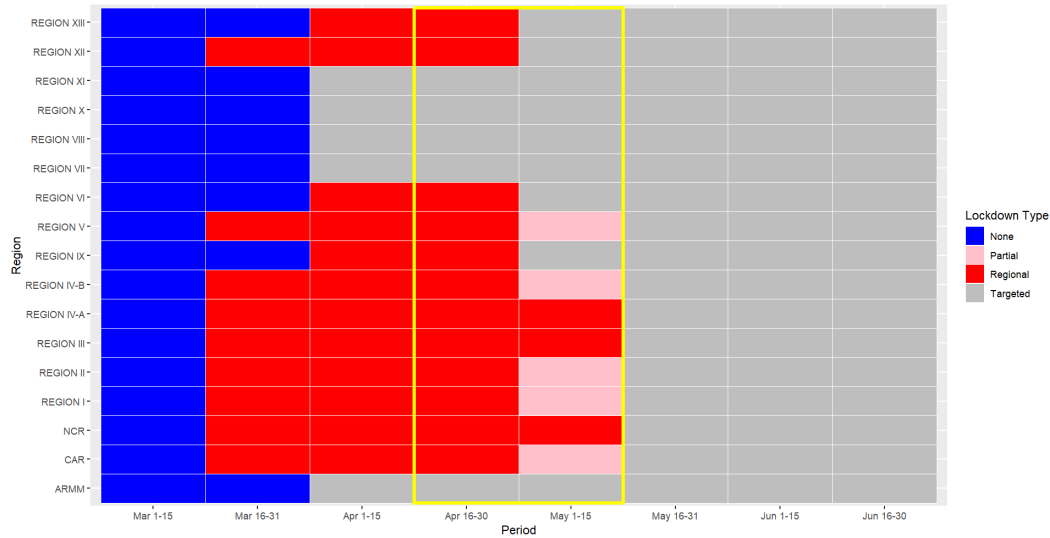
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Appendix A

Appendix for Chapter 2

A.1 Supplementary Figures

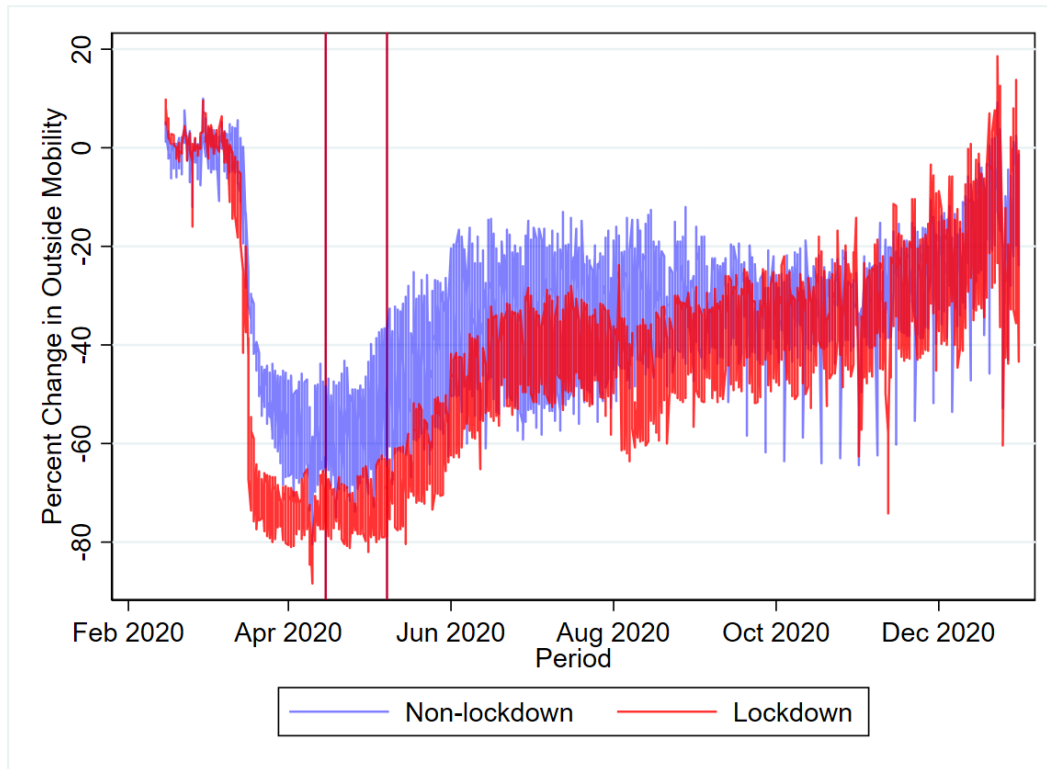
Figure A.1: Lockdown timeline in the Philippines at the onset of COVID



Source: Pronouncements of the Inter-Agency Task Force for COVID-19 (IATF) and online reports

Note: This timeline was developed from the pronouncements of the Inter-Agency Task Force for COVID-19 (IATF), local level executive orders and ordinances, and online news reports. A blue cell (“none”) implies that within that period (or the most part of it), there were no region-wide lockdowns. Pink cells (“partial”) denote a region-wide partial lockdown, which is a less stringent classification than the extreme lockdown. A region is considered partially locked down if the entire region was placed under a classification below the extreme form of lockdown (Enhanced Community Quarantine, or ECQ) including Modified Enhanced CQ, General CQ, and Modified General CQ. A grey cell (“targeted”) means that certain provinces or cities within that region were under extreme lockdowns while others were not. Finally, the red cells pertain to areas that underwent the most extreme form of lockdown (ECQ). As mentioned in the main text, our treatment units are regions that were consistently under extreme lockdown during the survey period (NCR, Regions III, and IV-A) and our control units are those that consistently did not impose regional lockdowns during the same period in favor of targeted ones (Regions 7, 8, 10, 11, and ARMM). The second quarter LFS was held between the second half of April and the first half of May (yellow box). In the latter part of the year, the government pursued more localized and targeted solutions rather than blanket and wide regional shutdowns.

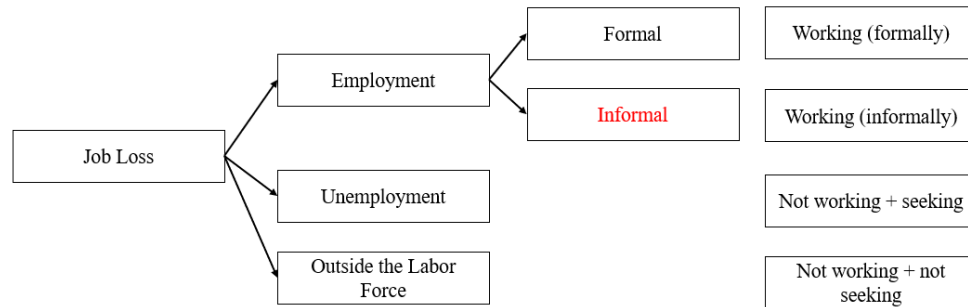
Figure A.2: Daily Average Mobility Changes by Lockdown Status, 2020



Source: Google (2021) accessed 05 December 2021

Note: Using data from Google LLC (2021) Community Mobility Reports Data, we plot the daily average mobility changes in 2020 in lockdown and non-lockdown regions to show that during the survey period, there is a clear and observable distinction in mobility changes between treatment and control units. It is evident that lockdown regions had a consistently larger decline in mobility relative to baseline during the survey period. Note that “non-lockdown” does not necessarily mean no mobility restrictions at all were present. It simply means that there was no regional-wide lockdown and any mobility restrictions imposed were targeted and localized. Consistent with what the timeline in Figure A1.1. shows, these mobility change differentials dissipate in the latter part of the year as lockdowns became more targeted and nuanced throughout the country.

Figure A.3: Labor Market Transitions Succeeding Job Loss Episodes



Note: In this framework, we illustrate potential pathways succeeding a job loss. In most of the labor literature on the implications and effects of the COVID-19 pandemic, the focus has been on the extensive and intensive margin—increase in unemployment and reduction in work hours, among others. In this paper, we focus specifically on the succeeding branch—whether those employed during the pandemic were either formally or informally employed.

Supplementary Tables

Table A.1: Analytical Sample Derivation from the LFS

	All Sample	Non-Lockdown Regions	Lockdown Regions
LFS Observations (16 quarters)	2,868,994	815,562	773,258
Retain regions of interest	1,427,254	815,562	773,258
Retain age =15	1,986,648	545,662	562,794
Retain employed with work	579,258	299,874	292,069
Number of periods	16	16	16
Number of groups (regions)	8	5	3

Note: In this analysis, we use 16 quarters of the Labour Force Survey, a quarterly survey representative of the Philippine population on matters pertaining to labor and employment. The final analytical sample has more than 500,000 observations from 8 regions, of which 3 were under extreme regional-wide lockdowns during the survey period.

Table A.2: Count of Formal and Informal Employment by lockdown status and gender, 2016 Q3 to 2020 Q2

Period	Lockdown Areas							Non-lockdown Areas						
	Formal	Males Informal	Sum	Formal	Females Informal	Sum	All Total	Formal	Males Informal	Sum	Formal	Females Informal	Sum	All Total
2016 Q3	0.855	0.145	10,808.29	0.885	0.115	12,106.41	22,914.69	0.760	0.240	7,627.67	0.735	0.265	7,153.61	14,781.28
2016 Q4	0.857	0.143	11,070.35	0.883	0.117	11,492.86	22,563.21	0.765	0.235	7,906.20	0.741	0.259	6,731.67	14,637.87
2017 Q1	0.856	0.144	10,757.71	0.891	0.109	11,789.33	22,547.04	0.763	0.237	7,286.55	0.758	0.242	6,460.23	13,746.78
2017 Q2	0.845	0.155	10,758.20	0.882	0.118	12,276.70	23,034.89	0.756	0.244	7,352.53	0.738	0.262	7,051.20	14,403.73
2017 Q3	0.864	0.136	11,181.82	0.893	0.107	11,675.42	22,857.23	0.762	0.238	7,302.94	0.759	0.241	6,742.52	14,045.46
2017 Q4	0.865	0.135	11,287.24	0.886	0.114	12,076.20	23,363.44	0.764	0.236	7,896.71	0.738	0.262	7,213.45	15,110.16
2018 Q1	0.856	0.144	11,103.06	0.881	0.119	12,273.99	23,377.05	0.753	0.247	7,646.07	0.735	0.265	7,217.99	14,864.05
2018 Q2	0.861	0.139	11,165.62	0.881	0.119	11,780.21	22,945.83	0.750	0.250	7,543.81	0.724	0.276	6,915.23	14,459.04
2018 Q3	0.867	0.133	11,205.00	0.887	0.113	11,849.66	23,054.66	0.766	0.234	7,775.11	0.749	0.251	6,752.36	14,527.47
2018 Q4	0.866	0.134	11,307.95	0.884	0.116	12,158.40	23,466.35	0.768	0.232	7,756.88	0.747	0.253	6,988.79	14,745.67
2019 Q1	0.850	0.150	11,435.66	0.879	0.121	12,000.30	23,435.96	0.764	0.236	7,942.35	0.739	0.261	7,071.97	15,014.32
2019 Q2	0.847	0.153	11,448.00	0.861	0.139	12,550.48	23,998.48	0.746	0.254	7,978.02	0.724	0.276	7,445.80	15,423.82
2019 Q3	0.851	0.149	11,660.32	0.877	0.123	12,411.49	24,071.82	0.743	0.257	8,191.46	0.744	0.256	7,628.52	15,819.98
2019 Q4	0.863	0.137	11,752.58	0.869	0.131	12,471.56	24,224.14	0.769	0.231	8,100.92	0.737	0.263	7,670.15	15,771.07
2020 Q1	0.862	0.138	11,924.46	0.776	0.224	12,230.98	24,155.44	0.881	0.119	8,389.94	0.718	0.282	7,470.82	15,860.75
2020 Q2	0.873	0.127	3,454.76	0.691	0.309	7,746.01	11,200.77	0.898	0.102	3,050.84	0.681	0.319	4,469.70	7,520.54
2020Q2 vs. 2020Q1 (%)	-70.64	-73.42	-71.03	-34.91	-49.31	-36.67	-53.63	-67.63	-49.80	-63.64	-42.33	-33.99	-70.64	-73.42
2020Q2 vs. average (%)	-68.76	-72.65	-69.31	-34.32	-47.24	-35.86	-52.00	-64.39	-49.34	-60.79	-39.29	-30.69	-68.76	-72.65

Note: The row 2020Q2 vs 2020Q1 refers to the percentage change of employment in 2020Q2 relative to 2020Q1 for each column. The row 2020Q2 vs average refers to the percentage change of employment in 2020Q2 relative to the average employment from 2016Q3 to 2020Q1. The estimate can be interpreted as follows: for informally employed males in lockdown regions, informal employment decreased by 73.4 percent relative to the previous quarter and by 72.7 percent relative to the average of previous quarters. These figures are weighted estimates from the Labour Force Survey.

Table A.3: Robustness Analysis across Varying Timeline Specifications

VARIABLES	(1) 2016 Q3+	(2) Q2s in 2016Q3+	(3) 2019-20 Q2s	(4) 2019 Q3+
Lockdown area	-0.052*** (0.002)	-0.080*** (0.005)	-0.090*** (0.008)	-0.062*** (0.005)
Lockdown period	0.000 (0.004)	-0.012** (0.004)	-0.015*** (0.004)	-0.003 (0.004)
Lockdown area*period	0.017** (0.006)	0.020** (0.006)	0.027*** (0.007)	0.025*** (0.006)
Age	0.004*** (0.000)	0.004*** (0.000)	0.003*** (0.000)	0.004*** (0.000)
Basic education	0.010*** (0.002)	0.012*** (0.003)	0.024*** (0.005)	0.023*** (0.003)
Some post-secondary ed	0.067*** (0.002)	0.056*** (0.004)	0.054*** (0.007)	0.070*** (0.005)
Secondary education	0.080*** (0.001)	0.090*** (0.003)	0.105*** (0.004)	0.097*** (0.003)
Female	0.130*** (0.001)	0.146*** (0.002)	0.156*** (0.004)	0.134*** (0.002)
Region FE	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes
Constant	-0.036*** (0.003)	0.009 (0.006)	0.005 (0.009)	-0.040*** (0.006)
Observations	579,258	128,380	54,953	132,780
R-squared	0.057	0.059	0.066	0.062

Robust standard errors in parentheses

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.1$.

Note: (1) is a replication of the preferred specification in the main results for the whole sample, including region and period fixed effects and a female dummy. Column (2) only contains observations from Q2 surveys from 2017-2020 to show that the estimates are robust to potential quarterly seasonality. Column (3) contains only the observations from 2019 and 2020 Q2 to show the effect sizes when the control unit is only the same quarter of the year prior, thereby eliminating most of the time-varying confounding that is present when considering multiple periods. Finally, column (4) retains observations after 2019Q3, the quarter immediately succeeding the 2019 national and local elections.

Table A.4: Robustness Analysis across Varying Age Restrictions

VARIABLES	(1) 15+	(2) 15-64	(3) 18+	(4) 18-64
Lockdown area	-0.052*** (0.002)	-0.060*** (0.002)	-0.053*** (0.002)	-0.061*** (0.002)
Lockdown period	0.000 (0.004)	0.003 (0.004)	0.000 (0.004)	0.003 (0.004)
Lockdown area*period	0.017** (0.006)	0.014* (0.006)	0.018** (0.006)	0.014* (0.006)
Age	0.004*** (0.000)	0.004*** (0.000)	0.004*** (0.000)	0.004*** (0.000)
Basic education	0.010*** (0.002)	0.012*** (0.002)	0.009*** (0.002)	0.011*** (0.002)
Some post-secondary ed	0.067*** (0.002)	0.068*** (0.002)	0.053*** (0.002)	0.054*** (0.002)
Secondary education	0.080*** (0.001)	0.079*** (0.001)	0.079*** (0.001)	0.078*** (0.001)
Female	0.130*** (0.001)	0.125*** (0.001)	0.129*** (0.001)	0.123*** (0.001)
Region FE	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes
Constant	-0.036*** (0.003)	-0.034*** (0.003)	-0.047*** (0.003)	-0.046*** (0.003)
Observations	579,258	553,051	568,952	542,745
R-squared	0.057	0.052	0.059	0.054

Robust standard errors in parentheses

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.1$.

Note: (1) is a replication of the preferred specification in the main results for the whole sample, including region and period fixed effects and a female dummy. The rest of the columns have varied age restrictions namely (2) 15-64; (3) 18 and above; and (4) 18-64. Since the national definition of the labor force includes individuals above 15, we adopt this definition in the main text.

A.2 Alternative Informal Employment Definitions

There exist various definitions of informal employment in the literature given that there are differences in country contexts, data availability, and research questions. Benanav (2019) offers a historical account of the evolving definitions of “informal employment” and “employment in the informal sector” led by the ILO and the contesting interpretations and operationalizations of these measures. Our definition of informal employment is largely based on the ILO guidance on informal employment agreed upon at the 17th International Conference of Labour Statisticians (Hussmanns 2004) and encompasses all non-professional, own-account, or contributing family workers outside agriculture and government. For context, below are examples of some definitions in the recent literature:

Author and Year	Journal/Outlet	Informal Employment Definition*	Country Context
(Villanueva & Lin, 2020)	Social Forces	Non-contributor to the pension system	5 Latin American countries, using household surveys
(Comola & De Mello, 2011)	Review of Income and Wealth	all individuals who are self-employed, employers, or family/unpaid workers.	Indonesia, using Labour Force Surveys
(Dabla-Norris et al., 2020)	IMF Working Paper Series	all employers working in non-registered firms, own-account workers working in the informal or non-farming household sectors; all family contributing workers; members of producer’s cooperatives working in the informal sector; and wage workers either not having a contract of more than three months or, if they have such a contract, not participating in social insurance	Vietnam, using Labour Force Surveys
(Mendoza & Jara, 2020)	Regional Science Policy and Practice	self-employed workers and employers in the informal sector, unpaid household and non-household workers, and unpaid worker or day laborer assistant, and, salaried workers who are not affiliated to social security.	Ecuador, using Labour Force Surveys
(Cabegin, 2008)	Philippine Journal of Labour and Industrial Relations	wage work with no employment contract or contracted only by verbal agreement; jobs that are outside the coverage of national labour legislation and social security regulation; or those that are in law or in practice not subject to income taxation	Philippines, using Informal Employment Survey
(Pecha Garzón, 2017)	IDB Working Paper Series	own-account, farm workers and contributing family workers (with some restrictions based on size of the employer)	Jamaica, using Labour Force Surveys

*Note: Some may be direct quotes from the publication

From the table above, it is evident that there is no consensus and universal

approach in defining informal employment. In many papers cited above, informal employment has a social security-based definition (whether one is a contributor to the social security system). Within mainland Southeast Asia, there are operational differences in the definition of informal employment with Thailand incorporating the social security-based definition, whereas Myanmar and Lao PDR incorporating both social security-based and worker-based definitions (Ducanes et al. 2023). This does not mean that one approach, more so one paper, is superior or necessarily correct. On the contrary, highlighting and scrutinizing alternative, context-based definitions is critical in advancing research on informal employment.

Our definition is different than most ones in the literature that are able to define informal employment not only from the workers' job type/class (as we do in this paper) but also based on the legal structure of their employer (formal/informal enterprise), their eligibility for paid leaves (eligible/ineligible), and their social security contributions (covered/uncovered). A more holistic and ideal definition of informal employment would account for all these factors. The PH Labor Force Survey unfortunately does not contain information about social security. Notwithstanding, our objective in the paper is not to provide the “closest to the true” estimate of informal employment in the Philippines—such an estimate does not exist in the literature nor in official reports to the best of our knowledge. Rather, we aim to **demonstrate the robustness of our positive, significant, and gendered lockdown effect across various definitions of informal employment.**

With this in mind, Cabegin (2008) argues that “[job-based] figures of informal employment are underestimated in that they do not include workers that are informally employed in the formal sector and in private households”. We agree with this argument. However, for the purposes of this paper, the preferred definition we adopt is both class- (or job-) and sector-based with an **explicit preference** for an **underestimate**. An underestimated informal employment rate means that the estimates we present are likely **lower bounds** such that the **true lockdown effect might even be larger.**

We demonstrate this empirically in the succeeding tables of this section of the Supplementary Material. In Table A3.1 and A3.2, we show that across all relevant and alternative definitions of informal employment, the lockdown effect is positive and strongly significant and across all definitions, the gendered informalization finding holds true. For ease of reference, we summarize these estimates below—definitions that yield a higher IER also yield higher magnitudes of the lockdown effect. Thus, our preferred

definition which is presented in the main text is in light of our aim to **demonstrate credible yet modest estimates of the lockdown effect**. Our results also hold even when merely using a class-based definition of informal employment and controlling for or omitting agriculture and government workers instead (see Table A3.3).

Table A.5: Alternative Sector- and Class-Based Definitions of Informal Employment

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Lockdown area	-0.052*** (0.002)	0.005* (0.002)	-0.437*** (0.003)	0.009*** (0.002)	-0.455*** (0.003)	-0.454*** (0.003)	-0.437*** (0.003)
Lockdown period	0.000 (0.004)	-0.006 (0.004)	0.071*** (0.005)	-0.007† (0.004)	0.065*** (0.005)	0.065*** (0.005)	0.071*** (0.005)
Lockdown area*period	0.017** (0.006)	0.058*** (0.006)	0.030*** (0.007)	0.058*** (0.007)	0.034*** (0.007)	0.034*** (0.007)	0.030*** (0.007)
Age	0.004*** (0.000)	0.004*** (0.000)	0.007*** (0.000)	0.004*** (0.000)	0.008*** (0.000)	0.008*** (0.000)	0.007*** (0.000)
Basic education	0.010*** (0.002)	0.093*** (0.002)	0.290*** (0.002)	0.074*** (0.002)	0.270*** (0.002)	0.270*** (0.002)	0.289*** (0.002)
Some post-secondary ed	0.067*** (0.002)	0.126*** (0.002)	0.214*** (0.002)	0.115*** (0.002)	0.204*** (0.002)	0.204*** (0.002)	0.214*** (0.002)
Secondary education	0.080*** (0.001)	0.149*** (0.001)	0.204*** (0.001)	0.137*** (0.001)	0.188*** (0.002)	0.188*** (0.002)	0.204*** (0.001)
Female	0.130*** (0.001)	0.228*** (0.001)	0.191*** (0.001)	0.228*** (0.001)	0.180*** (0.001)	0.180*** (0.001)	0.191*** (0.001)
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	-0.036*** (0.003)	-0.116*** (0.003)	0.191*** (0.004)	-0.121*** (0.003)	0.182*** (0.004)	0.182*** (0.004)	0.191*** (0.004)
Observations	579,258	579,258	579,258	579,258	579,258	579,258	579,258
R-squared	0.057	0.105	0.195	0.105	0.202	0.202	0.195

Robust standard errors in parentheses

*** p<0.001, ** p<0.01, * p<0.05, † p<0.1.

Note: The columns show the DiD regression results under various definitions of informal employment. (1) replicates the original and preferred model specification presented in the main text where informal employment is defined as non-professional own-account workers outside agriculture and government. The alternative definitions are as follows: (2) non-professional, own-account, and household workers outside agriculture and government; (3) non-professional own-account, and household workers outside government; (4) non-professional own-account, household, and farm workers outside agriculture and government; (5) non-professional own-account, household, and farm workers outside government; (6) non-professional, own-account, household and farm workers; (7) non-professional, own-account, and household workers. Note that these different definitions could be made based on one's assumptions about the nature of informal work across different occupational classes. In this paper, we adopt the most conservative definition of informal employment that may even understate the true lockdown effect. Even with this “lower-bound” estimate, however, we find a positive and significant effect.

Table A.6: Alternative Sector- and Class-Based Definitions of Informal Employment, by Gender

VARIABLES	(1)		(2)		(3)		(4)		(5)		(6)		(7)	
	M	F	M	F	M	F	M	F	M	F	M	F	M	F
Lockdown area	-0.031*** (0.002)	-0.108*** (0.005)	-0.009*** (0.002)	0.023*** (0.005)	-0.514*** (0.003)	-0.260*** (0.005)	-0.004 (0.003)	0.024*** (0.006)	-0.533*** (0.003)	-0.280*** (0.005)	-0.533*** (0.003)	-0.280*** (0.005)	-0.514*** (0.003)	-0.260*** (0.005)
Lockdown period	-0.017*** (0.004)	0.029*** (0.008)	-0.019*** (0.004)	0.015† (0.008)	0.075*** (0.006)	0.062*** (0.008)	-0.020*** (0.005)	0.013 (0.008)	0.069*** (0.006)	0.057*** (0.007)	0.069*** (0.006)	0.056*** (0.007)	0.075*** (0.006)	0.062*** (0.008)
Lockdown area*period	0.003 (0.007)	0.022* (0.011)	0.018* (0.007)	0.090*** (0.011)	0.004 (0.009)	0.052*** (0.010)	0.016* (0.008)	0.092*** (0.011)	0.009 (0.009)	0.054*** (0.010)	0.009 (0.009)	0.054*** (0.010)	0.004 (0.009)	0.052*** (0.010)
Age	0.003*** (0.000)	0.006*** (0.000)	0.003*** (0.000)	0.005*** (0.000)	0.006*** (0.000)	0.007*** (0.000)	0.003*** (0.000)	0.006*** (0.000)	0.007*** (0.000)	0.008*** (0.000)	0.007*** (0.000)	0.008*** (0.000)	0.006*** (0.000)	0.007*** (0.000)
Basic education	-0.032*** (0.002)	0.050*** (0.003)	-0.024*** (0.002)	0.244*** (0.003)	0.170*** (0.002)	0.449*** (0.003)	-0.046*** (0.002)	0.228*** (0.003)	0.148*** (0.002)	0.434*** (0.003)	0.148*** (0.002)	0.433*** (0.003)	0.170*** (0.002)	0.448*** (0.003)
Some post-secondary ed	0.018*** (0.002)	0.119*** (0.003)	0.029*** (0.002)	0.238*** (0.004)	0.130*** (0.003)	0.311*** (0.004)	0.014*** (0.003)	0.231*** (0.004)	0.116*** (0.003)	0.306*** (0.004)	0.116*** (0.003)	0.305*** (0.004)	0.130*** (0.003)	0.311*** (0.004)
Secondary education	0.024*** (0.002)	0.141*** (0.002)	0.035*** (0.002)	0.280*** (0.002)	0.091*** (0.002)	0.339*** (0.002)	0.018*** (0.002)	0.273*** (0.002)	0.069*** (0.002)	0.332*** (0.002)	0.069*** (0.002)	0.331*** (0.002)	0.090*** (0.002)	0.339*** (0.002)
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	0.047*** (0.004)	0.026*** (0.007)	0.040*** (0.004)	-0.047*** (0.007)	0.364*** (0.005)	0.123*** (0.006)	0.039*** (0.004)	-0.057*** (0.007)	0.345*** (0.005)	0.124*** (0.006)	0.345*** (0.005)	0.124*** (0.006)	0.364*** (0.005)	0.124*** (0.006)
Observations	353,121	226,137	353,121	226,137	353,121	226,137	353,121	226,137	353,121	226,137	353,121	226,137	353,121	226,137
R-squared	0.014	0.066	0.014	0.109	0.155	0.242	0.018	0.109	0.168	0.251	0.168	0.250	0.155	0.241

Robust standard errors in parentheses

*** p<0.001, ** p<0.01, * p<0.05, † p<0.1.

Note: The columns show the DiD regression results under various definitions of informal employment, estimated by gender. Similar to Table A3.1 (1) replicates the original and preferred model specification presented in the main text where informal employment is defined as non-professional own-account workers outside agriculture and government. The alternative definitions are as follows: (2) non-professional, own-account, and household workers outside agriculture and government; (3) non-professional own-account, and household workers outside government; (4) non-professional own-account, household, and farm workers outside agriculture and government; (5) non-professional own-account, household, and farm workers outside government; (6) non-professional, own-account, household and farm workers; (7) non-professional, own-account, and household workers. Note that these different definitions could be made based on one's assumptions about the nature of informal work across different occupational classes. In this paper, we adopt the most conservative definition of informal employment that may even understate the true lockdown effect. Even with this "lower-bound" estimate, however, we find a positive and significant effect.

Table A.7: Alternative Class-Based Definitions of Informal Employment, with Sector Controls

VARIABLES	Control for agriculture and public sector						Omit the entire agriculture sector					
	All	(1) Males	Females	All	(2) Males	Females	All	(3) Males	Females	All	(4) Males	Females
Lockdown area	-0.466*** (0.002)	-0.473*** (0.003)	-0.384*** (0.004)	-0.456*** (0.003)	-0.468*** (0.003)	-0.362*** (0.004)	-0.304*** (0.004)	-0.313*** (0.005)	-0.281*** (0.005)	-0.301*** (0.004)	-0.312*** (0.005)	-0.275*** (0.005)
Lockdown period	0.070*** (0.004)	0.061*** (0.006)	0.068*** (0.007)	0.077*** (0.005)	0.070*** (0.006)	0.073*** (0.007)	0.053*** (0.006)	0.038*** (0.008)	0.056*** (0.008)	0.054*** (0.006)	0.035*** (0.008)	0.059*** (0.008)
Lockdown area*period	0.046*** (0.007)	0.025** (0.009)	0.045*** (0.009)	0.040*** (0.007)	0.018* (0.009)	0.042*** (0.009)	0.032*** (0.008)	0.014 (0.011)	0.039*** (0.010)	0.032*** (0.008)	0.017 (0.011)	0.036*** (0.010)
Age	0.009*** (0.000)	0.008*** (0.000)	0.009*** (0.000)	0.007*** (0.000)	0.006*** (0.000)	0.008*** (0.000)	0.008*** (0.000)	0.007*** (0.000)	0.009*** (0.000)	0.007*** (0.000)	0.006*** (0.000)	0.008*** (0.000)
Basic education	0.161*** (0.002)	0.042*** (0.002)	0.329*** (0.003)	0.195*** (0.002)	0.081*** (0.002)	0.356*** (0.003)	0.153*** (0.002)	0.002 (0.003)	0.340*** (0.003)	0.181*** (0.002)	0.032*** (0.003)	0.365*** (0.003)
Some post-secondary ed	0.128*** (0.002)	0.050*** (0.003)	0.219*** (0.003)	0.148*** (0.002)	0.074*** (0.003)	0.232*** (0.003)	0.128*** (0.002)	0.030*** (0.003)	0.231*** (0.004)	0.145*** (0.002)	0.051*** (0.003)	0.242*** (0.004)
Secondary education	0.108*** (0.001)	0.007*** (0.002)	0.238*** (0.002)	0.132*** (0.001)	0.037*** (0.002)	0.254*** (0.002)	0.135*** (0.002)	0.013*** (0.002)	0.268*** (0.003)	0.153*** (0.002)	0.037*** (0.002)	0.281*** (0.003)
Female	0.199*** (0.001)			0.206*** (0.001)			0.260*** (0.001)			0.261*** (0.001)		
Agriculture	0.052*** (0.001)	0.163*** (0.002)	-0.156*** (0.002)	0.025*** (0.001)	0.129*** (0.002)	-0.167*** (0.002)						
Public sector	-0.432*** (0.001)	-0.331*** (0.002)	-0.508*** (0.002)	-0.400*** (0.001)	-0.297*** (0.002)	-0.478*** (0.002)	-0.420*** (0.001)	-0.316*** (0.002)	-0.490*** (0.002)	-0.391*** (0.001)	-0.287*** (0.001)	-0.462*** (0.002)
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	0.240*** (0.003)	0.323*** (0.004)	0.301*** (0.005)	0.254*** (0.004)	0.351*** (0.004)	0.296*** (0.006)	0.079*** (0.005)	0.240*** (0.007)	0.204*** (0.006)	0.085*** (0.005)	0.245*** (0.007)	0.210*** (0.006)
Observations	579,258	353,121	226,137	579,258	353,121	226,137	407,076	227,952	179,124	407,076	227,952	179,124
R-squared	0.266	0.229	0.353	0.248	0.202	0.335	0.259	0.104	0.368	0.251	0.095	0.353

Robust standard errors in parentheses

*** p<0.001, ** p<0.01, * p<0.05, † p<0.1.

Note: The columns show the DiD regression results under various definitions of informal employment with sector controls. In lieu of making sector-based assumptions of informality in the actual definition of informal employment, we define informal employment for cols (1) and (3) as non-professional, own-account, household and farm workers; and for cols (2) and (4) non-professional, own-account, and household workers. In the left-hand side of the table, columns (1) and (2), we include dummies for agriculture and public sector/government workers. In the right-hand side, however, we address the difficulty in classifying agricultural workers as formal or informal simply based on the variables included in the LFS. Thus so, for columns (3) and (4) we show estimates for models where all workers in the agricultural sector are omitted from the sample, thereby eliminating any sector-specific bias in our estimates. Our main results hold, and are even larger in magnitude, under these alternative specifications.

A.3 Testing the “Survivalist Motive” Mechanism

The survivalist motive of informal work, particularly relevant among developing countries, posits that absent social assistance mechanisms and precautionary savings, workers who are affected by the lockdown are unable to afford unemployment or inactivity and are thus constrained to seek gainful employment informally. To test this mechanism of whether lockdowns increase the probability of being in paid informal employment relative to unemployment, we employ a multinomial logistic regression model where workers take one of the following employment categories—formal employment, informal employment, inactivity (outside the labor force), and unemployment. This approach is useful to estimate the probability of these outcomes, expressing them relative to other outcomes (Hosmer et al. 2013). Retaining the same explanatory variables from equation (2) including regional and period fixed effects, the equation to test this mechanism is parsimoniously specified as:

$$\ln \frac{Pr(y = K - 1)}{Pr(y = K)} = \alpha + \beta LA_g + \gamma LP_t + \delta_{DD}(LA_g * LP_t) + \sum_{j=1}^n \theta_j X_{jigt} + \sigma_g + \tau_t + \epsilon_{igt}, \quad (\text{A.1})$$

where K is the base outcome and K-1 as the alternative outcomes. The relative risk ratios (RRR) are the exponentiated multinomial logit coefficients from the equation above and are reported in the table. RRRs compare the risk of individuals in lockdown relative to those in non-lockdown regions to be in one outcome relative to the base outcome, which is defined as “unemployment”.

Results suggest that for individuals in lockdown relative to non-lockdown regions, the relative risk of being in informal employment relative to unemployment is increased by 14 percent for males. Simply put, male workers in lockdown regions are more likely than those in non-lockdown regions to be informally employed relative to being unemployed. As for females, coefficients for the female dummy in the full sample are telling—the relative risk of females being in informal employment relative to unemployment is double relative to males. These findings partly support the hypothesized mechanism, and Table A4 shows the full results.

Table A.8: Survivalist Motive—Multinomial Logistic Regression Results

VARIABLES	All			Males			Females		
	Formal	Informal	Outside LF	Formal	Informal	Outside LF	Formal	Informal	Outside LF
Lockdown area	1.214*** (0.022)	0.844*** (0.019)	0.660*** (0.013)	0.883*** (0.020)	0.675*** (0.020)	1.514*** (0.056)	2.522*** (0.076)	1.241*** (0.042)	0.815*** (0.023)
Lockdown period	0.193*** (0.005)	0.239*** (0.008)	0.308*** (0.009)	0.199*** (0.006)	0.236*** (0.011)	0.288*** (0.015)	0.181*** (0.008)	0.242*** (0.012)	0.311*** (0.013)
Lockdown area*period	0.547*** (0.016)	1.059 (0.039)	1.070* (0.034)	0.505*** (0.018)	1.139* (0.058)	0.979 (0.055)	0.625*** (0.029)	0.994 (0.053)	1.067 (0.047)
Age	1.026*** (0.000)	1.054*** (0.000)	1.087*** (0.000)	1.022*** (0.000)	1.043*** (0.001)	1.132*** (0.001)	1.037*** (0.001)	1.068*** (0.001)	1.082*** (0.001)
Basic education	1.216*** (0.017)	1.260*** (0.021)	1.336*** (0.021)	1.403*** (0.024)	1.027 (0.023)	0.541*** (0.013)	0.872*** (0.021)	1.266*** (0.034)	1.556*** (0.039)
Some post-secondary ed	1.019 (0.017)	1.627*** (0.033)	1.339*** (0.025)	1.135*** (0.024)	1.330*** (0.037)	0.693*** (0.022)	0.836*** (0.022)	1.748*** (0.052)	1.506*** (0.041)
Secondary education	0.976* (0.011)	1.699*** (0.023)	1.554*** (0.019)	1.116*** (0.016)	1.373*** (0.025)	0.653*** (0.013)	0.785*** (0.014)	1.874*** (0.037)	1.845*** (0.033)
Female	1.003 (0.009)	2.503*** (0.027)	9.325*** (0.099)						
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	2.667*** (0.073)	0.115*** (0.004)	0.036*** (0.001)	3.538*** (0.121)	0.267*** (0.012)	0.004*** (0.000)	1.187*** (0.053)	0.120*** (0.006)	0.338*** (0.015)
Observations	864,932	864,932	864,932	436,566	436,566	436,566	428,366	428,366	428,366

*** p<0.001, ** p<0.01, * p<0.05, † p<0.1.

Note: The table shows relative risk ratios, with **unemployment as the base outcome**. An RRR greater than 1 for the coefficient of interest implies that an individual in a lockdown region is more likely to be in that outcome relative to unemployment. All specifications include age and education controls (where higher education is the base category), a female dummy in the case of full samples, as well as region and period fixed-effects. Sample interpretation in the full sample: in lockdown regions, the risk of being formally employed relative to being unemployed is decreased by 45 percent relative to non-lockdown regions ((0.547-1.000)/1.000). Standard errors are reported in parentheses.

A.4 Supplementary Contextual Information

Table A.9: Social Protection and ALMP Coverage during the COVID-19 Pandemic (in %)

Program	All	Non-Lockdown	Lockdown
Social Protection Programs (all HHs)			
Social Amelioration Program	46.7	45.75	42.91
TUPAD	0.73	0.48	0.34
CAMP	0.64	0.4	0.9
AKAP	0.09	0.09	0.08
DTI Livelihood Seeding Program	0.05	0.06	0.02
DA Rice Farmers Support	2.88	2.01	0.61
Active Labor Market Policies (all HHs with non-working head)			
Micro-enterprise development	0.13	0.21	0.02
Employment facilitation	0.07	0	0.18
Skills training	0.02	0.05	0
Integrated Livelihood	0.16	0.18	0.16

Source: Own weighted estimates from the Annual Poverty Income Survey conducted in the first half of 2020

Note: TUPAD is a cash-for-work program for displaced and disadvantaged workers; CAMP is a one-time PHP 5,000 (USD 90) financial assistance for displaced workers; AKAP is a one-time PHP 10,000 (USD 180) financial assistance for displaced migrant workers; Department of Trade and Industry (DTI) Livelihood Seeding Program provides in-kind capital/inputs for micro, small, and medium enterprises. Department of Agriculture (DA) Rice Farmers Support Program is a PHP 5,000 (USD 90) cash aid for rice farmers. These estimates show extremely low coverage with some programs covering less than one-tenth of one percent of households. Notably, programs such as the Social Amelioration Program and TUPAD have slightly lower coverage in fully lockdown areas where arguably the effects of COVID are stronger. These own estimates are consistent with estimates presented in a World Bank report showing low coverage of social protection programs (Cho and Johnson 2022). It is also interesting to note that among households with a non-working head, coverage of ALMPs is likewise low—all programs have a coverage of less than 0.2 percent.

Table A.10: Survivalist Motive: Coping Mechanisms during COVID-19 (in %)

Coping Mechanism	National	Non-Lockdown	Lockdown
Sold assets / property	8.7	7.6	9.5
Engaged in additional income-generating activities	39.7	32.7	44.3
Engaged in online selling	20	13.5	29
Took advanced payment from employer	8.4	8.1	10

Source: Own weighted estimates from the World Bank (2022) COVID-19 High-Frequency Social Monitoring Survey Round 1 microdata collected 28 Aug-01 Sep 2020. Estimates are expressed as percent of all households.

Table A.11: Effects of the Lockdown on Working Hours

VARIABLES	All Employed			Formally Employed			Informally Employed		
	All	Males	Females	All	Males	Females	All	Males	Females
Lockdown area	6.407*** (0.085)	6.388*** (0.093)	6.652*** (0.188)	7.647*** (0.086)	7.199*** (0.095)	9.318*** (0.192)	0.895*** (0.234)	0.562† (0.300)	1.522*** (0.364)
Lockdown period	-5.014*** (0.183)	-6.106*** (0.212)	-3.213*** (0.336)	-5.057*** (0.184)	-5.634*** (0.216)	-3.867*** (0.343)	-4.722*** (0.536)	-9.128*** (0.773)	-1.950** (0.721)
Lockdown area*period	-0.040 (0.294)	-0.229 (0.373)	-0.425 (0.477)	0.058 (0.305)	-0.435 (0.382)	0.482 (0.510)	-1.119 (0.794)	1.238 (1.323)	-2.888** (0.993)
Age	-0.035*** (0.002)	-0.043*** (0.002)	-0.020*** (0.003)	-0.118*** (0.002)	-0.074*** (0.002)	-0.190*** (0.003)	0.172*** (0.005)	0.108*** (0.008)	0.224*** (0.008)
Basic education	-1.888*** (0.072)	-1.486*** (0.082)	-2.952*** (0.143)	-2.589*** (0.071)	-1.611*** (0.082)	-4.368*** (0.146)	0.267 (0.227)	0.902** (0.304)	-0.612† (0.331)
Some post-secondary ed	-1.264*** (0.087)	-1.974*** (0.108)	-0.445** (0.147)	-1.552*** (0.085)	-1.798*** (0.109)	-0.911*** (0.145)	-2.745*** (0.287)	-2.949*** (0.400)	-2.684*** (0.403)
Secondary education	1.494*** (0.058)	1.166*** (0.070)	1.991*** (0.099)	0.776*** (0.057)	1.079*** (0.069)	0.806*** (0.099)	1.299*** (0.193)	1.459*** (0.268)	1.124*** (0.271)
Female	-0.544*** (0.050)			-2.828*** (0.050)			3.579*** (0.140)		
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	42.323*** (0.133)	42.861*** (0.153)	40.650*** (0.257)	44.779*** (0.133)	43.031*** (0.155)	43.433*** (0.257)	39.394*** (0.396)	43.128*** (0.531)	39.633*** (0.582)
Observations	579,258	353,121	226,137	475,039	307,208	167,831	104,219	45,913	58,306
R-squared	0.048	0.057	0.043	0.082	0.069	0.103	0.043	0.041	0.040

Robust standard errors in parentheses

*** p<0.001, ** p<0.01, * p<0.05, † p<0.1.

Note: Coefficients are expressed in hourly units per week. Sample interpretation of the last column: among informally employed females, the lockdown has an effect of a 2.9-hour reduction in average working hours per week, ceteris paribus.

Appendix B

Appendix for Chapter 3

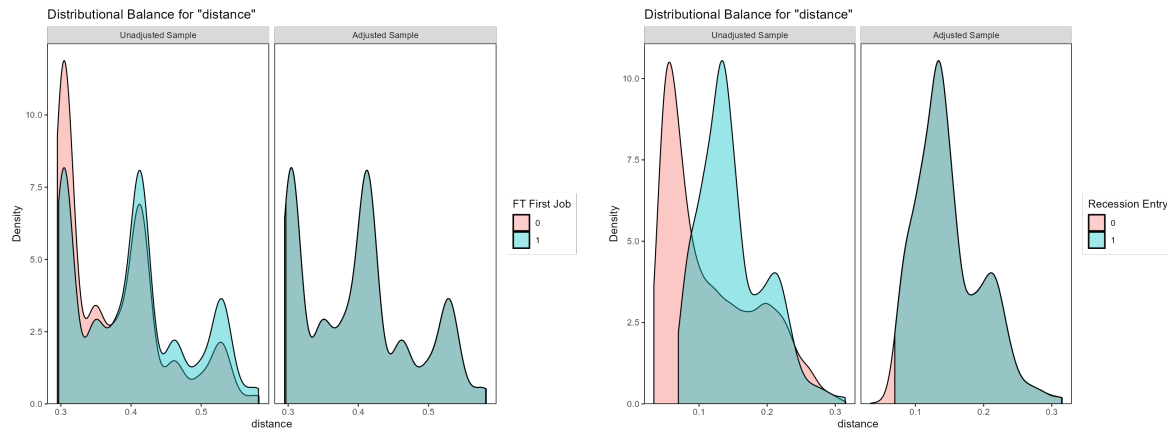
B.1 Supplementary Figures

Figure B.1: Sample selection criteria examples in the SOEP



Note: Individuals with green bars are retained in the sample for having valid information on labor market entry while individuals with black bars are dropped from the sample either for being a parent upon labor market entry or for being a censored case because of recent labor market entry.

Figure B.2: Covariate Balance Plots Pre- and Post-Matching

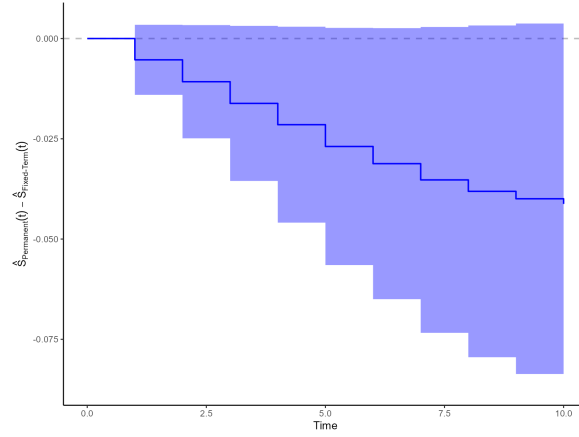


((a)) Fixed-Term vs. Permanent ATT

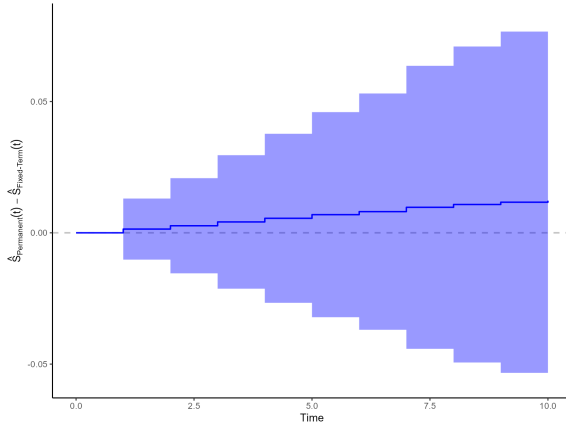
((b)) Recession vs. Non-Recession ATT

Note: The optimal full matching procedure using propensity scores was able to balance the means and standard deviations of the treatment and control groups with respect to the propensity to receive treatment, making both groups more comparable. These plots were generated using the cobalt R package (Greifer 2024a).

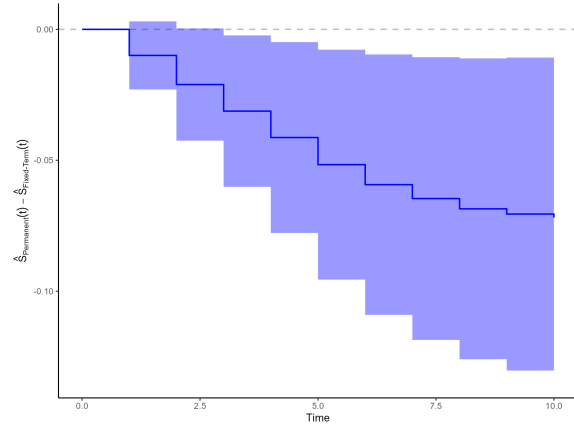
Figure B.3: Difference in Survival Probabilities for Fixed-Term Entry using Matched Sample



((a)) All



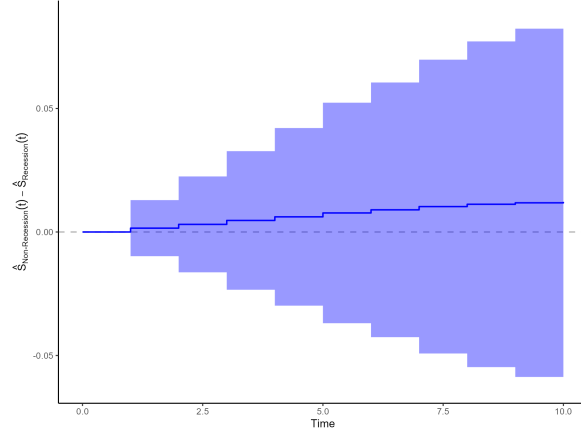
((b)) Males



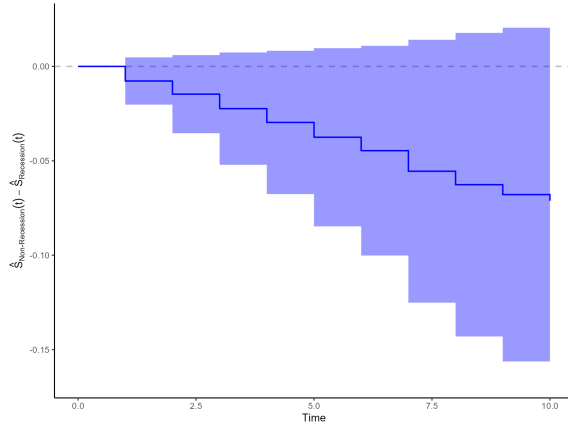
((c)) Females

Note: These plots show the difference in survival probabilities of treated and control groups over the first 10 years after LM entry using the ARTKALEN measure. A statistically significant difference between the groups can be inferred if the shaded blue area (confidence interval width) does not include zero. Panels (b) and (c) are group-specific estimates for males and females, respectively. These plots were generated using the R `adjustedCurves` package (Denz 2023).

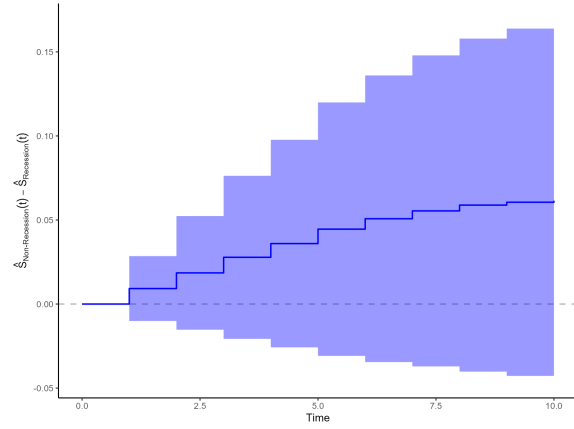
Figure B.4: Difference in Survival Probabilities for Recession Entry using Matched Sample



((a)) All



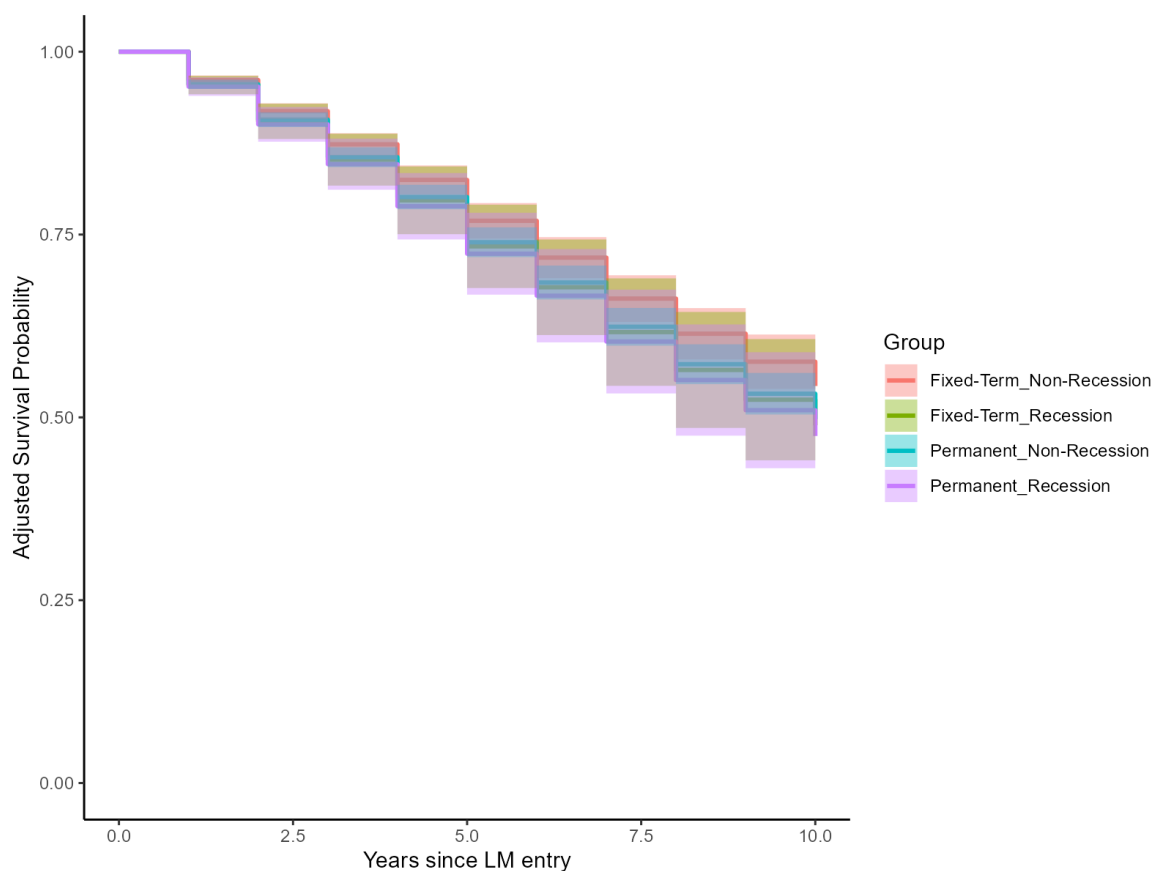
((b)) Males



((c)) Females

Note: These plots show the difference in survival probabilities of treated and control groups over the first 10 years after LM entry using the ARTKALEN measure. A statistically significant difference between the groups can be inferred if the shaded blue area (confidence interval width) does not include zero. Panels (b) and (c) are group-specific estimates for males and females, respectively. These plots were generated using the R `adjustedCurves` package (Denz 2023).

Figure B.5: Adjusted Survival Curves from Interaction of Fixed-Term and Recession Entry



Note: This plot shows the four adjusted survival curves for each combination of fixed-term/permanent and recession/non-recession entrants. These curves were fitted in post-estimation of Cox PH interaction models, controlling for the covariates shown in Appendix Table A3.1. Overlapping confidence intervals suggest no statistically significant difference between group-specific survival probabilities. These plots were generated using the R `adjustedCurves` package (Denz 2023).

B.2 Supplementary Tables

Table B.1: Deriving the Analytical Sample

	PBIO measure		ARTKALEN measure	
	Observations	Person-Yrs	Observations	Person-Yrs
Info on Fixed-Term First Job	9,083	80,846	9,164	75,497
Drop First Job = Unemployed	5,663		6,419	
Retain Childless Upon Entry	4,954		5,741	
Drop Missing Educ and Sex	4,795		5,500	
First Employment Status				
Fixed-term	1,718 (36%)		2,111 (38%)	
Permanent	3,077 (64%)		3,389 (62%)	
Recession Entry				
Recession Year	493 (10%)		583 (11%)	
Non-Recession Year	4,302 (90%)		4,917 (89%)	
Final Analytical Sample	4,795		5,500	
10-year post-attrition sample*	3,265		3,704	

Note: *In each model estimated for year-specific ATTs (see Figures 1 and 2), right-censored individuals are removed from the sample (e.g. those whose labor market entry is recent so there is no information yet about their fertility for that year). For illustration, consider a respondent who entered the labor market in 2015, a 2021 survey year only contains information about her fertility behavior within 6 years of entry, not more. Thus, this respondent is dropped from models estimating first birth probabilities for years 7-onwards. Note that in this analysis, we only focus on the first 10 years succeeding entry because expanding this would cause higher sample attrition.

Table B.2: Standardized Differences Pre- and Post-Matching on Fixed-Term Entry

Variables	ARTKALEN		PBIO	
	Unadjusted	Adjusted	Unadjusted	Adjusted
Distance	0.313	0.000	0.337	0.000
Female	0.008	0.011	0.003	0.038
Low Education	0.021	0.000	0.028	0.000
Mid Education	-0.106	0.000	-0.117	-0.005
High Education	0.084	0.000	0.089	0.005
Age at LM Entry	0.098	0.019	0.070	-0.025
East Germany	0.022	0.000	0.022	-0.023
Full-Time First Job	-0.114	0.000	-0.115	0.007
Migration Background	-0.014	-0.012	0.023	0.020

Note: These differences were calculated using the R `cobalt` package (Greifer 2024a).

Table B.3: Standardized Differences Pre- and Post-Matching on Recession Entry

	ARTKALEN		PBIO	
	Unadjusted	Adjusted	Unadjusted	Adjusted
Distance	0.646	0.000	0.596	0.001
Female	0.048	0.041	0.072	0.039
Low Education	-0.002	-0.017	-0.004	-0.003
Mid Education	0.020	0.026	-0.019	0.014
High Education	-0.018	-0.009	0.023	-0.011
Year of LM entry	-0.986	-0.048	-0.916	-0.029
Birth year	-0.849	-0.014	-0.757	0.010
State Unemployment Rate	0.386	-0.095	0.337	-0.077

Note: These differences were calculated using the R `cobalt` package (Greifer 2024a).

Table B.4: Cox PH Results on Matched Sample: Fixed-Term Entry (Interaction Models)

	<i>Dependent variable:</i>			
	TIME			
	(1)	(2)	(3)	(4)
Fixed-Term Entrant	0.906 (0.057)	0.879* (0.058)	0.839** (0.066)	1.035 (0.093)
Female		1.825*** (0.057)	1.827*** (0.057)	2.001*** (0.072)
Education-Mid		0.987 (0.097)	0.985 (0.097)	0.989 (0.096)
Education-High		1.475** (0.118)	1.470** (0.118)	1.485*** (0.118)
East Germany		1.368*** (0.068)	1.277** (0.084)	1.366*** (0.068)
LM Entry Age 21-25		1.275** (0.086)	1.276** (0.086)	1.277** (0.086)
LM Entry Age ≥ 26		1.689*** (0.111)	1.690*** (0.111)	1.689*** (0.111)
LM Entry Full-Time		1.178** (0.062)	1.177** (0.062)	1.183** (0.062)
Migration Background		1.419*** (0.068)	1.416*** (0.068)	1.420*** (0.068)
Fixed-Term X East Germany			1.212 (0.134)	
Fixed-Term X Female				0.770* (0.119)
Observations	5,500	5,500	5,500	5,500
R ²	0.001	0.051	0.052	0.052

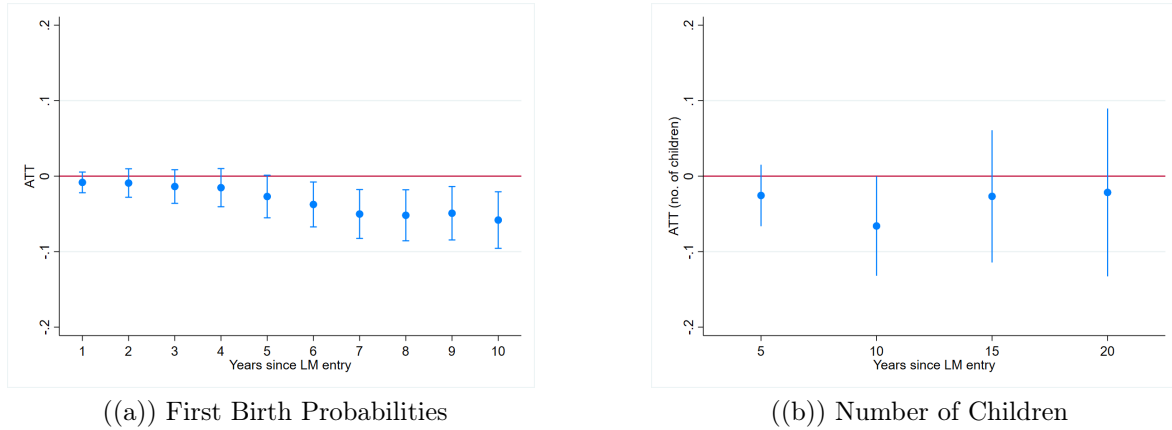
Note:

*p<0.1; **p<0.05; ***p<0.01

B.3 Supplementary Analyses using an Alternative LM Entry Measure

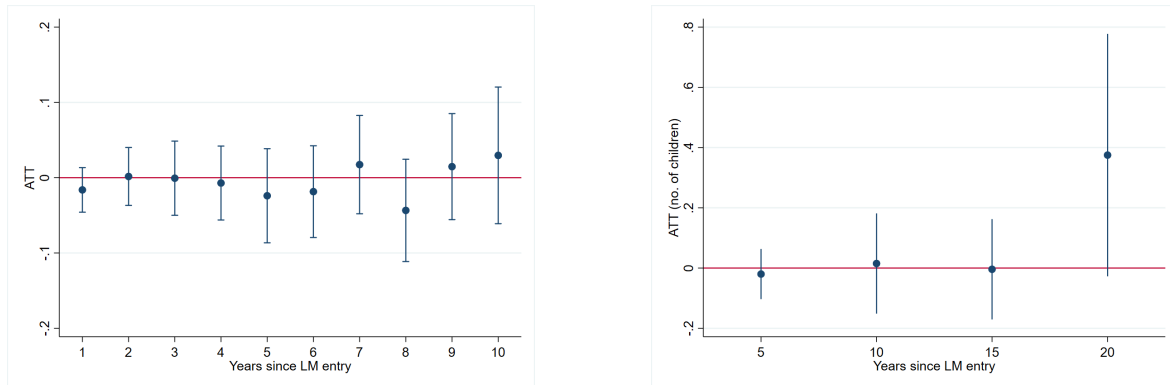
In this section, we present the replications of the figures in the main text (Figures 3.1 to 3.4 using an alternative measure of labor market entry present in the German Socio-Economic Panel. This alternative measure (PBIO) identifies the year of entry based on respondents' answers to the personal biography questionnaire. Appendix Table B.1 shows that while the shares of fixed-term first entrants and recession entrants of both measures are similar, the PBIO measure has a slightly lower sample size. Results are consistent for annual ATTs of fixed-term first jobs and recession entry and the gendered difference in the effects of recession entry.

Figure B.6: Fixed-Term First Jobs and Subsequent Fertility Outcomes using PBIO



Note: The matching variables used in these models are socio-demographic pre-treatment characteristics including sex, educational attainment at entry, East Germany, age at entry, and migration background.

Figure B.7: Recession Entry and Subsequent Fertility Outcomes using PBIO

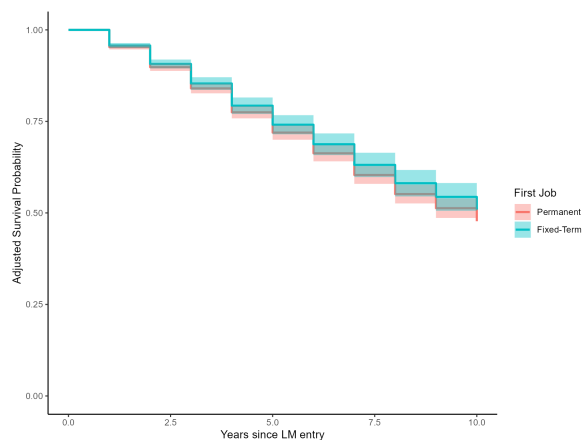


((a)) First Birth Probabilities

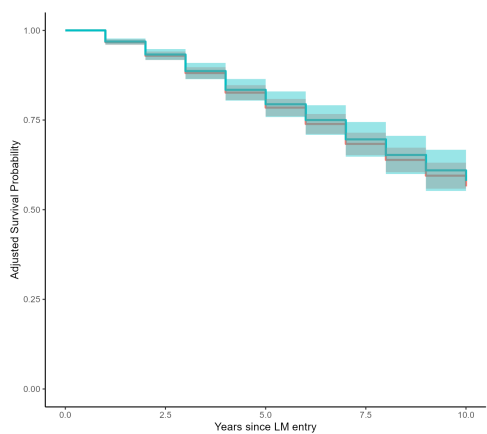
((b)) Number of Children

Note: The matching variables used in these models include the state-level (BuLa) unemployment rate and pre-treatment socio-demographic and temporal characteristics including sex, educational attainment at entry, and age at entry.

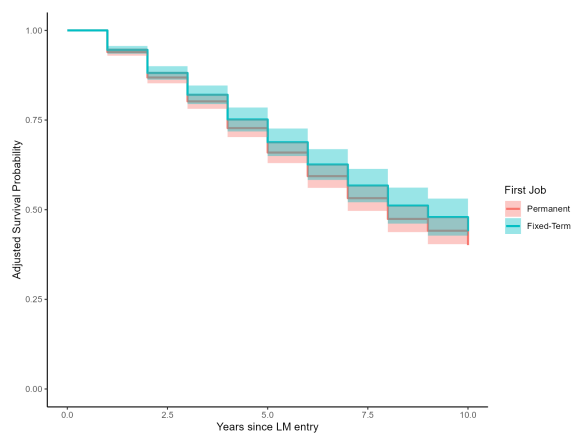
Figure B.8: Adjusted Survival Curves for Fixed-Term Entry using Matched Sample (PBIO)



((a)) All



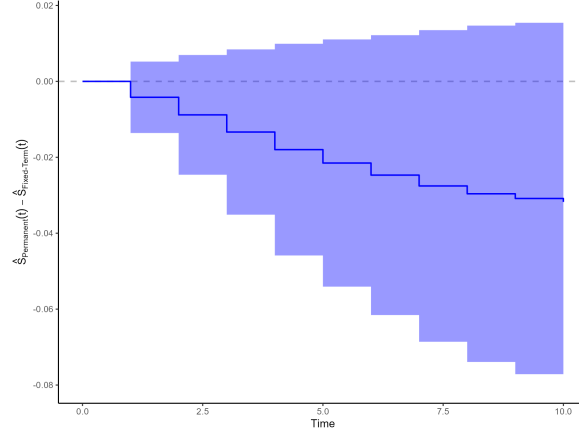
((b)) Males



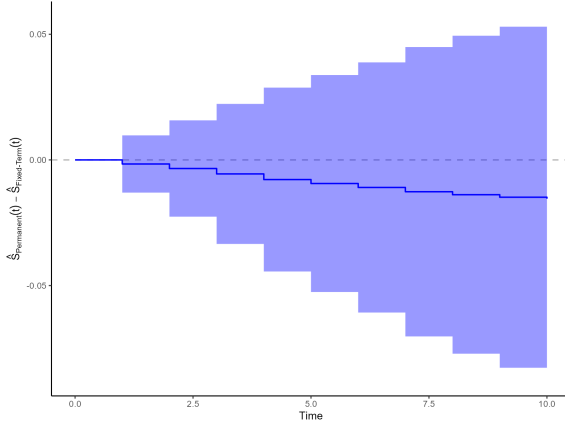
((c)) Females

Note: These survival curves are estimated with a maximum exit time capped at 10 years after LM entry. Panels (b) and (c) are group-specific estimates for males and females, respectively.

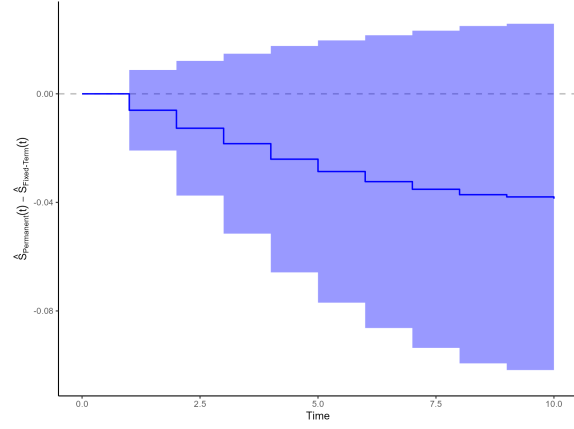
Figure B.9: Difference in Survival Probabilities for Fixed-Term Entry using Matched Sample (PBIO)



((a)) All



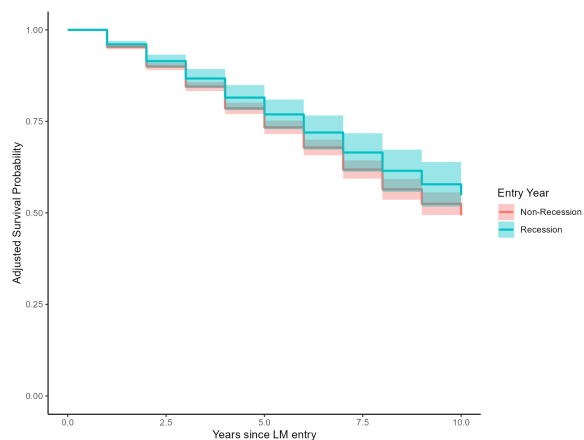
((b)) Males



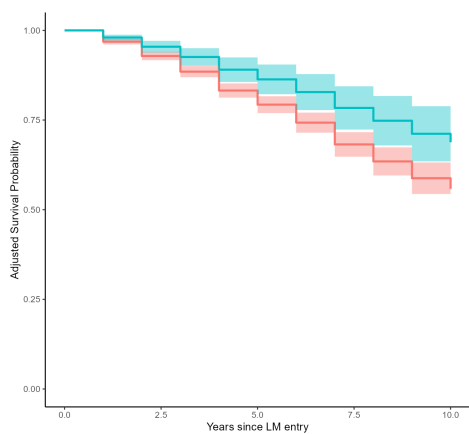
((c)) Females

Note: These plots show the difference in survival probabilities of treated and control groups over the first 10 years after LM entry. A statistically significant difference between the groups can be inferred if the shaded blue area (confidence interval width) does not include zero. Panels (b) and (c) are group-specific estimates for males and females, respectively. These plots were generated using the R `adjustedCurves` package (Denz 2023).

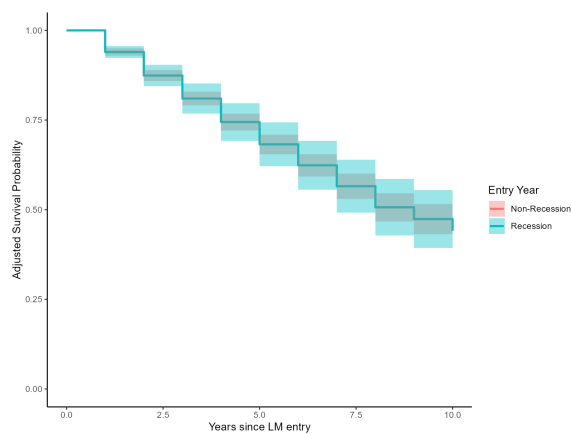
Figure B.10: Adjusted Survival Curves for Recession Entry using Matched Sample (PBIO)



((a)) All



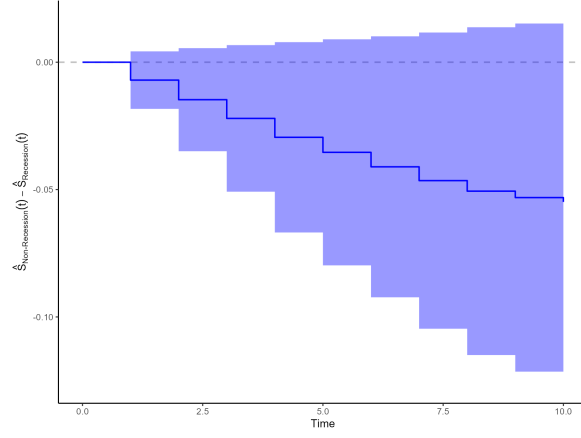
((b)) Males



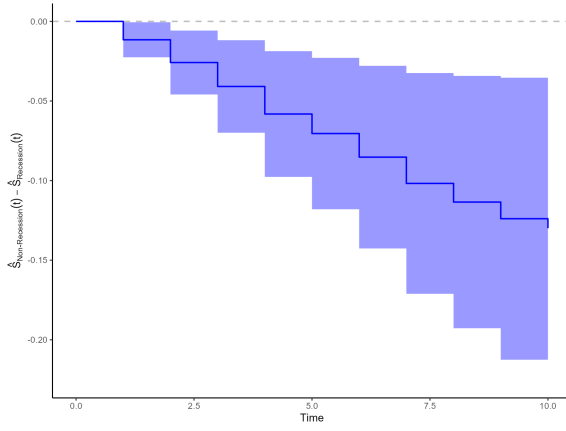
((c)) Females

Note: These survival curves are estimated with a maximum exit time capped at 10 years after LM entry. Panels (b) and (c) are group-specific estimates for males and females, respectively.

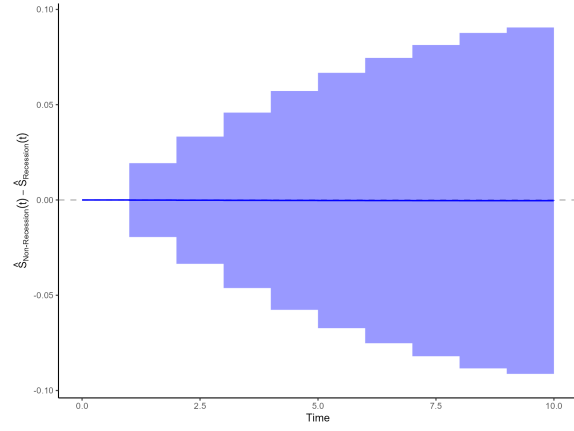
Figure B.11: Difference in Survival Probabilities for Recession Entry using Matched Sample (PBIO)



((a)) All



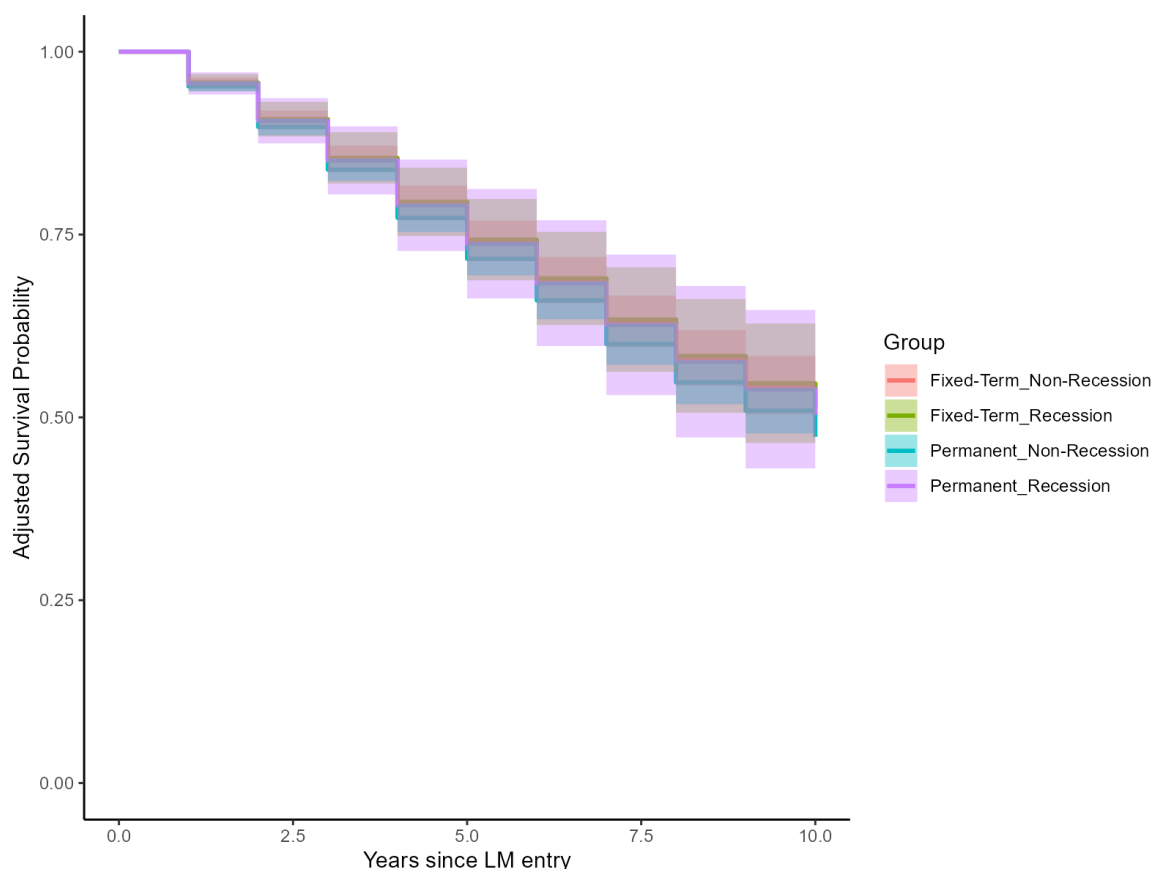
((b)) Males



((c)) Females

Note: These plots show the difference in survival probabilities of treated and control groups over the first 10 years after LM entry. A statistically significant difference between the groups can be inferred if the shaded blue area (confidence interval width) does not include zero. Panels (b) and (c) are group-specific estimates for males and females, respectively. These plots were generated using the R `adjustedCurves` package (Denz 2023).

Figure B.12: Adjusted Survival Curves from Interaction of Fixed-Term and Recession Entry (PBIO)



Note: This plot shows the four adjusted survival curves for each combination of fixed-term/permanent and recession/non-recession entrants. These curves were fitted in post-estimation of Cox PH interaction models, controlling for the covariates shown in Table 3. Overlapping confidence intervals suggest no statistically significant difference between group-specific survival probabilities. These plots were generated using the R `adjustedCurves` package (Denz 2023).

Table B.5: Cox PH Results on Matched Sample (PBIO): Fixed-Term by Recession Entry

	<i>Dependent variable:</i>			
	TIME			
	Full	with SUR	RY	Non-RY
Fixed-Term Entrant	0.907 (0.060)	0.935 (0.064)	0.959 (0.169)	0.929 (0.070)
Recession Entrant	0.938 (0.084)	0.953 (0.086)		
Female	1.562*** (0.057)	1.632*** (0.063)	1.888*** (0.179)	1.603*** (0.068)
Education-Mid	0.937 (0.098)	0.971 (0.119)	0.762 (0.296)	1.023 (0.131)
Education-High	1.385** (0.120)	1.475** (0.137)	1.588 (0.349)	1.491** (0.151)
East Germany	1.273** (0.068)	1.145 (0.102)	2.530** (0.350)	1.025 (0.108)
LM Entry Age 21-25	1.161 (0.090)	1.339** (0.113)	1.437 (0.284)	1.391** (0.125)
LM Entry Age 26-	1.558*** (0.111)	1.773*** (0.132)	1.609 (0.358)	1.892*** (0.145)
LM Entry Full-Time	1.139 (0.063)	1.094 (0.067)	0.732 (0.178)	1.163 (0.073)
Migration Background	1.472*** (0.069)	1.415*** (0.078)	2.118** (0.243)	1.363*** (0.082)
State UR at Entry		1.022* (0.010)	1.030 (0.034)	1.018 (0.010)
Observations	4,795	4,147	493	3,654
R ²	0.041	0.049	0.134	0.046

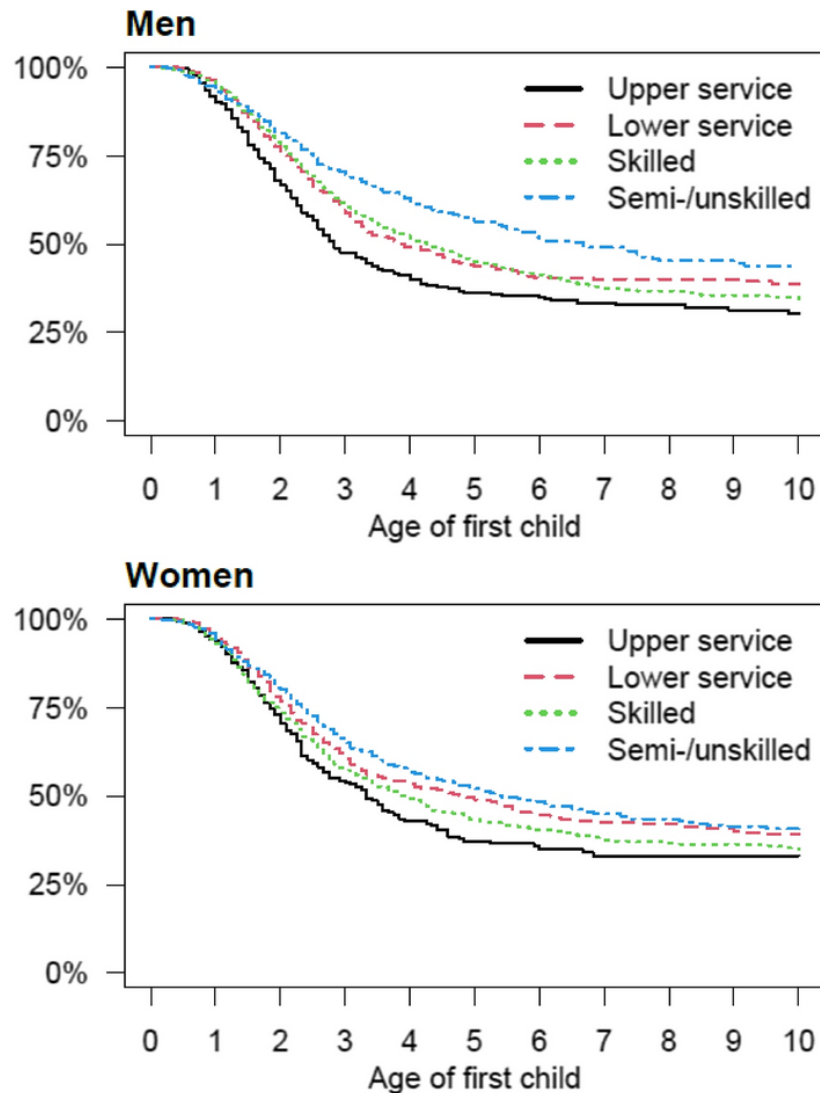
Note:

*p<0.1; **p<0.05; ***p<0.01

Appendix C

Appendix for Chapter 4

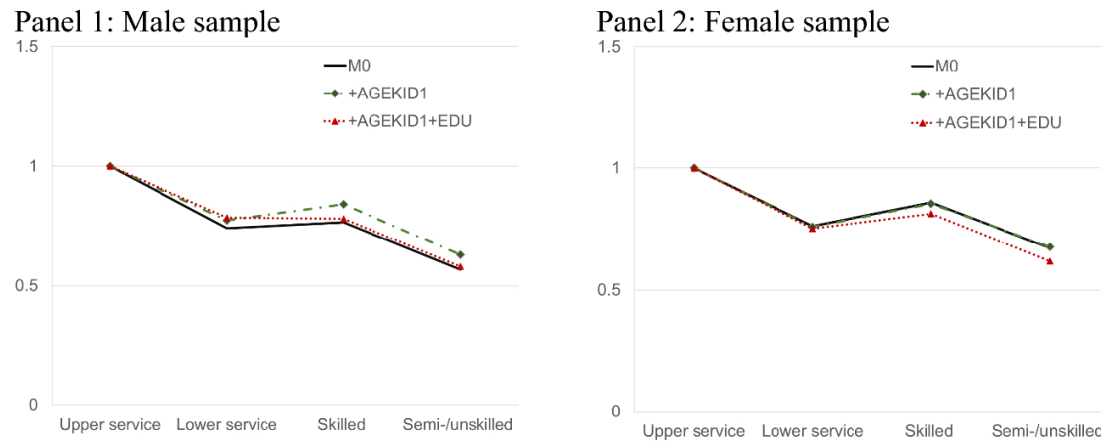
Figure C.1: Kaplan–Meier survival functions to the second birth by social class at the first birth and gender



Note: It should be noted that the survivals are not weighted. As the GSOEP oversamples certain groups, such as migrant populations and East Germans, caution is advised when considering the descriptive results, as they do not control for migration status and region (as is done in the multiple regression). Individuals who were unemployed when they had their first child are not included, as they represent a small fraction of the men, and including them would have resulted in unstable estimates of the survival curves.

Source: SOEP, v37, 1990–2020. Own unweighted estimates)

Figure C.2: Robustness Checks: Piecewise constant event history model. Relative second birth risks (hazard ratios).



Note: ($n = 108,241$ person-months for male sample; $n = 132,348$ person-months for female sample). The hazard ratio for “not employed” is not displayed in the figures due to the small sample size in the male sample. M0: Further variables in the model are the age of the first child, region, migration status, and social mobility. M0 + AGEKID1: M0 + age at first birth. M0 + AGEKID1 + EDU: M0 + age at first birth + education.

Source: SOEP, v37, 1990–2020. Own unweighted estimates

Table C.1: Sample statistics, time-constant covariates, column %

	Men	Women
<i>Social class in the year of the first batch</i>		
Upper service	19	10
Lower service	19	26
Skilled	36	24
Semi-/unskilled	19	23
Not employed	7	18
<i>Partner's social class at first batch</i>		
Upper service	8	13
Lower service	20	12
Skilled	19	22
Semi-/unskilled	16	11
Not employed	13	4
No partner/missing	25	38
<i>Migration background</i>		
Native	70	71
Migration background	30	29
<i>Age at first birth</i> (mean)	30.8	28.1
<i>Sample size</i>		
Persons	2282	2819
Second births	1161	1371
<i>Source: SOEPv37, 1990-2020, unweighted estimates</i>		

Table C.2: Sample statistics, time-varying covariates, column %

	Men	Women
<i>Mobility</i>		
Stable	67	38
Upward	9	5
Downward	7	5
Not employed	14	52
Other	2	1
<i>Education</i>		
Low	11	12
Medium	63	62
High	24	23
Missing	2	3
<i>Financial worries</i>		
No worries	20	22
Stable	53	56
Great worries	27	21
<i>Region</i>		
West Germany	78	77
East Germany	22	23
<i>Sample size</i>		
Person-months	108,241	132,384
Second-births	1161	1371

There are some few (< 1%) missings for financial worries

Source: SOEPv37, 1990-2020, unweighted estimates

Table C.3: Social class of woman by partner's social class, column %

Social class woman	Social class man				
	Upper service	Lower service	Skilled	Semi-/unskilled	Not employed
Upper service	48	24	17	7	10
Lower service	26	26	18	13	14
Skilled	15	33	44	46	35
Semi-unskilled	9	14	18	30	30
Not employed	2	3	3	5	11

n = 83,654 person-months, Spearman's correlation coefficient: 0.34

Source: SOEPv37, 1990-2020, unweighted estimates

Table C.4: Social class of man by partner's social class, column %

Social class woman	Social class men				
	Upper service	Lower service	Skilled	Semi-/unskilled	Not employed
Upper service	24	13	4	4	3
Lower service	36	39	25	20	13
Skilled	22	24	31	23	18
Semi-unskilled	10	15	29	33	25
Not employed	8	9	12	20	40

n = 81,669 person-months, Spearman's correlation coefficient: 0.34

Source: SOEPv37, 1990-2020, unweighted estimates

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Appendix D

Appendix for Chapter 5

Table D.1: Descriptive Statistics

Variables	Observations	%
Future care uncertainty		
High	2,176	24.9
Some	2,210	25.3
Low	2,179	24.9
None	2,185	25.0
Future employment uncertainty		
Both	2,177	24.9
Male	2,181	24.9
Female	2,250	25.7
None	2,142	24.5
Sex		
Male	4,315	49.3
Female	4,435	50.7
Lena's age		
25	2,905	33.2
30	2,760	31.5
35	3,085	35.3
Household's Financial Worries		
Major worries	850	9.7
Some worries	3,605	41.2
No worries	3,935	45.0
Don't know	345	3.9
Marital Status		
Married	5,195	59.4
Single	1,920	21.9
Divorced/Widowed	1,590	18.2
Others/Don't Know	45	0.5
Age Group		
18-35	1,405	16.1
36-50	1,930	22.1
51-65	2,585	29.5
66+	2,830	32.3

Note: These statistics are tabulated from observations, not individual respondents. Each respondent assesses 5 vignettes and thus yield 5 observations.

Table D.2: Full Regression Results

Variables	Base	Full	Males	Females
Future Care Uncertainty- High	Ref	Ref	Ref	Ref
Some	0.673*** (0.075)	0.683*** (0.075)	0.582*** (0.103)	0.770*** (0.109)
Low	1.493*** (0.083)	1.492*** (0.083)	1.186*** (0.118)	1.784*** (0.118)
None	2.800*** (0.086)	2.806*** (0.086)	2.582*** (0.121)	3.025*** (0.121)
Future Employment Uncertainty- Both	Ref	Ref	Ref	Ref
Male	0.291*** (0.079)	0.287*** (0.079)	0.248* (0.106)	0.332** (0.115)
Female	0.578*** (0.077)	0.580*** (0.077)	0.633*** (0.105)	0.531*** (0.11)
None	1.907*** (0.085)	1.903*** (0.085)	1.906*** (0.115)	1.915*** (0.122)
Lena's age (vignette)- 25		Ref	Ref	Ref
30		0.122 (0.114)	0.011 (0.155)	0.232 (0.168)
35		0.0247 (0.109)	0.0179 (0.15)	0.0518 (0.159)
Household's Financial Worries- Major		Ref	Ref	Ref
Some		0.119 (0.172)	-0.0681 (0.247)	0.311 (0.238)
None		0.172 (0.172)	0.188 (0.245)	0.18 (0.24)
Don't Know		0.0689 (0.286)	0.351 (0.362)	-0.216 (0.442)
Age Group- 18-35		Ref	Ref	Ref
46-50		0.149 (0.137)	0.299 (0.186)	-0.004 (0.199)
51-65		-0.164 (0.14)	-0.0939 (0.184)	-0.278 (0.208)
66 and above		-0.168 (0.152)	-0.0075 (0.201)	-0.36 (0.226)
Marital Status- Married		Ref	Ref	Ref
Single		-0.301** (0.116)	-0.253 (0.153)	-0.362* (0.175)
Divorced/Widowed		-0.216 (0.136)	0.358 (0.21)	-0.101 (0.18)
Don't Know		-1.144 (0.802)	-0.923* (0.405)	-1.433 (1.772)
Female Dummy		0.139 (0.093)		
Constant	2.342*** (0.0861)	2.271*** (0.236)	2.415*** (0.318)	2.252*** (0.328)
Observations	8501	8490	4168	4322
Adjusted R squared	0.193	0.198	0.19	0.209

Individual level-clustered errors in parentheses

* p<0.05, ** p<0.01, *** p<0.001

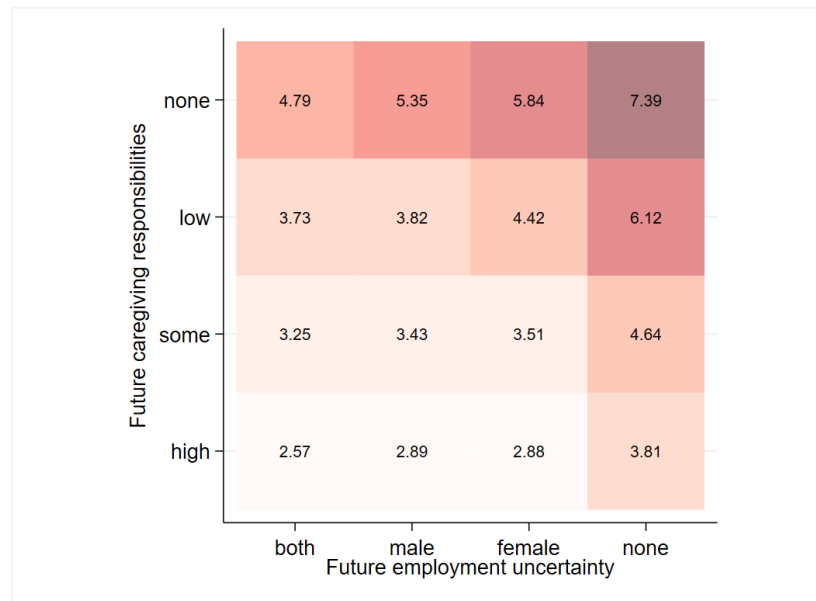


Figure D.1: Mean Fertility Intentions (0-10 scale) by future caregiving responsibilities and employment uncertainties

Appendix E

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Appendix F

List of Papers

This cumulative dissertation includes introductory (Chapter 1) and concluding (Chapter 6) sections and four stand-alone articles (Chapters 2 to 5), namely:

1. **CH 2.** Extreme Lockdowns and the Gendered Informalization of Employment: Evidence from the Philippines
 - sole-authored article **published** at *Work, Employment, and Society* available online: <https://doi.org/10.1177/09500170241247121>
2. **CH 3.** Initial Employment Conditions and Subsequent Fertility Behavior
 - sole-authored article **under review (first round)**
3. **CH 4.** Second Birth Fertility in Germany: Social Class, Gender, and the Role of Economic Uncertainty
 - co-authored article with Michaela Kreyenfeld, Dirk Konietzka, and Philippe Lambert **published** at the *European Journal of Population* available online: <https://doi.org/10.1007/s10680-023-09656-5>
4. **CH 5.** On Ascribed Fertility Intentions Under Future Caregiving Responsibilities and Employment Uncertainties: Experimental Evidence from Germany
 - co-authored article with Michaela Kreyenfeld, Julie O’Sullivan, Enrique Alonso-Perez, Jan Paul Heisig, and Paul Gellert **under review (revise & resubmit)**